SET 23 • HSA-REI.B.4.b

FORWARD 1

Find the zeros of a polynomial from its factors • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

Find all zeros of $p(x) = (x - 2)(x + 1)(x - 5)$.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 23 • FORWARD 1 • Q1/5

Find all zeros of $p(x) = x^3 - 4x$.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 23 • FORWARD 1 • Q2/5

Which polynomial has zeros -3, 0, and 4?

A. $x(x + 3)(x - 4)$

B. $(x + 3)(x - 4)$

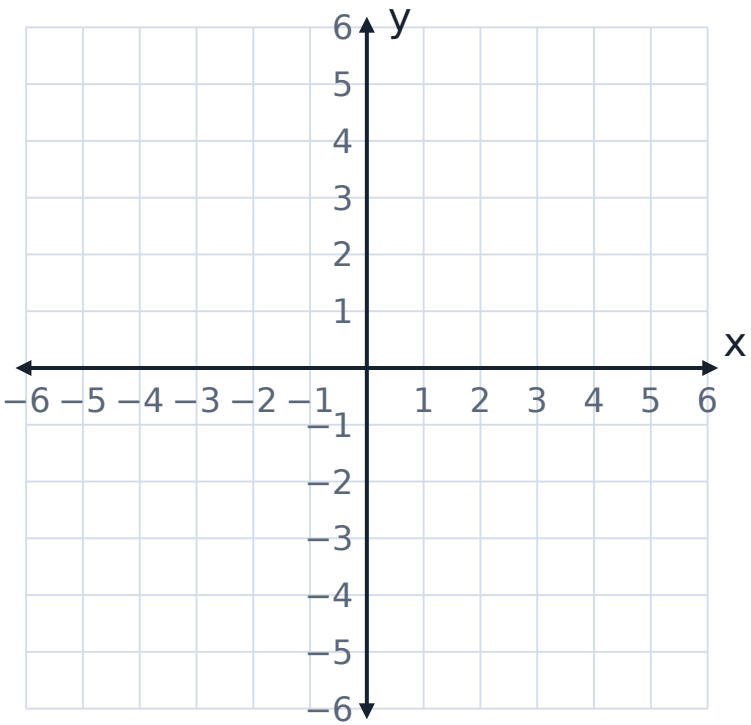
C. $x(x - 3)(x + 4)$

D. $x(x - 3)(x - 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F1-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 23 • FORWARD 1 • Q3/5

Sketch a rough graph of $p(x) = -(x + 3)(x - 1)(x - 4)$. Mark the zeros and the y-intercept, and show the end behavior. (Use the grid for x; the y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 23 • FORWARD 1 • Q4/5

True or false?

$p(x) = x^2(x - 3)$ has three different zeros.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 23 • FORWARD 1 • Q5/5

SET 23 • HSA-REI.B.4.b

FORWARD 2

Use the Fundamental Theorem of Algebra to count complex zeros • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9

How many complex zeros, counted with multiplicity, does a polynomial of degree 5 have?

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 23 • FORWARD 2 • Q1/5

$p(x) = x^3 - x^2 + 4x - 4 = (x - 1)(x^2 + 4)$.

Find all complex zeros.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 23 • FORWARD 2 • Q2/5

A polynomial with real coefficients has zeros 3 and 2i.

Which number must also be a zero?

A. 3i

B. 2

C. −3

D. −2i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F2-Q3**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 23 • FORWARD 2 • Q3/5

True or false?

$x^2 + 1 = (x + i)(x - i)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 23 • FORWARD 2 • Q4/5

True or false?

Every polynomial of degree 3 has three real zeros.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F2-Q5**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 23 • FORWARD 2 • Q5/5

Solve quadratic equations with the quadratic formula and use the discriminant

MAIN	Algebra I main question	**HSA-REI.B.4.b**
BACKWARD 1	Complete the square to solve a quadratic equation	**HSA-REI.B.4.a**
BACKWARD 2	Estimate square roots with rational approximations	**8.NS.A.2**
FORWARD 1	Solve right-triangle problems that lead to quadratic equations	**HSG-SRT.C.8**
FORWARD 2	Solve quadratic equations with complex solutions	**HSN-CN.C.7**

SET 24 • HSA-REI.B.4.b

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve $2x^2 - 3x - 4 = 0$ with the quadratic formula.

Give exact answers in simplest radical form.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 24 • MAIN • Q1/5

Solve $x^2 + 4x - 1 = 0$. Round each solution to the nearest hundredth.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 24 • MAIN • Q2/5

Use the discriminant to find how many real solutions $3x^2 - 2x + 5 = 0$ has.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 24 • MAIN • Q3/5

Solve $x^2 + 2x + 7 = 0$ over the real numbers.

A. No real solution

B. $x = -1 + \sqrt{6}$ or $x = -1 - \sqrt{6}$

C. $x = -7$ or $x = 1$

D. $x = 1 + \sqrt{6}$ or $x = 1 - \sqrt{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 24 • MAIN • Q4/5

True or false?

The equation $4x^2 - 12x + 9 = 0$ has exactly one real solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 24 • MAIN • Q5/5

SET 24 • HSA-REI.B.4.b

BACKWARD 1

Complete the square to solve a quadratic equation • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a

Solve $x^2 - 4x = 5$ by completing the square.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 24 • BACKWARD 1 • Q1/5

Solve $x^2 + 2x - 2 = 0$ by completing the square. Give exact answers.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 24 • BACKWARD 1 • Q2/5

Which equation results from completing the square on $x^2 - 6x + 4 = 0$?

A. $(x + 3)^2 = 5$

B. $(x - 3)^2 = 5$

C. $(x - 3)^2 = 13$

D. $(x - 6)^2 = 32$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-B1-Q3**

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 24 • BACKWARD 1 • Q3/5

True or false?

Adding 25 to both sides of $x^2 + 10x = 3$ gives $(x + 5)^2 = 28$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 24 • BACKWARD 1 • Q4/5

True or false?

Completing the square on $x^2 + 6x = 1$ gives $(x + 3)^2 = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 24 • BACKWARD 1 • Q5/5

SET 24 • HSA-REI.B.4.b

BACKWARD 2

Estimate square roots with rational approximations • Grade 8 • The Number System • 8.NS.A.2

Between which two consecutive whole numbers is $\sqrt{41}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 24 • BACKWARD 2 • Q1/5

Round $\sqrt{5}$ to the nearest hundredth.

$\sqrt{5} \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 24 • BACKWARD 2 • Q2/5

Which is the best estimate of $\sqrt{30}$?

A. 6.2

B. 5.0

C. 15

D. 5.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 24 • BACKWARD 2 • Q3/5

True or false?

$\sqrt{2} \approx 1.41$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 24 • BACKWARD 2 • Q4/5

True or false?

$\sqrt{90}$ is between 8 and 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 24 • BACKWARD 2 • Q5/5

SET 24 • HSA-REI.B.4.b

FORWARD 1

Solve right-triangle problems that lead to quadratic equations • High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

A right triangle's legs are x and x + 7, and its hypotenuse is 13.

Write an equation and find the lengths of the legs.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 24 • FORWARD 1 • Q1/5

A 10-ft ladder leans against a wall and makes a 70° angle with the ground.

How high up the wall does it reach, to the nearest tenth of a foot?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 24 • FORWARD 1 • Q2/5

A rectangle's diagonal is 10, and its length is 2 more than its width.

What is the width?

A. -8

B. 6

C. 8

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F1-Q3**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 24 • FORWARD 1 • Q3/5

True or false?

A ramp rises 1 ft over a horizontal run of 12 ft. Its angle with the ground is about 4.8°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 24 • FORWARD 1 • Q4/5

True or false?

A right triangle with legs 1 and 1 has a hypotenuse of 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F1-Q5**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 24 • FORWARD 1 • Q5/5

SET 24 • HSA-REI.B.4.b

FORWARD 2

Solve quadratic equations with complex solutions • High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7

Solve $x^2 - 4x + 13 = 0$. Write the solutions in the form $a + bi$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q1

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 24 • FORWARD 2 • Q1/5

Solve $x^2 + 25 = 0$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q2

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 24 • FORWARD 2 • Q2/5

What are the solutions of $x^2 + 2x + 5 = 0$?

A. $1 + 2i$ and $1 - 2i$

B. $-1 + 4i$ and $-1 - 4i$

C. $-2 + 4i$ and $-2 - 4i$

D. $-1 + 2i$ and $-1 - 2i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q3

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 24 • FORWARD 2 • Q3/5

True or false?

If a quadratic equation with real coefficients has a negative discriminant, its two solutions are complex conjugates.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 24 • FORWARD 2 • Q4/5

True or false?

The solutions of $x^2 + 2x + 2 = 0$ are $1 + i$ and $1 - i$.

A. True

B. False

Create quadratic and exponential equations in one variable and use them to solve problems

MAIN	Algebra I main question	**HSA-CED.A.1**
BACKWARD 1	Solve quadratic equations	**HSA-REI.B.4.b**
BACKWARD 2	Use properties of integer exponents	**8.EE.A.1**
FORWARD 1	Use volume formulas to write and solve equations for a dimension	**HSG-GMD.A.3**
FORWARD 2	Solve exponential models with logarithms	**HSF-LE.A.4**

SET 25 • HSA-CED.A.1

MAIN

Algebra I main question • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1

A ball's height is $h = -16t^2 + 48t + 4$ feet after t seconds.

Write and solve an equation to find when the ball is 36 feet high.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 25 • MAIN • Q1/5

A rectangle's length is 5 cm more than its width. Its area is 84 cm².

Write an equation and find the dimensions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 25 • MAIN • Q2/5

A culture of 200 bacteria doubles every hour.

Write an equation for the number of hours t until there are 6,400 bacteria, and solve it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 25 • MAIN • Q3/5

A $2,000 investment earns 5% interest, added once at the end of each year.

Which inequality gives the whole numbers of years n after which it is worth at least $3,000?

A. $2{,}000(1.5)^n \geq 3{,}000$

B. $2{,}000(0.05)^n \geq 3{,}000$

C. $2{,}000 + 0.05n \geq 3{,}000$

D. $2{,}000(1.05)^n \geq 3{,}000$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 25 • MAIN • Q4/5

True or false?

A square garden of side x ft has a 2-ft path around it, and the garden and path cover 144 ft². Then $(x + 4)^2 = 144$, so $x = 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 25 • MAIN • Q5/5

SET 25 • HSA-CED.A.1

BACKWARD 1

Solve quadratic equations • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve $t^2 - 3t + 2 = 0$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 25 • BACKWARD 1 • Q1/5

Solve $w^2 + 5w = 84$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 25 • BACKWARD 1 • Q2/5

Solve $(x + 4)^2 = 144$.

A. $x = 12$ or $x = -12$

B. $x = 140$ or $x = -148$

C. $x = 8$ only

D. $x = 8$ or $x = -16$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 25 • BACKWARD 1 • Q3/5

True or false?

$x = -2$ is a solution of $x^2 + 5x + 6 = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 25 • BACKWARD 1 • Q4/5

True or false?

The only solution of $x^2 = 3x$ is $x = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 25 • BACKWARD 1 • Q5/5

SET 25 • HSA-CED.A.1

BACKWARD 2

Use properties of integer exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

Write 32 as a power of 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 25 • BACKWARD 2 • Q1/5

Simplify $2^3 \cdot 2^4$. Write the answer as a power of 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 25 • BACKWARD 2 • Q2/5

Which power of 2 equals 64?

A. 2^{32}

B. 2^{8}

C. 2^{5}

D. 2^{6}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 25 • BACKWARD 2 • Q3/5

True or false?

$3^4 = 81$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 25 • BACKWARD 2 • Q4/5

True or false?

$2^{-3} = \frac{1}{8}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 25 • BACKWARD 2 • Q5/5

SET 25 • HSA-CED.A.1

FORWARD 1

Use volume formulas to write and solve equations for a dimension • High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A cylinder has height 10 cm and volume 250π cm³.

Write an equation and find its radius.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 25 • FORWARD 1 • Q1/5

A square pyramid has height 9 in and volume 300 in³.

Find the side length of its base.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 25 • FORWARD 1 • Q2/5

A cone has volume 12π and height 4.

What is its radius?

A. 9

B. 6

C. $\sqrt{3}$

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F1-Q3**

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 25 • FORWARD 1 • Q3/5

True or false?

A sphere of radius 3 has volume 36π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 25 • FORWARD 1 • Q4/5

True or false?

Doubling the radius of a cylinder doubles its volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F1-Q5**

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 25 • FORWARD 1 • Q5/5

SET 25 • HSA-CED.A.1

FORWARD 2

Solve exponential models with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $200 \cdot 2^{t} = 5{,}000$. Give the exact answer using logarithms, then round it to the nearest hundredth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 25 • FORWARD 2 • Q1/5

Solve $2{,}000(1.05)^n = 3{,}000$. Round to the nearest tenth.

$n \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 25 • FORWARD 2 • Q2/5

Solve $10^x = 500$. (log means base 10.)

A. $x = 2.5$

B. $x = 5{,}000$

C. $x = 50$

D. $x = \log 500 \approx 2.70$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 25 • FORWARD 2 • Q3/5

True or false?

The solution of $e^x = 7$ is $x = \ln 7$. (ln means base e.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 25 • FORWARD 2 • Q4/5

True or false?

The solution of $5^x = 30$ is $x = 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 25 • FORWARD 2 • Q5/5

Show that replacing an equation by its sum with a multiple of the other keeps the solutions of a system

MAIN	Algebra I main question	**HSA-REI.C.5**
BACKWARD 1	Justify steps with the properties of equality	**HSA-REI.A.1**
BACKWARD 2	Recognize the solution of a system as a point on both lines	**8.EE.C.8.a**
FORWARD 1	Write a system as a matrix equation AX = B and use row operatio..	**HSA-REI.C.8**
FORWARD 2	Use determinants and inverse matrices	**HSN-VM.C.10**

SET 26 • HSA-REI.C.5

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5

System: $x + y = 7$ and $x - y = 1$.

Replace the second equation by the sum of the two equations. Write the new system, solve it, and show that the solution also satisfies the original system.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 26 • MAIN • Q1/5

Suppose (a, b) is a solution of the system $2x + y = 5$ and $x - 3y = 6$.

Explain why (a, b) is also a solution of $(2x + y) + 2(x - 3y) = 5 + 2(6)$.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 26 • MAIN • Q2/5

System: $3x + 2y = 12$ and $x - y = -1$.

Which system has the same solution?

A. $3x + 2y = 12$ and $4x + y = 13$

B. $3x + 2y = 12$ and $2x + 3y = 11$

C. $3x + 2y = 12$ and $5x = 14$

D. $3x + 2y = 12$ and $5x = 10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 26 • MAIN • Q3/5

True or false?

Replacing one equation of a system by the sum of that equation and 3 times the other equation can change the solutions of the system.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 26 • MAIN • Q4/5

True or false?

In the system $2x + 3y = 7$ and $4x - 3y = 5$, adding the equations eliminates y and gives $6x = 12$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 26 • MAIN • Q5/5

SET 26 • HSA-REI.C.5

BACKWARD 1

Justify steps with the properties of equality • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

If a = b and c = d, explain why a + c = b + d.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 26 • BACKWARD 1 • Q1/5

Name the property used to go from $x = 5$ to $2x = 10$.

Property: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 26 • BACKWARD 1 • Q2/5

If $p = q$, which equation must also be true?

A. $4p = q$

B. $p = q + 4$

C. $p + 4 = q + 4$

D. $p + 4 = q$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 26 • BACKWARD 1 • Q3/5

True or false?

If a = b, then 5a = 5b.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 26 • BACKWARD 1 • Q4/5

True or false?

If $a = b$ and $c = d$, then $a - c = b + d$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 26 • BACKWARD 1 • Q5/5

SET 26 • HSA-REI.C.5

BACKWARD 2

Recognize the solution of a system as a point on both lines • Grade 8 • Expressions & Equations • 8.EE.C.8.a

Is (2, 5) a solution of the system $y = 2x + 1$ and $y = x + 3$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 26 • BACKWARD 2 • Q1/5

The lines $y = x + 1$ and $y = -x + 5$ meet at $(2, 3)$.

What is the solution of the system?

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 26 • BACKWARD 2 • Q2/5

Which point is a solution of both $x + y = 6$ and $x - y = 2$?

A. (2, 4)

B. (4, 2)

C. (3, 3)

D. (5, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 26 • BACKWARD 2 • Q3/5

True or false?

The solution of a system of two linear equations is the point where their graphs intersect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 26 • BACKWARD 2 • Q4/5

True or false?

(1, 1) is a solution of the system y = 2x and y = x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 26 • BACKWARD 2 • Q5/5

SET 26 • HSA-REI.C.5

FORWARD 1

Write a system as a matrix equation AX = B and use row operations • High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8

Write the system $2x + y = 7$ and $x - 3y = 0$ as a matrix equation $AX = B$. Give A, X, and B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q1

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q1/5

In the matrix equation $AX = B$, A has rows $[1\ 2]$ and $[3\ -1]$, X is the column $[x, y]$, and B is the column $[5, 1]$.

Write the system it represents.

System: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q2

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q2/5

On the rows [1 2 | 5] and [3 −1 | 1], replace row 2 by (row 2) − 3 × (row 1).

What is the new row 2?

A. [0 −7 | 14]

B. [2 −3 | −4]

C. [0 −7 | −14]

D. [0 5 | 16]

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F1-Q3**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q3/5

True or false?

Adding a multiple of one row of an augmented matrix to another row does the same thing as replacing that equation by its sum with a multiple of the other equation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q4/5

True or false?

The coefficient matrix of $x + y = 3$ and $2x - y = 0$ has the rows [1 1] and [2 1].

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q5/5

SET 26 • HSA-REI.C.5

FORWARD 2

Use determinants and inverse matrices • High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10

Find the determinant of the matrix with rows [3 1] and [5 2].

Determinant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q1/5

Show that the matrix with rows [2 −1] and [−5 3] is the inverse of the matrix with rows [3 1] and [5 2].

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q2/5

Which matrix has no inverse?

A. Rows [2 4] and [1 2]

B. Rows [1 0] and [0 1]

C. Rows [2 1] and [1 1]

D. Rows [3 1] and [5 2]

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q3/5

True or false?

If the coefficient matrix of a system of two linear equations has an inverse, the system has exactly one solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q4/5

True or false?

The inverse of the matrix with rows [2 0] and [0 4] has rows [−2 0] and [0 −4].

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q5/5

Prove that replacing an equation by its sum with a multiple of the other gives an equivalent system, in both directions

MAIN	Algebra I main question	**HSA-REI.C.5**
BACKWARD 1	Use the properties of equality to justify a step	**HSA-REI.A.1**
BACKWARD 2	Solve simple systems by adding equations	**8.EE.C.8.b**
FORWARD 1	Use row operations on an augmented matrix as elimination steps	**HSA-REI.C.8**
FORWARD 2	Solve the same system with an inverse matrix	**HSA-REI.C.9**

SET 72 • HSA-REI.C.5

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5

System A: $x + 2y = 7$ and $3x - y = 7$. System B: $x + 2y = 7$ and $7x = 21$, where $7x = 21$ is the first equation plus 2 times the second.

Show that every solution of A solves B and every solution of B solves A. Then give the solution.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 72 • MAIN • Q1/5

Let P = c and Q = d be the two equations of a system (P and Q are expressions in x and y). Prove that the system P = c and Q + kP = d + kc has exactly the same solutions, for any number k.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 72 • MAIN • Q2/5

Why must the new system keep one of the original equations unchanged? Give an example.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 72 • MAIN • Q3/5

System: $2x + y = 5$ and $x - y = 1$. Which step can produce a system with DIFFERENT solutions?

A. Replacing both equations by their sum, $3x = 6$, alone

B. Multiplying the first equation by 3

C. Writing the equations in the other order

D. Replacing the second equation by the sum of the two equations

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REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 72 • MAIN • Q4/5

True or false?

If (2, 1) solves $x + y = 3$ and $x - y = 1$, then (2, 1) also solves $(x + y) + 4(x - y) = 3 + 4(1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 72 • MAIN • Q5/5

SET 72 • HSA-REI.C.5

BACKWARD 1

Use the properties of equality to justify a step • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

If P = c and Q = d are true, explain why P + Q = c + d is true.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 72 • BACKWARD 1 • Q1/5

If P + 3Q = c + 3d and Q = d, explain why P = c.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 72 • BACKWARD 1 • Q2/5

Which property justifies going from $x - y = 1$ to $4x - 4y = 4$?

A. Addition property of equality

B. Commutative property

C. Substitution of y

D. Multiplication property of equality

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 72 • BACKWARD 1 • Q3/5

True or false?

Subtracting equal quantities from both sides of a true equation gives a true equation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 72 • BACKWARD 1 • Q4/5

True or false?

If a + b = 10 and a = 4, then b = 14.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 72 • BACKWARD 1 • Q5/5

SET 72 • HSA-REI.C.5

BACKWARD 2

Solve simple systems by adding equations • Grade 8 • Expressions & Equations • 8.EE.C.8.b

Solve by adding the equations.

$x + y = 9$

$x - y = 3$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 72 • BACKWARD 2 • Q1/5

Solve.

$2x + y = 11$

$y = 3$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 72 • BACKWARD 2 • Q2/5

Solve.

$x + 2y = 8$

$x = 2$

A. (2, 5)

B. (8, 0)

C. (2, 3)

D. (3, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 72 • BACKWARD 2 • Q3/5

True or false?

Adding $x + y = 5$ and $x - y = 1$ gives $2x = 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 72 • BACKWARD 2 • Q4/5

True or false?

The system $x + y = 4$ and $2x + 2y = 8$ has exactly one solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 72 • BACKWARD 2 • Q5/5

SET 72 • HSA-REI.C.5

FORWARD 1

Use row operations on an augmented matrix as elimination steps • High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8

Write the augmented matrix of the system $x + 2y = 7$ and $3x - y = 7$.

Rows: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F1-Q1

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 72 • FORWARD 1 • Q1/5

Apply R2 → 2R2 + R1 to the rows [1 2 | 7] and [3 −1 | 7]. What is the new second row, and which equation does it represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F1-Q2

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 72 • FORWARD 1 • Q2/5

Which row operation matches "replace equation 2 by itself minus 3 times equation 1"?

A. $R2 \to R1$

B. $R2 \to R2 - 3R1$

C. $R2 \to 3R2 - R1$

D. $R1 \to R1 - 3R2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F1-Q3**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 72 • FORWARD 1 • Q3/5

True or false?

A row operation R2 → R2 + kR1 does not change the solutions of the system.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F1-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 72 • FORWARD 1 • Q4/5

True or false?

Replacing R2 by 0 · R2 + R1 keeps the solutions of every system.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F1-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 72 • FORWARD 1 • Q5/5

SET 72 • HSA-REI.C.5

FORWARD 2

Solve the same system with an inverse matrix • High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9

For $x + 2y = 7$ and $3x - y = 7$, the coefficient matrix A has rows [1 2] and [3 −1]. Find the determinant of A.

Determinant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F2-Q1

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 72 • FORWARD 2 • Q1/5

A^{-1} is $(1 \div (-7))$ times the matrix with rows $[-1\ -2]$ and $[-3\ 1]$. Use $X = A^{-1}B$, with B the column $[7, 7]$, to solve the system.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F2-Q2

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 72 • FORWARD 2 • Q2/5

If the determinant of the coefficient matrix of a 2-by-2 linear system is 0, which is true?

A. $X = A^{-1}B$ gives the solution.

B. The system has exactly one solution.

C. The system has no solution or infinitely many solutions.

D. The two equations must be identical.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F2-Q3**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 72 • FORWARD 2 • Q3/5

True or false?

Solving $AX = B$ with $X = A^{-1}B$ gives the same solution as elimination.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F2-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 72 • FORWARD 2 • Q4/5

True or false?

For all 2-by-2 matrices A and B, $A^{-1}B = BA^{-1}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F2-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 72 • FORWARD 2 • Q5/5

Solve systems of linear equations exactly by substitution and elimination

MAIN	Algebra I main question	**HSA-REI.C.6**
BACKWARD 1	Eliminate a variable by adding a multiple of an equation	**HSA-REI.C.5**
BACKWARD 2	Solve simple systems of linear equations algebraically	**8.EE.C.8.b**
FORWARD 1	Use slopes and systems to find where parallel or perpendicular li..	**HSG-GPE.B.5**
FORWARD 2	Multiply a matrix by a column to check a solution	**HSN-VM.C.8**

SET 27 • HSA-REI.C.6

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

Solve by substitution.

$y = 3x - 5$

$2x + y = 15$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q1/5

Solve by elimination.

$3x + 4y = 10$

$5x - 2y = 34$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q2/5

Solve.

$\frac{1}{2}x + \frac{1}{3}y = 4$

$x - y = 3$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q3/5

Solve.

$2x - 3y = -1$

$4x + y = 19$

A. $(-4, -3)$

B. $(4, 3)$

C. $(3, 4)$

D. $(5, -1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q4/5

True or false?

The system $2x + y = 5$ and $4x + 2y = 9$ has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q5/5

SET 27 • HSA-REI.C.6

BACKWARD 1

Eliminate a variable by adding a multiple of an equation • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5

Add the equations $x + 2y = 9$ and $3x - 2y = 3$.

What equation results, and what is x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q1/5

System: $2x + y = 8$ and $6x - 5y = 8$.

What multiple of the first equation should you subtract from the second to eliminate x?

Multiple: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q2/5

System: $x + y = 5$ and $2x + 3y = 12$.

Which equation can replace the second equation without changing the solution?

A. $x = 2$

B. $y = 7$

C. $y = 2$

D. $3x + 4y = 12$

True or false?

Multiplying one equation of a system by a nonzero number does not change the solutions of the system.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q4/5

True or false?

Adding the equations $x + y = 4$ and $x - y = 2$ gives $2x = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q5/5

SET 27 • HSA-REI.C.6

BACKWARD 2

Solve simple systems of linear equations algebraically • Grade 8 • Expressions & Equations • 8.EE.C.8.b

Solve.

$y = x + 2$

$y = 3x$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q1/5

Solve.

$x + y = 10$

$y = 4$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q2/5

Solve.

$y = 2x$

$x + y = 12$

A. (4, 8)

B. (6, 6)

C. (8, 4)

D. (3, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q3/5

True or false?

The system $y = 2x + 3$ and $y = 2x - 1$ has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q4/5

True or false?

The solution of $y = x$ and $y = -x + 6$ is $(2, 4)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q5/5

SET 27 • HSA-REI.C.6

FORWARD 1

Use slopes and systems to find where parallel or perpendicular lines meet • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Line ℓ is $y = 2x + 1$. Write an equation of the line through $(5, 1)$ that is perpendicular to ℓ, and find where the two lines meet.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q1/5

Are the lines $2x + 3y = 6$ and $3x - 2y = 4$ perpendicular? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q2/5

Which line is parallel to $y = 3x - 2$ and passes through $(0, 5)$?

A. $y = 5x + 3$

B. $y = -3x + 5$

C. $y = 3x + 5$

D. $y = -\frac{1}{3}x + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations

HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q3/5

True or false?

The lines $y = 4x + 1$ and $y = 4x - 7$ have no point in common.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q4/5

True or false?

The lines $y = 2x$ and $y = -2x$ are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q5/5

SET 27 • HSA-REI.C.6

FORWARD 2

Multiply a matrix by a column to check a solution • High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8

Multiply the matrix with rows [2 1] and [1 −3] by the column [4, 1].

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q1/5

The matrix A has rows [2 1] and [1 −3].

Use multiplication by A to decide whether x = 4, y = 1 solves 2x + y = 9 and x − 3y = 1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q2/5

What is the product of the matrix with rows [1 2] and [3 4] and the column [1, 1]?

A. The column [3, 7]

B. The column [2, 6]

C. The column [10]

D. The column [4, 6]

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F2-Q3**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q3/5

True or false?

Multiplying a 2-by-2 matrix by a 2-by-1 column gives a 2-by-1 column.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q4/5

True or false?

For all 2-by-2 matrices A and B, AB = BA.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q5/5

Solve systems of linear equations in context, including systems with no solution or infinitely many solutions

MAIN	Algebra I main question	**HSA-REI.C.6**
BACKWARD 1	Write and solve an equation for one quantity in context	**HSA-CED.A.1**
BACKWARD 2	Solve real-world problems with two linear equations	**8.EE.C.8.c**
FORWARD 1	Use matrices to organize data in context	**HSN-VM.C.6**
FORWARD 2	Solve a system in context with an inverse matrix	**HSA-REI.C.9**

SET 28 • HSA-REI.C.6

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

A theater sold 340 tickets. Adult tickets cost $12 and student tickets cost $7. Sales totaled $3,255.

Write a system and find how many of each kind were sold.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q1/5

How many solutions does the system $6x - 4y = 10$ and $3x - 2y = 5$ have? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q2/5

A chemist mixes a 10% acid solution with a 30% acid solution to make 20 liters of a 25% acid solution.

How many liters of each are needed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q3/5

For which value of k does the system $y = 3x + 2$ and $6x - 2y = k$ have infinitely many solutions?

A. $k = 2$

B. $k = -2$

C. $k = 4$

D. $k = -4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q4/5

True or false?

A system of two linear equations in two variables can have exactly two solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q5/5

SET 28 • HSA-REI.C.6

BACKWARD 1

Write and solve an equation for one quantity in context • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1

A gym has 340 members. There are 40 more adults than teens.

Write and solve an equation for the number of teens t.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q1/5

Pens cost $2 and notebooks cost $5. Ana buys 3 notebooks and some pens and spends $27.

Write and solve an equation for the number of pens p.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q2/5

A 20-liter mix has x liters of a 10% acid solution and the rest a 30% acid solution.

Which expression gives the liters of acid?

A. $0.40(20)$

B. $0.10x + 0.30x$

C. $0.10x + 0.30(20 - x)$

D. $10x + 30(20 - x)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q3/5

True or false?

25 coins are all dimes and quarters. If d coins are dimes, then 25 − d coins are quarters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q4/5

True or false?

"3 more than 5 times a number is 48" gives $5n + 3 = 48$, and its solution is $n = 10$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q5/5

SET 28 • HSA-REI.C.6

BACKWARD 2

Solve real-world problems with two linear equations • Grade 8 • Expressions & Equations • 8.EE.C.8.c

Two numbers have a sum of 30 and a difference of 8.

Find the numbers.

Numbers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q1/5

Plan A costs $10 plus $2 per hour. Plan B costs $4 per hour.

After how many hours do they cost the same?

Hours: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q2/5

3 pencils and 2 erasers cost $1.30. 1 pencil and 2 erasers cost $0.70.

What does one pencil cost?

A. $0.60

B. $0.20

C. $0.35

D. $0.30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q3/5

True or false?

Two cars leave the same place at the same time in the same direction, at 50 mi/h and 60 mi/h. After 2 hours the faster car is 20 miles ahead.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q4/5

True or false?

One company charges $y = 2x + 5$ dollars and another charges $y = 2x + 8$ dollars for x units. They charge the same for 3 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q5/5

SET 28 • HSA-REI.C.6

FORWARD 1

Use matrices to organize data in context • High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6

Friday: 120 adult and 90 student tickets. Saturday: 150 adult and 60 student tickets.

Write a 2-by-2 matrix with one row per day (adult first).

Rows: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q1/5

Adult tickets cost $12 and student tickets $7.

Multiply the matrix with rows [120 90] and [150 60] by the column [12, 7]. What do the entries mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q2/5

A matrix has rows [5 2] and [3 4]. Each row is a store and each column is a product.

What does the entry in row 2, column 1 represent?

A. Store 2's amount of product 1

B. Store 1's amount of product 2

C. Store 2's amount of product 2

D. The total of product 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F1-Q3**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q3/5

True or false?

A matrix with 3 rows and 2 columns is a 3-by-2 matrix.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q4/5

True or false?

Any two matrices can be added, whatever their dimensions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q5/5

SET 28 • HSA-REI.C.6

FORWARD 2

Solve a system in context with an inverse matrix • High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9

Write the system $a + s = 340$ and $12a + 7s = 3{,}255$ as a matrix equation $AX = B$. Give A, X, and B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q1

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q1/5

The matrix with rows [1 1] and [12 7] has determinant −5.

Does the ticket system have exactly one solution? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q2

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q2/5

The inverse of the matrix with rows [1 1] and [12 7] is $(1 \div (-5))$ times which matrix?

A. Rows [7 −1] and [−12 1]

B. Rows [7 1] and [12 1]

C. Rows [−7 1] and [12 −1]

D. Rows [1 −1] and [−12 7]

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q3

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q3/5

True or false?

If the coefficient matrix of a system has determinant 0, the system does not have exactly one solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q4/5

True or false?

The determinant of the matrix with rows [2 3] and [4 6] is 24.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q5/5

Solve a system of a linear equation and a quadratic equation algebraically

MAIN	Algebra I main question	**HSA-REI.C.7**
BACKWARD 1	Solve quadratic equations by factoring	**HSA-REI.B.4.b**
BACKWARD 2	Solve systems of linear equations	**HSA-REI.C.6**
FORWARD 1	Decide whether points lie on a circle	**HSG-GPE.B.4**
FORWARD 2	Find complex solutions when a line and a parabola do not meet	**HSN-CN.C.7**

SET 29 • HSA-REI.C.7

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7

Solve the system.

$y = x^2 - 3$

$y = 2x$

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q1/5

Solve the system.

$y = x^2 + 4x + 1$

$y = x - 1$

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q2/5

How many real solutions does the system $y = x^2 + 2$ and $y = x$ have?

A. Infinitely many

B. One

C. Two

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q3/5

Solve the system.

$x^2 + y^2 = 25$

$y = x + 1$

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q4/5

True or false?

The line $y = 2x - 1$ meets the parabola $y = x^2$ at exactly one point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q5/5

SET 29 • HSA-REI.C.7

BACKWARD 1

Solve quadratic equations by factoring • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve $x^2 - 2x - 3 = 0$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q1/5

Solve $x^2 + x - 12 = 0$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q2/5

Solve $x^2 + 3x + 2 = 0$.

A. $x = -1$ or $x = 2$

B. $x = 1$ or $x = 2$

C. $x = 1$ or $x = -2$

D. $x = -1$ or $x = -2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q3/5

True or false?

$x^2 - x + 2 = 0$ has no real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q4/5

True or false?

The solutions of $x^2 = 2x$ are $x = 2$ and $x = -2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q5/5

SET 29 • HSA-REI.C.7

BACKWARD 2

Solve systems of linear equations • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

Solve.

$y = 2x$

$y = -x + 9$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 29 • BACKWARD 2 • Q1/5

Solve.

$x + y = 1$

$y = x + 1$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 29 • BACKWARD 2 • Q2/5

Solve.

$2x + y = 7$

$x - y = 2$

A. $(4, -1)$

B. $(3, 1)$

C. $(1, 3)$

D. $(2, 3)$

True or false?

Substitution and elimination give the same solution of a system.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 29 • BACKWARD 2 • Q4/5

True or false?

(2, 1) is a solution of the system $x + y = 3$ and $x - y = 3$.

A. True

B. False

SET 29 • HSA-REI.C.7

FORWARD 1

Decide whether points lie on a circle • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

A circle is centered at the origin and passes through (0, 5).

Does the point (3, −4) lie on the circle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q1/5

Find the points where the line $y = 3$ meets the circle $x^2 + y^2 = 25$.

Points: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q2/5

A circle has center (1, 2) and radius 5.

Which point is on the circle?

A. (1, 6)

B. (4, 6)

C. (3, 3)

D. (6, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q3/5

True or false?

The line $y = 6$ does not meet the circle $x^2 + y^2 = 25$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q4/5

True or false?

The point (2, 2) lies on the circle $x^2 + y^2 = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q5/5

SET 29 • HSA-REI.C.7

FORWARD 2

Find complex solutions when a line and a parabola do not meet • High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7

Substituting $y = 2x$ into $y = x^2 + 2$ gives $x^2 - 2x + 2 = 0$.

Solve it over the complex numbers.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q1

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q1/5

Solve $x^2 + 4 = 0$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q2

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q2/5

Solve $x^2 - 6x + 10 = 0$.

A. $3 + i$ and $3 - i$

B. $6 + i$ and $6 - i$

C. $3 + 2i$ and $3 - 2i$

D. $-3 + i$ and $-3 - i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F2-Q3**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q3/5

True or false?

When a line and a parabola do not intersect, the equation from substitution has no real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q4/5

True or false?

$i^3 = i$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F2-Q5**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q5/5

Understand the graph of an equation as the set of all its solutions

MAIN	Algebra I main question	**HSA-REI.D.10**
BACKWARD 1	Use tables and graphs to relate two variables	**6.EE.C.9**
BACKWARD 2	Understand a function's graph as its input-output pairs	**8.F.A.1**
FORWARD 1	Derive a parabola from its focus and directrix	**HSG-GPE.A.2**
FORWARD 2	Use the unit circle as the set of points (cos θ, sin θ)	**HSF-TF.A.2**

SET 30 • HSA-REI.D.10

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10

Is $(3, -2)$ on the graph of $2x + 5y = -4$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q1/5

The graph of $y = x^2 - 2x - 3$ is shown.

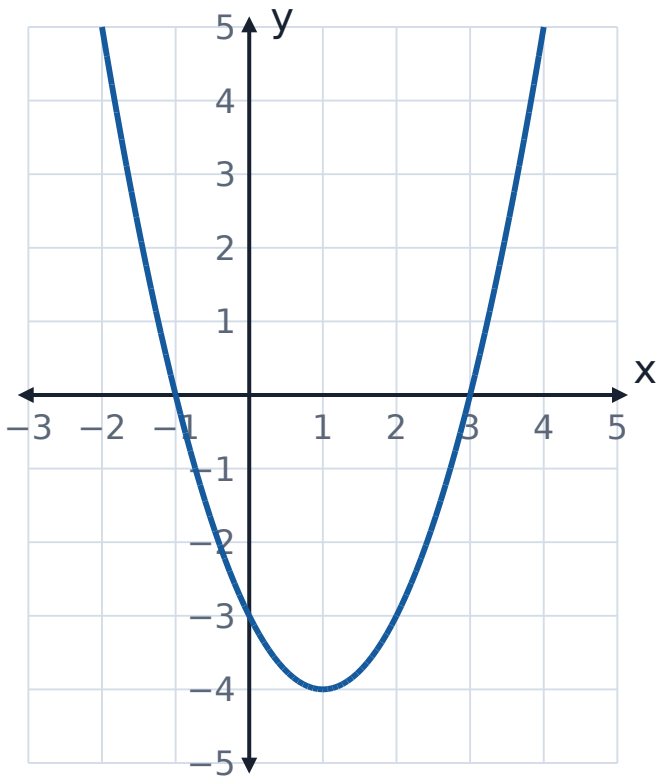
Which point is NOT a solution of the equation?

A. (2, 3)

B. (3, 0)

C. (0, −3)

D. (1, −4)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q2/5

The point (k, 7) is on the graph of $y = 3x - 2$.

Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q3/5

The graph of an equation in x and y passes through (4, 1) and (−2, 5).

What does that tell you about the equation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q4/5

True or false?

A point that is not on the graph of $y = x^2$ can still be a solution of $y = x^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q5/5

SET 30 • HSA-REI.D.10

BACKWARD 1

Use tables and graphs to relate two variables • Grade 6 • Expressions & Equations • 6.EE.C.9

$y = 3x$. Find y for $x = 0, 1, 2,$ and 3.

Values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q1/5

A car travels at 40 miles per hour.

Write an equation for the distance d after t hours.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q2/5

Which point is on the graph of y = x + 4?

A. (2, 6)

B. (1, 3)

C. (4, 0)

D. (6, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q3/5

True or false?

In y = 5x, y is the dependent variable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q4/5

True or false?

The point (2, 8) is on the graph of $y = 3x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q5/5

SET 30 • HSA-REI.D.10

BACKWARD 2

Understand a function's graph as its input-output pairs • Grade 8 • Functions • 8.F.A.1

The function f pairs each x with $2x - 1$.

What output does it pair with $x = 4$?

Output: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q1

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q1/5

Is {(1, 2), (2, 3), (1, 5)} a function? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q2

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q2/5

Which point is on the graph of $y = x^2$?

A. $(9, -3)$

B. $(-3, -9)$

C. $(-3, 9)$

D. $(3, 6)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q3

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q3/5

True or false?

A function assigns exactly one output to each input.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q4

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q4/5

True or false?

The vertical line $x = 2$ is the graph of a function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q5

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q5/5

SET 30 • HSA-REI.D.10

FORWARD 1

Derive a parabola from its focus and directrix • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2

A parabola is the set of points equally far from the focus $(0, 2)$ and the directrix $y = -2$.

Derive its equation.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q1/5

Is (4, 2) on the parabola with focus (0, 2) and directrix $y = -2$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q2/5

Where is the focus of the parabola $y = x^2 \div 4$?

A. (0, 4)

B. $(0, \frac{1}{4})$

C. (1, 0)

D. (0, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q3/5

True or false?

Every point on a parabola is the same distance from its focus as from its directrix.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q4/5

True or false?

The vertex of a parabola lies on its directrix.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q5/5

SET 30 • HSA-REI.D.10

FORWARD 2

Use the unit circle as the set of points (cos θ, sin θ) • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2

Explain why the point $(\cos \theta, \sin \theta)$ lies on the circle $x^2 + y^2 = 1$ for every angle θ.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q1/5

Which point on the unit circle corresponds to $\theta = \pi \div 2$?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q2/5

Which point is on the unit circle $x^2 + y^2 = 1$?

A. (0.5, 0.5)

B. (1, 1)

C. (0.6, 0.8)

D. (0.8, 0.8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q3/5

True or false?

The point $(-1, 0)$ on the unit circle corresponds to an angle of π radians.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q4/5

True or false?

The point (cos 0, sin 0) is (0, 1).

A. True

B. False

Explain why intersections of y = f(x) and y = g(x) solve f(x) = g(x), and approximate solutions with graphs and tables

MAIN	Algebra I main question	**HSA-REI.D.11**
BACKWARD 1	Check whether points are on a graph	**HSA-REI.D.10**
BACKWARD 2	Read the solution of a system from the graphs of two lines	**8.EE.C.8.a**
FORWARD 1	Solve exponential equations exactly with logarithms	**HSF-LE.A.4**
FORWARD 2	Solve trigonometric equations with graphs and technology	**HSF-TF.B.7**

SET 31 • HSA-REI.D.11

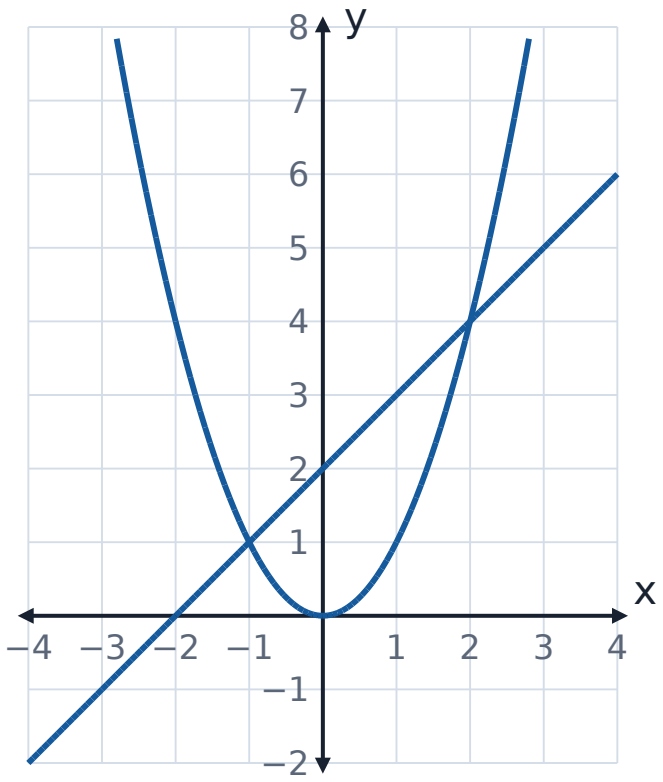
MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

The graphs of $f(x) = x^2$ and $g(x) = x + 2$ are shown.

Use the graphs to solve $x^2 = x + 2$.

Solutions: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q1/5

Use a table of values to find the solution of $2^x = 3x$ between $x = 3$ and $x = 4$, to the nearest tenth.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q2/5

The table shows $f(x) = |x - 1|$ and $g(x) = 3 - x$.

For which x is $f(x) = g(x)$?

x	0	1	2	3	4
f(x)	1	0	1	2	3
g(x)	3	2	1	0	−1

A. $x = 2$ **B.** $x = 3$ **C.** $x = 0$ **D.** $x = 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q3/5

Explain why the x-coordinates of the points where the graphs of $y = f(x)$ and $y = g(x)$ meet are the solutions of $f(x) = g(x)$.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q4/5

True or false?

The graphs of $y = 3x + 1$ and $y = 3x - 4$ never meet, so $3x + 1 = 3x - 4$ has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q5/5

SET 31 • HSA-REI.D.11

BACKWARD 1

Check whether points are on a graph • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10

Is (2, 4) on both $y = x^2$ and $y = x + 2$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q1/5

The point $(-1, y)$ is on the graph of $y = x + 2$.

Find y.

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q2/5

Which point is on the graph of $y = |x - 1|$?

A. $(-2, -3)$

B. $(-2, 3)$

C. $(3, -2)$

D. $(1, 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q3/5

True or false?

If (a, b) is on the graph of y = f(x), then b = f(a).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q4/5

True or false?

(0, 1) is on the graph of $y = 2^x + 1$.

A. True

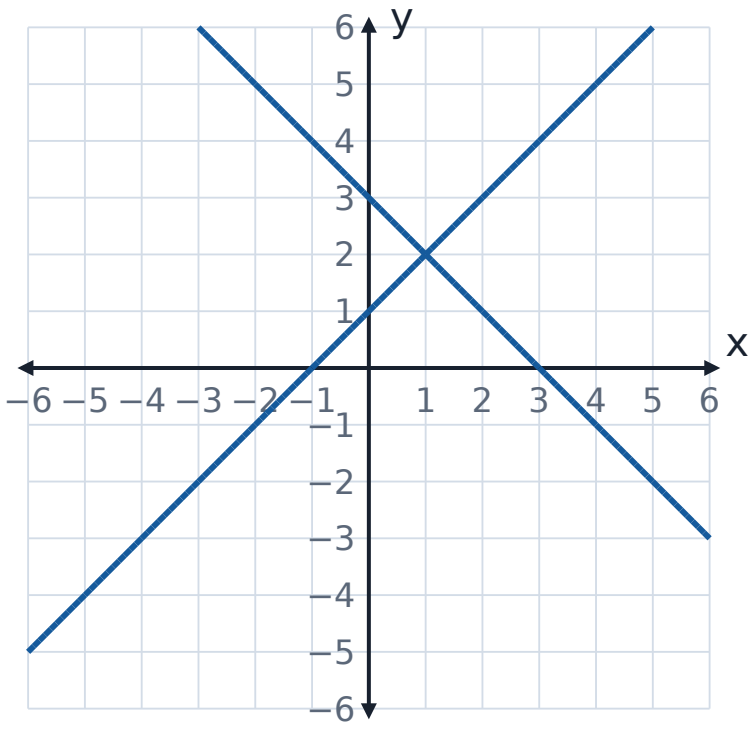
B. False

SET 31 • HSA-REI.D.11

BACKWARD 2

Read the solution of a system from the graphs of two lines • Grade 8 • Expressions & Equations • 8.EE.C.8.a

Use the graph to solve the system $y = x + 1$ and $y = -x + 3$.



Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q1/5

At what x-value do the graphs of $y = 2x - 3$ and $y = 5$ meet?

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q2/5

The graphs of two linear equations are the same line.

How many solutions does the system have?

A. One

B. Infinitely many

C. Two

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q3/5

True or false?

Two lines with different slopes meet at exactly one point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q4/5

True or false?

Two lines with the same slope always meet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q5/5

SET 31 • HSA-REI.D.11

FORWARD 1

Solve exponential equations exactly with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

The graphs of $y = 2^x$ and $y = 10$ meet at one point.

Find its x-coordinate exactly using logarithms, and to the nearest hundredth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q1/5

Solve $5e^{x} = 20$ exactly.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q2/5

A graph shows $y = 3^x$ meeting $y = 15$ at about $x = 2.5$.

Which expression is the exact solution of $3^x = 15$? (log means base 10.)

A. $15 \div 3$

B. $\log 5$

C. $\log 15 \div \log 3$

D. $\sqrt{15}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q3/5

True or false?

$\log_2 8 = 3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q4/5

True or false?

The solution of $10^x = 0.01$ is $x = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q5/5

SET 31 • HSA-REI.D.11

FORWARD 2

Solve trigonometric equations with graphs and technology • High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7

Solve $\sin x = \frac{1}{2}$ for $0 \leq x \leq \pi$. Give exact answers.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q1/5

Use a calculator to solve $\cos x = 0.3$ for $0 \leq x \leq \pi$, to the nearest hundredth of a radian.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q2/5

How many solutions does $\sin x = 2$ have?

A. Two

B. One

C. Infinitely many

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q3/5

True or false?

The graphs of $y = \sin x$ and $y = 0.5$ meet at infinitely many points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q4/5

True or false?

For $0 \leq x \leq \pi$, the equation $\cos x = 0$ has two solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q5/5

Solve linear-quadratic systems by graphing and check the solutions algebraically

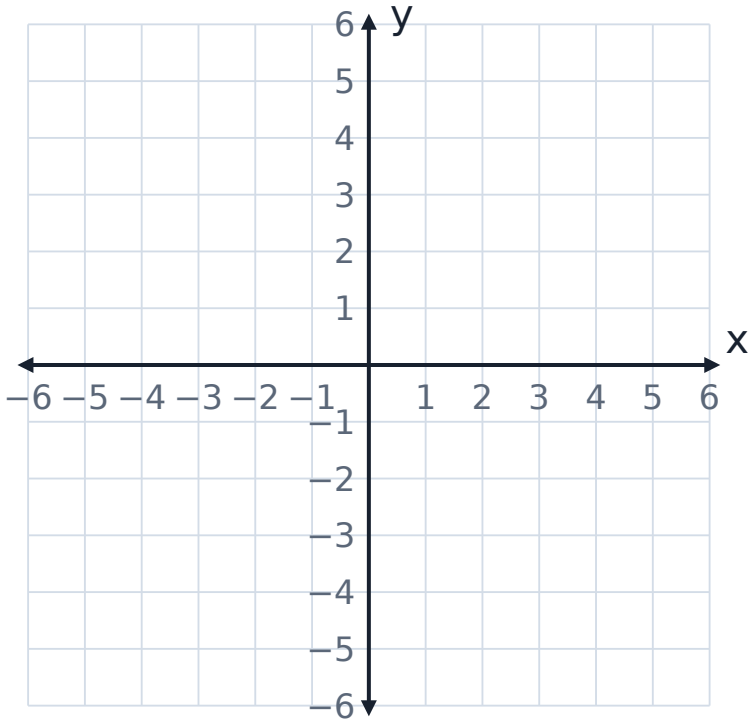
MAIN	Algebra I main question	**HSA-REI.C.7**
BACKWARD 1	Read solutions of f(x) = g(x) from intersections	**HSA-REI.D.11**
BACKWARD 2	Check whether a point is on a graph	**HSA-REI.D.10**
FORWARD 1	Find where a line meets a circle with coordinates	**HSG-GPE.B.4**
FORWARD 2	Find where a line meets an ellipse	**HSG-GPE.A.3**

SET 73 • HSA-REI.C.7

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7

Graph $y = x^2 - 2$ and $y = x$ on the grid. Read the solutions of the system from the intersections, then check them algebraically.



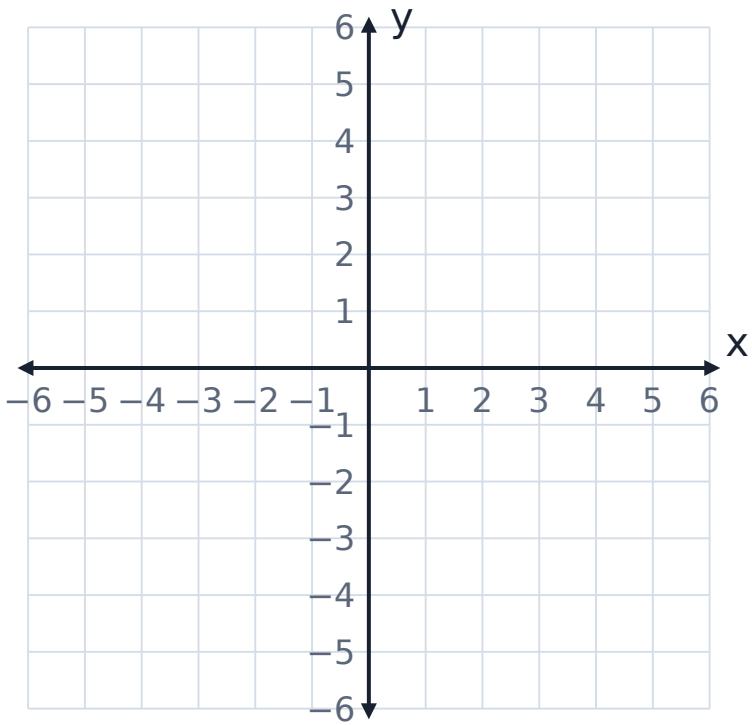
Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 73 • MAIN • Q1/5

Graph the circle $x^2 + y^2 = 25$ and the line $y = x + 1$. Read the intersection points, then check one of them algebraically.



Points: _____

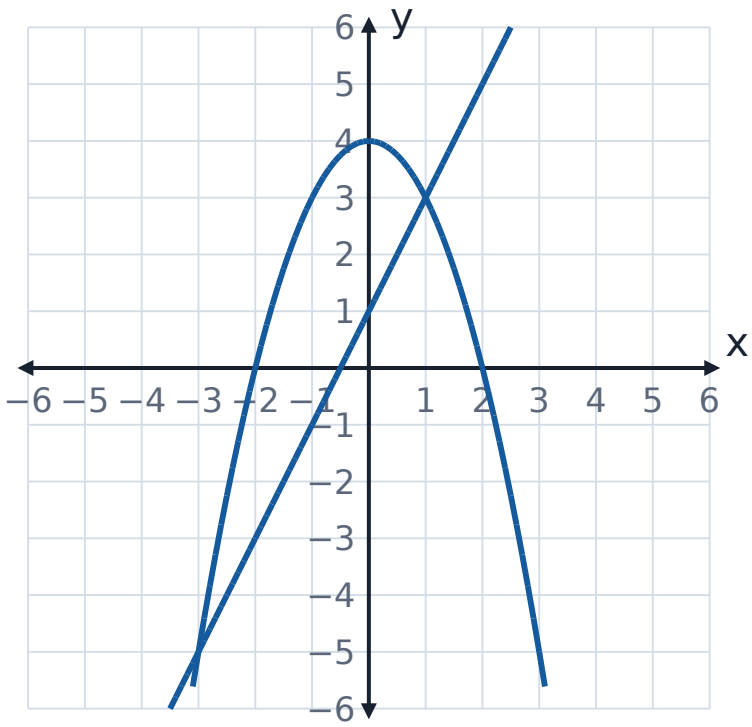
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 73 • MAIN • Q2/5

The graphs of $y = -x^2 + 4$ and $y = 2x + 1$ are shown.

Read the solutions of the system from the graph, then verify them algebraically.



Solutions: _____

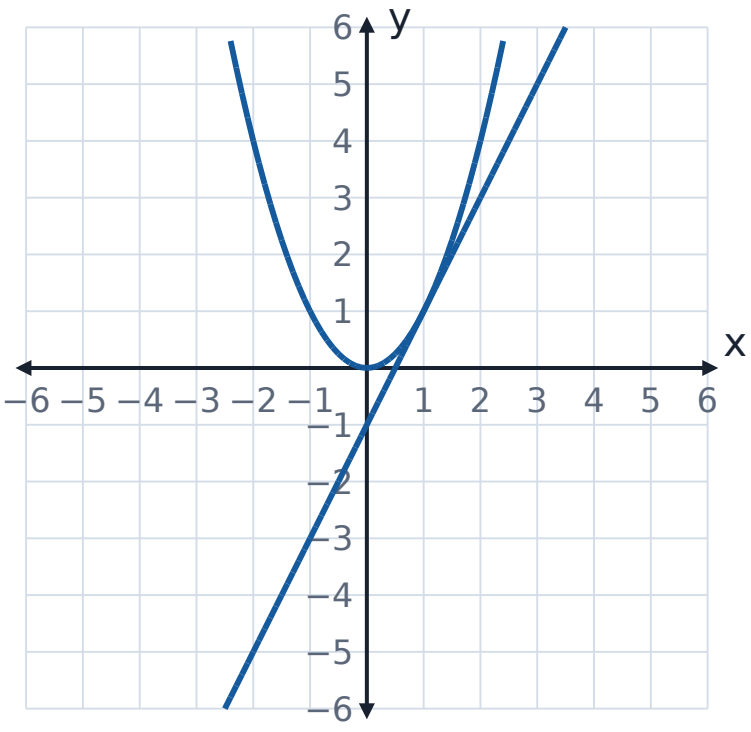
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 73 • MAIN • Q3/5

The graphs of $y = x^2$ and $y = 2x - 1$ are shown.

How many solutions does the system have?



A. One

B. Two

C. Infinitely many

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 73 • MAIN • Q4/5

True or false?

The line $y = 5$ and the circle $x^2 + y^2 = 9$ intersect at two points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 73 • MAIN • Q5/5

SET 73 • HSA-REI.C.7

BACKWARD 1

Read solutions of f(x) = g(x) from intersections • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

The graphs of $y = x^2$ and $y = 4$ meet at two points.

What are the solutions of $x^2 = 4$?

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 73 • BACKWARD 1 • Q1/5

The graphs of $y = 2x + 1$ and $y = x^2 + 1$ meet at (0, 1) and (2, 5). Which x-values solve $2x + 1 = x^2 + 1$? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 73 • BACKWARD 1 • Q2/5

The graphs of $y = f(x)$ and $y = g(x)$ meet only at $(3, 7)$. Which statement is true?

A. $f(x) = g(x)$ for every x

B. $f(7) = g(7) = 3$

C. $f(3) = 3$

D. $f(3) = g(3) = 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 73 • BACKWARD 1 • Q3/5

True or false?

If the graphs of $y = f(x)$ and $y = g(x)$ never meet, then $f(x) = g(x)$ has no real solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 73 • BACKWARD 1 • Q4/5

True or false?

The solutions of f(x) = g(x) are the y-coordinates of the intersection points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 73 • BACKWARD 1 • Q5/5

SET 73 • HSA-REI.C.7

BACKWARD 2

Check whether a point is on a graph • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10

Is (3, 4) on the circle $x^2 + y^2 = 25$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 73 • BACKWARD 2 • Q1/5

Is (2, 3) on the parabola $y = x^2 - 2$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 73 • BACKWARD 2 • Q2/5

Which point is on both $y = x$ and $y = x^2 - 2$?

A. (2, 2)

B. (−2, −2)

C. (1, 1)

D. (0, −2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 73 • BACKWARD 2 • Q3/5

True or false?

A point is a solution of a system exactly when it lies on the graphs of both equations.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 73 • BACKWARD 2 • Q4/5

True or false?

(0, 5) is on both the circle $x^2 + y^2 = 25$ and the line $y = x + 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 73 • BACKWARD 2 • Q5/5

SET 73 • HSA-REI.C.7

FORWARD 1

Find where a line meets a circle with coordinates • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Find the points where the line $y = 7 - x$ meets the circle $x^2 + y^2 = 25$.

Points: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 73 • FORWARD 1 • Q1/5

Show that the line $y = 5$ touches the circle $x^2 + y^2 = 25$ at exactly one point.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 73 • FORWARD 1 • Q2/5

How many points do the line $y = 2x$ and the circle $x^2 + y^2 = 20$ share?

A. One

B. Four

C. Two

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 73 • FORWARD 1 • Q3/5

True or false?

The line $x = 6$ does not meet the circle $x^2 + y^2 = 25$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 73 • FORWARD 1 • Q4/5

True or false?

The point (1, 5) lies on the circle $x^2 + y^2 = 25$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 73 • FORWARD 1 • Q5/5

SET 73 • HSA-REI.C.7

FORWARD 2

Find where a line meets an ellipse • High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

Find where the line $y = 2$ meets the ellipse $x^2 \div 16 + y^2 \div 4 = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F2-Q1

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations HSG-GPE.A.3 • Set 73 • FORWARD 2 • Q1/5

Find where the x-axis meets the ellipse $x^2 \div 9 + y^2 \div 16 = 1$.

Points: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F2-Q2

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 73 • FORWARD 2 • Q2/5

Where does the line $x = 0$ meet the ellipse $x^2 \div 25 + y^2 \div 9 = 1$?

A. (0, 5) and (0, −5)

B. (5, 0) and (−5, 0)

C. Nowhere

D. (0, 3) and (0, −3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-F2-Q3**

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 73 • FORWARD 2 • Q3/5

True or false?

A line can meet an ellipse in at most two points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F2-Q4

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 73 • FORWARD 2 • Q4/5

True or false?

The line $y = 4$ meets the ellipse $x^2 \div 25 + y^2 \div 9 = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F2-Q5

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 73 • FORWARD 2 • Q5/5

Solve systems of linear equations by graphing, including approximate solutions

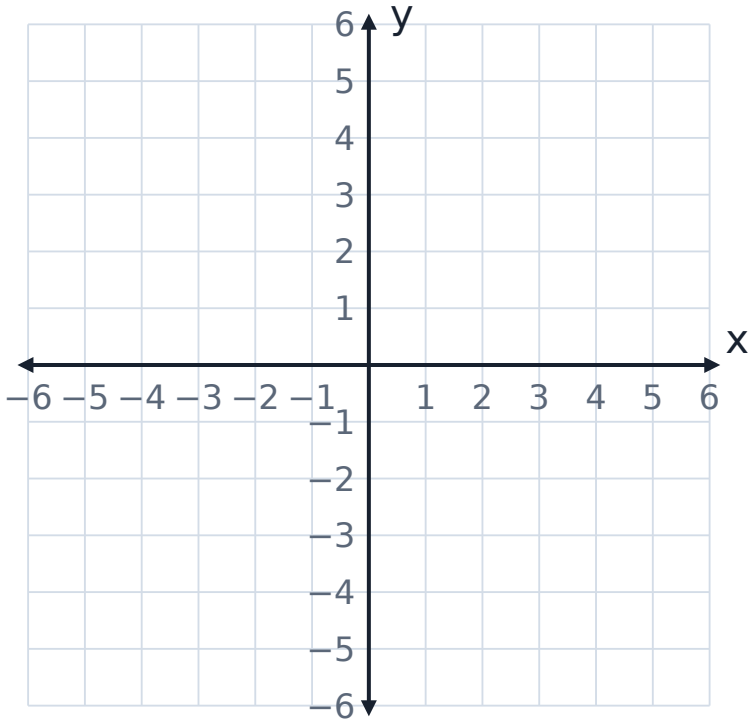
MAIN	Algebra I main question	**HSA-REI.C.6**
BACKWARD 1	Find the solution of a system where two lines cross	**8.EE.C.8.a**
BACKWARD 2	Decide whether points are on a line	**HSA-REI.D.10**
FORWARD 1	Graph parallel and perpendicular lines and find where they meet	**HSG-GPE.B.5**
FORWARD 2	Check a graphical estimate with an inverse matrix	**HSA-REI.C.9**

SET 84 • HSA-REI.C.6

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

Graph $y = 2x - 1$ and $y = -x + 5$ on the grid. What is the solution of the system?



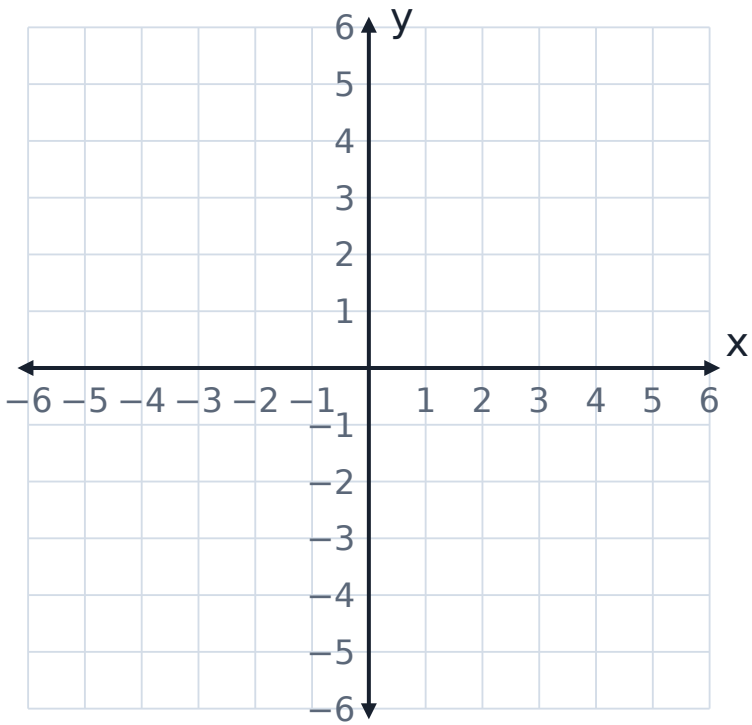
Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 84 • MAIN • Q1/5

Graph $y = 0.5x + 2$ and $y = -2x + 4$. Estimate the solution to the nearest half unit, then solve algebraically to check.



Answer: _____

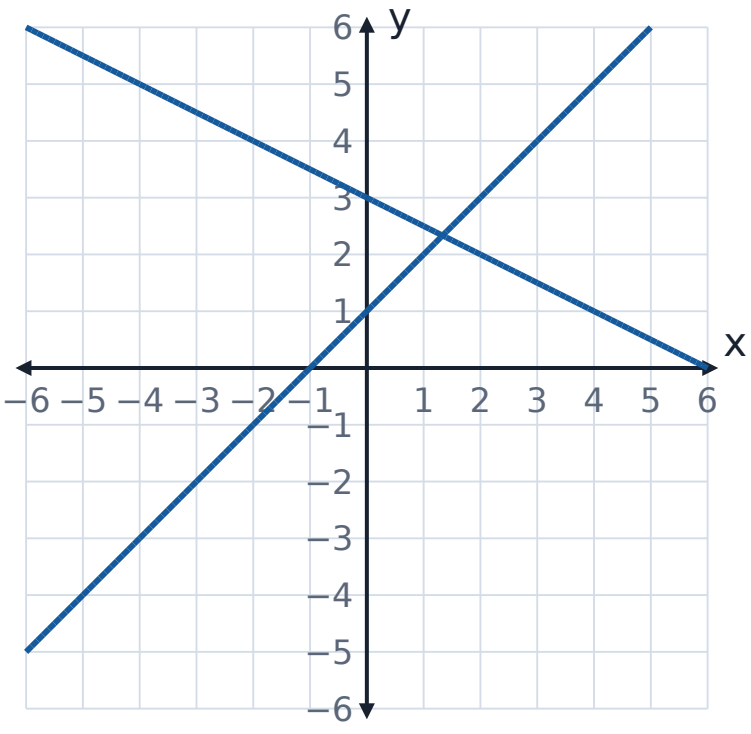
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 84 • MAIN • Q2/5

The graphs of $y = x + 1$ and $y = -0.5x + 3$ are shown.

Estimate the solution of the system to the nearest tenth, and explain how you made the estimate.



Answer: _____

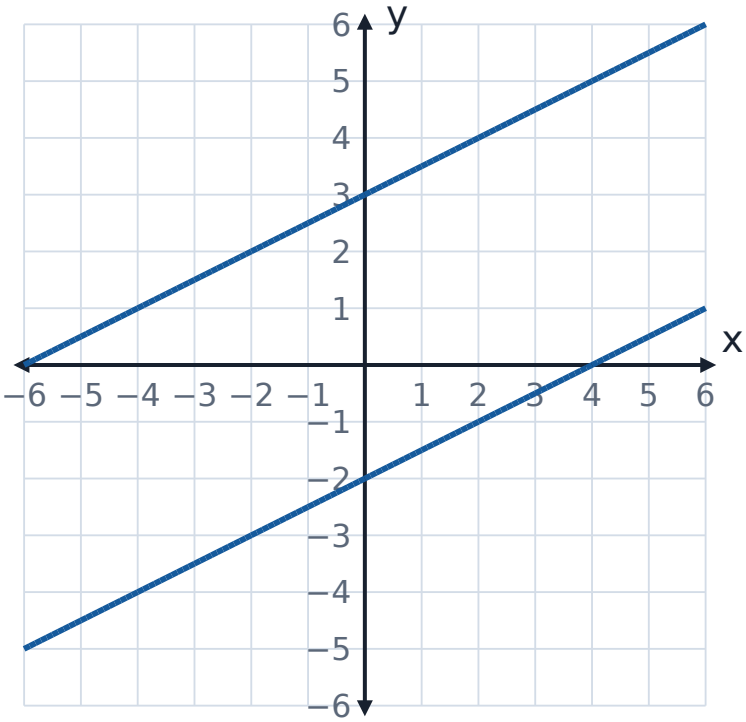
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 84 • MAIN • Q3/5

The graphs of the two equations of a system are shown.

How many solutions does the system have?



A. Two

B. One

C. Infinitely many

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 84 • MAIN • Q4/5

True or false?

If the graphs of two linear equations lie on top of each other, the system has infinitely many solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-M-Q5

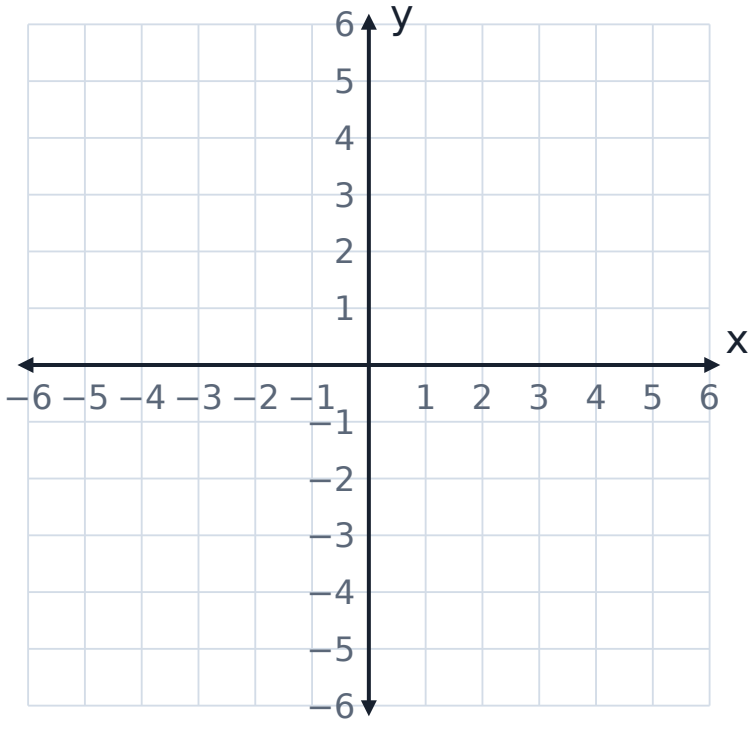
High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 84 • MAIN • Q5/5

SET 84 • HSA-REI.C.6

BACKWARD 1

Find the solution of a system where two lines cross • Grade 8 • Expressions & Equations • 8.EE.C.8.a

Graph $y = x$ and $y = 4 - x$, and mark the point where they meet.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 84 • BACKWARD 1 • Q1/5

Two lines meet at $(-1, 3)$. What is the solution of their system?

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 84 • BACKWARD 1 • Q2/5

The graphs of two lines cross at (4, −2). Which equations could they be?

A. $y = x - 6$ and $y = x + 2$

B. $y = x - 6$ and $y = -x + 2$

C. $y = x + 6$ and $y = -x + 2$

D. $y = 2x$ and $y = -2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 84 • BACKWARD 1 • Q3/5

True or false?

A point that lies on both lines is a solution of the system.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 84 • BACKWARD 1 • Q4/5

True or false?

Two lines with different slopes can have no point in common.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 84 • BACKWARD 1 • Q5/5

SET 84 • HSA-REI.C.6

BACKWARD 2

Decide whether points are on a line • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10

Is (2, 3) on the line $y = 2x - 1$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 84 • BACKWARD 2 • Q1/5

Is (2, 3) on the line $y = -x + 4$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 84 • BACKWARD 2 • Q2/5

Which point is on the line $y = 0.5x + 2$?

A. (2, 3)

B. (3, 2)

C. (0, 0.5)

D. (4, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 84 • BACKWARD 2 • Q3/5

True or false?

The graph of a linear equation in two variables is the set of all its solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 84 • BACKWARD 2 • Q4/5

True or false?

(0, 4) is on the graph of $y = -2x + 2$.

A. True

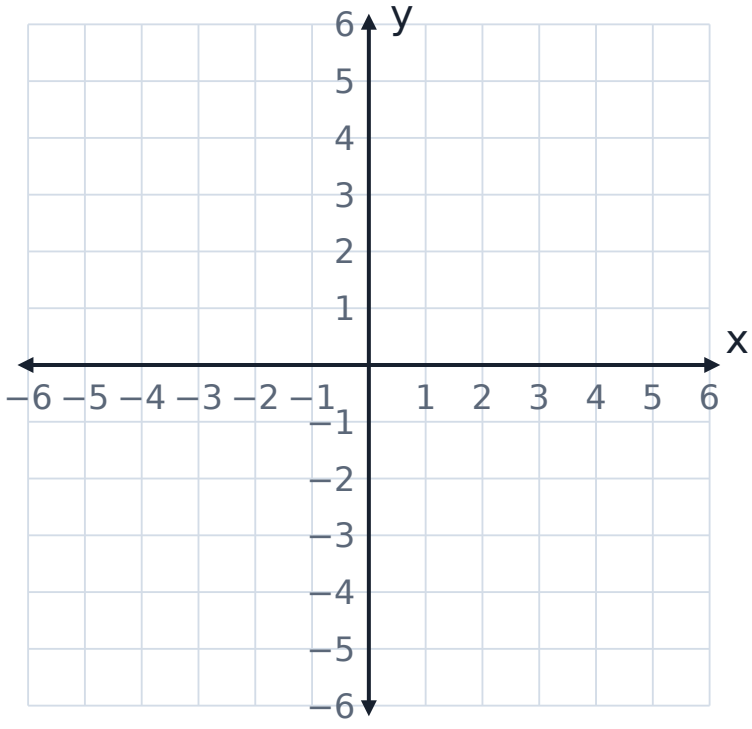
B. False

SET 84 • HSA-REI.C.6

FORWARD 1

Graph parallel and perpendicular lines and find where they meet • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Graph y = 2x + 1 and the line through (4, 1) that is perpendicular to it. Mark the point where they meet.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 84 • FORWARD 1 • Q1/5

Two lines have slopes 3 and $-\frac{1}{3}$. What angle do they form?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 84 • FORWARD 1 • Q2/5

Which system has no solution?

A. $y = 3x + 1$ and $y = -3x + 1$

B. $y = 3x + 1$ and $y = x + 1$

C. $y = 3x + 1$ and $y = 3x - 4$

D. $y = 3x$ and $y = 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 84 • FORWARD 1 • Q3/5

True or false?

The lines $y = -x + 2$ and $y = x - 2$ are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 84 • FORWARD 1 • Q4/5

True or false?

The lines $y = 2x + 5$ and $y = 2x + 5$ have no points in common.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 84 • FORWARD 1 • Q5/5

SET 84 • HSA-REI.C.6

FORWARD 2

Check a graphical estimate with an inverse matrix • High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9

The system $y = 0.5x + 2$ and $y = -2x + 4$ is written $-0.5x + y = 2$ and $2x + y = 4$. Find the determinant of the matrix A with rows $[-0.5\ 1]$ and $[2\ 1]$.

Determinant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F2-Q1

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 84 • FORWARD 2 • Q1/5

A^{-1} is $(1 \div (-2.5))$ times the matrix with rows $[1\ -1]$ and $[-2\ -0.5]$. Use $X = A^{-1}B$, with B the column $[2, 4]$, to find the exact solution.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F2-Q2

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 84 • FORWARD 2 • Q2/5

A graph shows two lines crossing near (1.3, 2.3). The inverse-matrix solution is (4/3, 7/3). Which statement is true?

A. The two results contradict each other.

B. The system has two solutions.

C. The graph is exact; the matrix method is approximate.

D. The graph gives an estimate; the matrix method gives the exact solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S84-F2-Q3**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 84 • FORWARD 2 • Q3/5

True or false?

If the coefficient matrix of a 2-by-2 linear system has a nonzero determinant, the system has exactly one solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F2-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 84 • FORWARD 2 • Q4/5

True or false?

If the graphs of two lines coincide, the determinant of the coefficient matrix is not 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F2-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 84 • FORWARD 2 • Q5/5

Graph the solutions of linear inequalities and systems of linear inequalities in two variables as half-planes

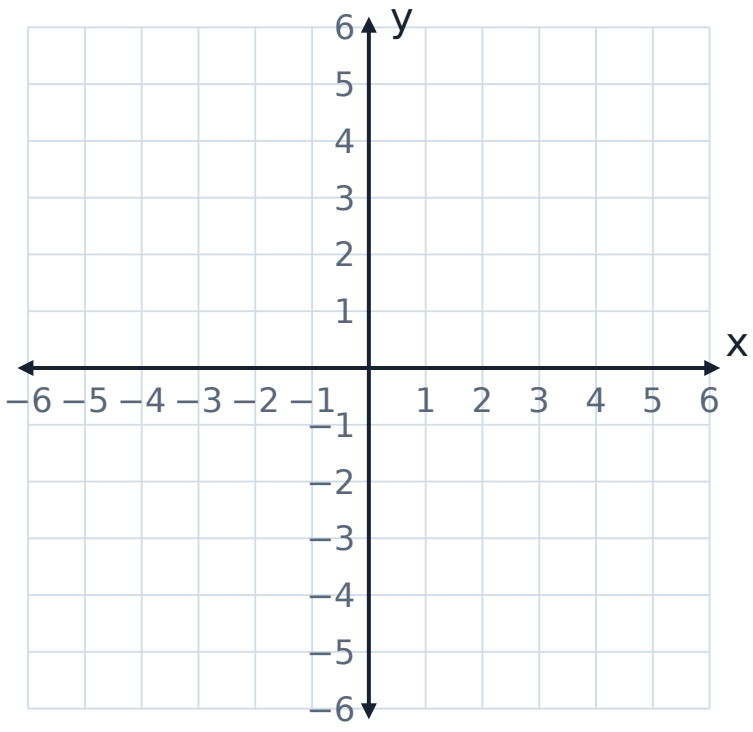
MAIN	Algebra I main question	**HSA-REI.D.12**
BACKWARD 1	Solve an inequality in two variables for y	**HSA-REI.B.3**
BACKWARD 2	Plot points and find distances in the coordinate plane	**6.NS.C.8**
FORWARD 1	Find the area of a region described by inequalities	**HSG-GPE.B.7**
FORWARD 2	Use zeros to find where a polynomial is positive or negative	**HSA-APR.B.3**

SET 32 • HSA-REI.D.12

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12

Graph $y < 2x - 1$ on the coordinate plane.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q1/5

Is (4, 1) a solution of the system $y \leq -x + 6$ and $y > \frac{1}{2}x - 2$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q2/5

A graph has a solid boundary line through (0, 2) and (3, 0), and the side that contains the origin is shaded.

Which inequality does it show?

A. $2x + 3y < 6$

B. $2x + 3y \geq 6$

C. $3x + 2y \leq 6$

D. $2x + 3y \leq 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

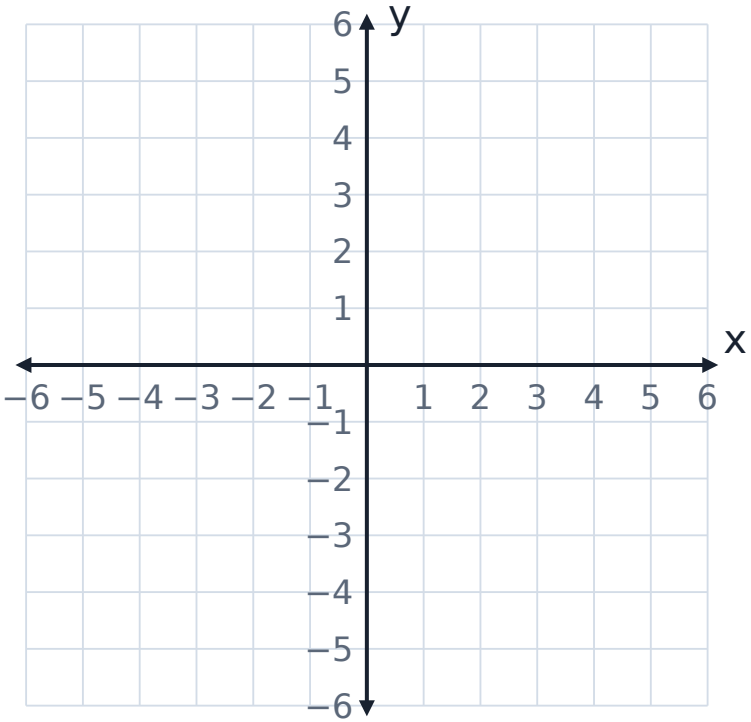
ID S32-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q3/5

Graph the system $y \geq x - 2$ and $x + y < 4$ on the coordinate plane.

Name one point that is a solution of the system.

Point: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q4/5

True or false?

Points on the boundary line $y = 3x$ are solutions of $y > 3x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q5/5

SET 32 • HSA-REI.D.12

BACKWARD 1

Solve an inequality in two variables for y • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve $2x + 3y \leq 6$ for y.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q1/5

Solve $x - 2y > 4$ for y.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q2/5

Solve $-y \geq x - 5$ for y.

A. $y \geq x + 5$

B. $y \geq -x + 5$

C. $y \leq -x + 5$

D. $y \leq x - 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q3/5

True or false?

Solving $4x + y < 3$ for y gives $y < -4x + 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q4/5

True or false?

Solving $-3y < 9$ for y gives $y < -3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q5/5

SET 32 • HSA-REI.D.12

BACKWARD 2

Plot points and find distances in the coordinate plane • Grade 6 • The Number System • 6.NS.C.8

Find the distance between $(2, -3)$ and $(2, 5)$.

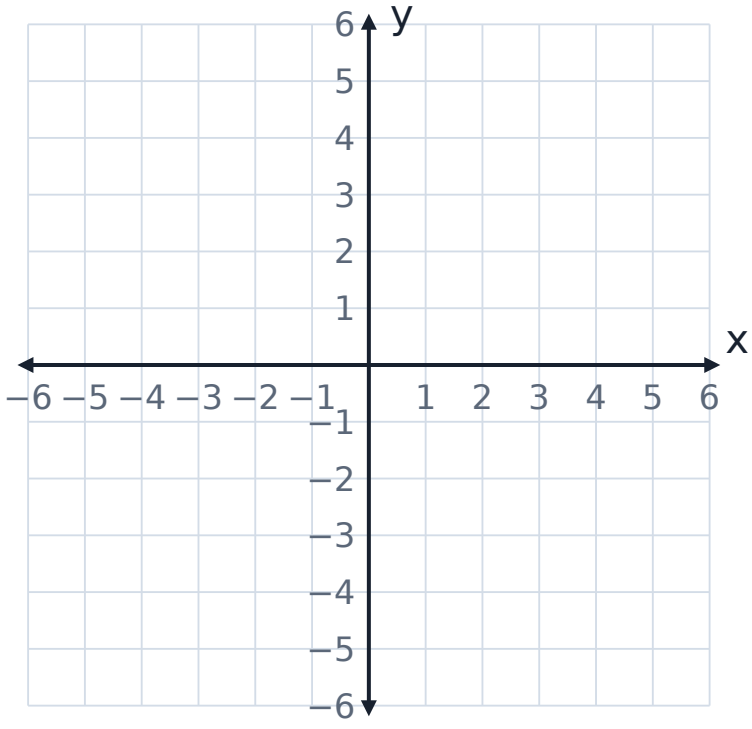
Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q1/5

Plot the points $(-3, 2)$ and $(4, -1)$.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q2/5

Which point is in Quadrant III?

A. (2, −5)

B. (−2, −5)

C. (2, 5)

D. (−2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q3/5

True or false?

The points (3, 4) and (3, −4) are reflections of each other across the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q4/5

True or false?

The distance between $(-4, 1)$ and $(6, 1)$ is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q5/5

SET 32 • HSA-REI.D.12

FORWARD 1

Find the area of a region described by inequalities • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

The system $x \geq 0$, $y \geq 0$, and $x + y \leq 6$ describes a triangle.

Find its vertices and its area.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q1/5

Find the perimeter of the rectangle with vertices (1, 1), (5, 1), (5, 4), and (1, 4).

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q2/5

The region $0 \leq x \leq 4$ and $0 \leq y \leq 2x$ is a triangle.

What is its area?

A. 32

B. 16

C. 8

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q3/5

True or false?

The triangle with vertices (0, 0), (3, 0), and (0, 4) has perimeter 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q4/5

True or false?

The square with vertices (0, 0), (2, 0), (2, 2), and (0, 2) has area 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q5/5

SET 32 • HSA-REI.D.12

FORWARD 2

Use zeros to find where a polynomial is positive or negative • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

$p(x) = (x - 1)(x - 4)$.

For which x is $p(x) < 0$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q1

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q1/5

$p(x) = x(x + 2)(x - 3)$.

Is $p(1)$ positive or negative? Explain using its factors.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q2

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q2/5

For which x is $(x + 3)(x - 2) > 0$?

A. $x < -3$ only

B. $x > 2$ only

C. $x < -3$ or $x > 2$

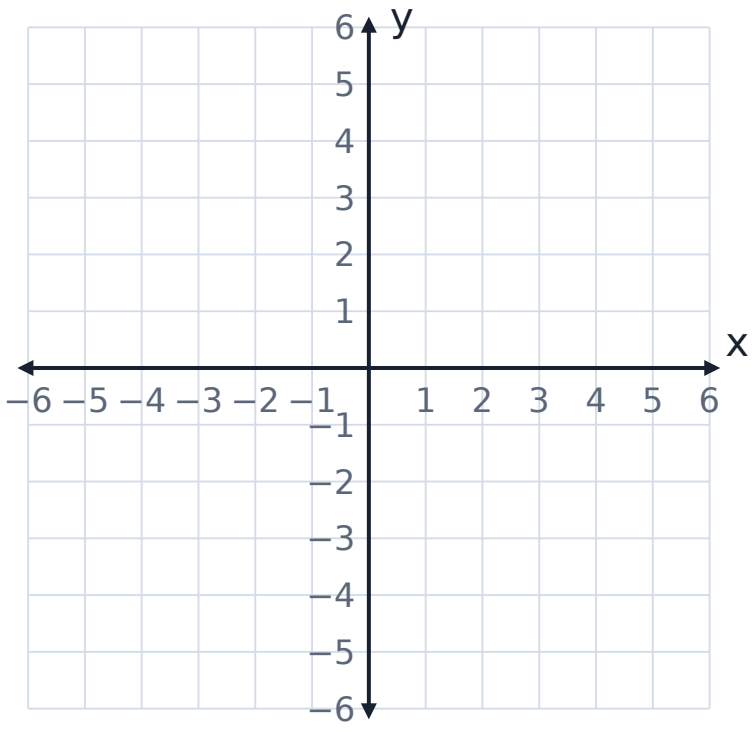
D. $-3 < x < 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q3

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q3/5

Sketch a rough graph of $y = x(x + 2)(x - 3)$, and mark the x-values where $y < 0$. (Use the grid for x; the y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q4/5

True or false?

$(x - 5)(x + 1) \geq 0$ for every real number x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q5/5

Create equations in two or more variables and graph them on labeled axes

MAIN	Algebra I main question	**HSA-CED.A.2**
BACKWARD 1	Write and solve an equation in one variable	**HSA-CED.A.1**
BACKWARD 2	Write the equation of a linear function	**8.F.B.4**
FORWARD 1	Write the equation of a circle from a situation	**HSG-GPE.A.1**
FORWARD 2	Write and interpret polynomial models	**HSF-IF.C.7.c**

SET 33 • HSA-CED.A.2

MAIN

Algebra I main question • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2

A savings account has $150 and grows by $25 each week.

Write an equation for the balance B after w weeks.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

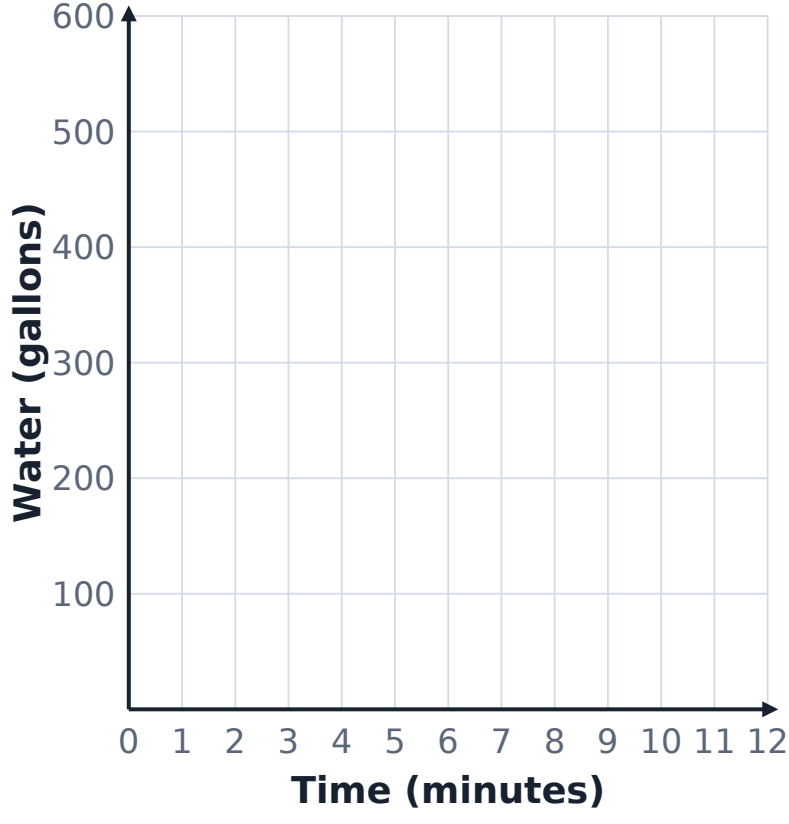
ID S33-M-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q1/5

A pool holds 600 gallons and drains at 50 gallons per minute.

Write an equation for the gallons G left after t minutes, and graph it for $0 \leq t \leq 12$.

Equation: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q2/5

A rectangle's length is 3 more than twice its width w.

Write an equation for its area A.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q3/5

A town has 8,000 people, and its population grows 3% each year.

Which equation gives the population P after t years?

A. $P = 8{,}000(0.03)^t$

B. $P = 8{,}000 + 0.03t$

C. $P = 8{,}000(1.03)^t$

D. $P = 8{,}000(1.3)^t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q4/5

True or false?

Student tickets cost $4 and adult tickets cost $9. The equation 4s + 9a = 300 relates the numbers of tickets when sales total $300.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q5/5

SET 33 • HSA-CED.A.2

BACKWARD 1

Write and solve an equation in one variable • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1

Solve 150 + 25w = 400 for w.

w = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q1/5

A pool has 600 gallons and drains at 50 gallons per minute.

Write and solve an equation for when it is empty.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q2/5

A plan costs $12 per month plus a $30 fee.

Which equation gives the number of months m for a total of $186?

A. $12m - 30 = 186$

B. $30m + 12 = 186$

C. $12(m + 30) = 186$

D. $12m + 30 = 186$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q3/5

True or false?

$w = 2.5$ is a solution of $w(2w + 3) = 20$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q4/5

True or false?

Solving $4s + 9(20) = 300$ gives $s = 40$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q5/5

SET 33 • HSA-CED.A.2

BACKWARD 2

Write the equation of a linear function • Grade 8 • Functions • 8.F.B.4

A line passes through (0, 5) and (2, 11).

Write its equation.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q1/5

A plumber charges $50 plus $30 per hour.

Write a function for the cost C(h).

C(h) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q2/5

A table has x-values 0, 1, 2 and y-values 4, 7, 10.

Which equation fits the table?

A. $y = 4x + 3$

B. $y = x + 4$

C. $y = 3x$

D. $y = 3x + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q3/5

True or false?

The rate of change of $y = -2x + 7$ is -2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q4/5

True or false?

The initial value of $y = 6x - 1$ is 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q5/5

SET 33 • HSA-CED.A.2

FORWARD 1

Write the equation of a circle from a situation • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

A sprinkler at (3, −2) waters everything within 4 meters.

Write an equation of the boundary of the watered region in standard form.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q1/5

A cell tower at the origin reaches 10 km. Is a phone at (6, 7) within range? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q2/5

Which equation describes all the points 5 units from (0, 0)?

A. $x^2 - y^2 = 25$

B. $x^2 + y^2 = 25$

C. $x + y = 5$

D. $x^2 + y^2 = 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q3/5

True or false?

The circle $(x - 1)^2 + (y - 1)^2 = 4$ passes through $(3, 1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q4/5

True or false?

The circle $x^2 + y^2 = 9$ passes through (3, 3).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q5/5

SET 33 • HSA-CED.A.2

FORWARD 2

Write and interpret polynomial models • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

An open box is made from a 10 in by 8 in sheet by cutting a square of side x from each corner.

Write an equation for its volume V.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q1/5

An open box is made from a 10 in by 8 in sheet by cutting a square of side x from each corner and folding up the sides. Its volume is $V = x(10 - 2x)(8 - 2x)$.

Which values of x make sense in the situation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q2/5

What are the zeros of $V = x(10 - 2x)(8 - 2x)$?

A. 5 and 4 only

B. 0, 10, and 8

C. 0, −5, and −4

D. 0, 5, and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q3/5

True or false?

The graph of $y = x^3 - x$ crosses the x-axis three times.

A. True

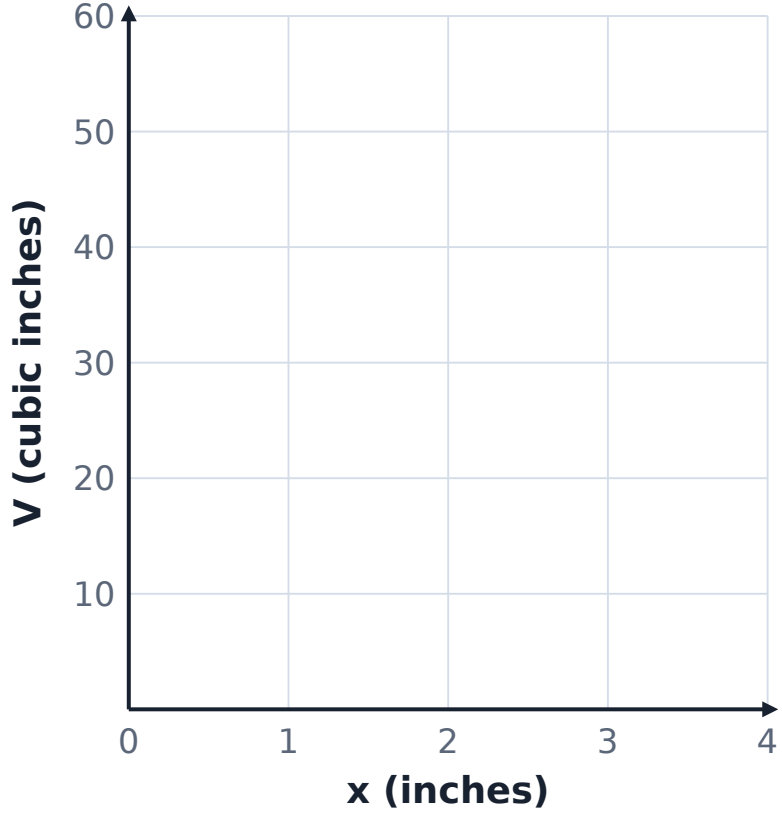
B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q4/5

An open box is made from a 10 in by 8 in sheet by cutting a square of side x from each corner. Its volume is $V = x(10 - 2x)(8 - 2x)$. Graph V for $0 \leq x \leq 4$.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q5/5

Represent constraints with equations and inequalities and decide whether solutions are viable

MAIN	Algebra I main question	**HSA-CED.A.3**
BACKWARD 1	Test points in a system of inequalities	**HSA-REI.D.12**
BACKWARD 2	Solve a system of linear equations	**HSA-REI.C.6**
FORWARD 1	Use constraints to optimize a design	**HSG-MG.A.3**
FORWARD 2	Use expected values to compare decisions — nearest related	**HSS-MD.B.7**

SET 34 • HSA-CED.A.3

MAIN

Algebra I main question • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3

A bakery makes m muffins and b loaves. A muffin uses 1 cup of flour and a loaf uses 3 cups; there are 60 cups. A muffin takes 2 minutes and a loaf takes 5 minutes; there are 140 minutes.

Write the system of constraints.

Constraints: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q1/5

A bakery makes m muffins and b loaves. A muffin uses 1 cup of flour and a loaf uses 3 cups; there are 60 cups. A muffin takes 2 minutes and a loaf takes 5 minutes; there are 140 minutes.

Is making 40 muffins and 10 loaves possible? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q2/5

A club needs at least $500 from selling shirts at $15 and hats at $10, and it can sell at most 45 items.

Write the constraints for s shirts and h hats.

Constraints: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q3/5

A club sells shirts for $15 and hats for $10. It needs at least $500 and can sell at most 45 items.

Which combination meets both constraints?

A. 15 shirts and 25 hats

B. 25 shirts and 15 hats

C. 30 shirts and 20 hats

D. 20 shirts and 10 hats

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q4/5

True or false?

If x counts people, x = 12.5 can be a viable solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q5/5

SET 34 • HSA-CED.A.3

BACKWARD 1

Test points in a system of inequalities • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12

Is (10, 10) a solution of $x + 3y \leq 60$ and $2x + 5y \leq 140$? Explain.

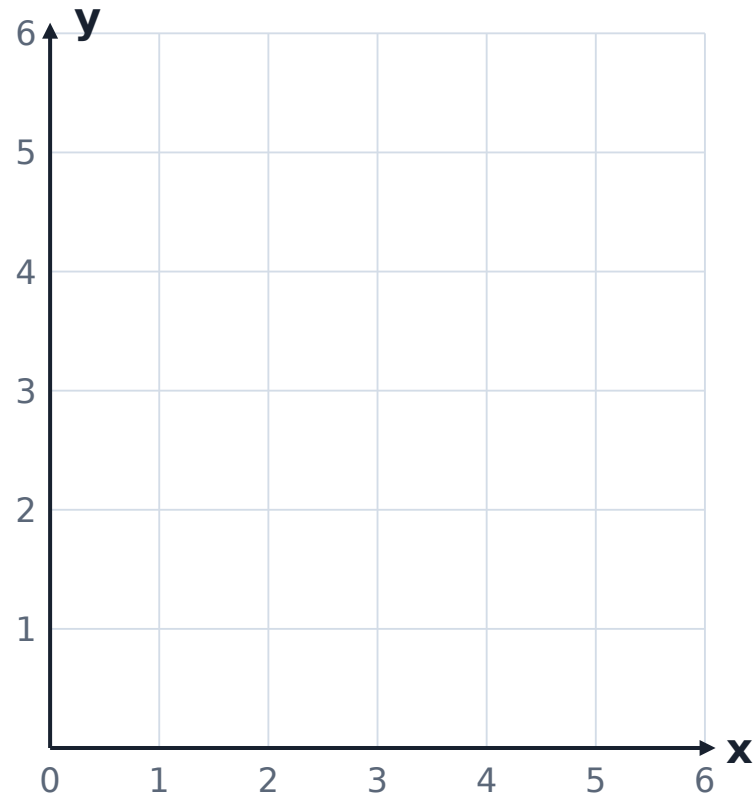
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q1/5

Graph the system $x + y \leq 5$, $x \geq 0$, and $y \geq 0$.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q2/5

Which point is in the solution region of $y \geq 2x$ and $y \leq 6$?

A. (4, 1)

B. (1, 7)

C. (3, 5)

D. (1, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q3/5

True or false?

The point (0, 0) satisfies $x \geq 0$ and $y \geq 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q4/5

True or false?

The point (2, 5) satisfies $x + y < 7$.

A. True

B. False

SET 34 • HSA-CED.A.3

BACKWARD 2

Solve a system of linear equations • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

Solve.

$x + y = 45$

$15x + 10y = 500$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 34 • BACKWARD 2 • Q1/5

Solve.

$2x + y = 12$

$x + y = 8$

Solution: _____

Solve.

$x + 3y = 60$

$x + y = 40$

A. (30, 10)

B. (15, 15)

C. (10, 30)

D. (20, 20)

True or false?

(10, 35) is a solution of $x + y = 45$.

A. True

B. False

True or false?

The system $x + y = 10$ and $x + y = 12$ has one solution.

A. True

B. False

SET 34 • HSA-CED.A.3

FORWARD 1

Use constraints to optimize a design • High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3

A feasible region has vertices (0, 0), (60, 0), and (0, 20).

Find the greatest value of P = 2m + 7b on the region and where it occurs.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q1/5

A rectangular pen uses at most 40 m of fence.

What dimensions give the greatest area?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q2/5

The cost $C = 3x + 2y$ is checked at the vertices (0, 6), (2, 2), and (6, 0) of a region.

Where is the cost least?

A. (2, 2)

B. (0, 0)

C. (6, 0)

D. (0, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-F1-Q3**

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q3/5

True or false?

For a linear expression on a polygonal region, the greatest value occurs at a vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q4/5

True or false?

Of all rectangles with area 36, a square has the smallest perimeter, so its side is 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q5/5

FORWARD 2

Use expected values to compare decisions • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: Choosing viable options under constraints extends to choosing between options by expected value.

Plan A earns $100 with probability 0.6 and $0 otherwise.
Plan B earns $70 for sure.

Which plan has the greater expected value?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q1/5 • NEAREST RELATED

Guessing on a 4-choice question earns 1 point if right and loses 0.25 point if wrong.

What is the expected score from guessing?

Expected score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q2/5 • NEAREST RELATED

A $5 warranty pays $100 if the item breaks (probability 0.04) and $0 otherwise.

What is the expected net value of buying it?

A. $-\$5$

B. $\$1$

C. $\$4$

D. $-\$1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

A choice with the higher expected value can still lose money on a single trial.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

If a game has an expected value of 0, a player cannot lose money playing it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q5/5 • NEAREST RELATED

SET 35 • HSF-IF.A.1 • INTERPRETING FUNCTIONS

Understand a function as a rule that assigns exactly one output to each input, with a domain and a range

MAIN	Algebra I main question	**HSF-IF.A.1**
BACKWARD 1	Decide whether a relation is a function	**8.F.A.1**
BACKWARD 2	Check whether points are on the graph of an equation	**HSA-REI.D.10**
FORWARD 1	Describe transformations of the plane as functions of points	**HSG-CO.A.2**
FORWARD 2	Restrict a domain to make a function invertible	**HSF-BF.B.4.d**

SET 35 • HSF-IF.A.1

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1

Does the relation $\{(-2, 4), (0, 1), (2, 4), (3, -1)\}$ define y as a function of x? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q1/5

Give the domain and the range of $f(x) = x^2 + 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q2/5

Which equation does NOT define y as a function of x?

A. $y = x^2$

B. $x = y^2$

C. $y = 2^x$

D. $y = |x|$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q3/5

$f(x) = \sqrt{(x - 3)}$. Over the real numbers, what is the domain of f?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q4/5

True or false?

If f is a function and f(2) = 5, then f(2) can also equal 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q5/5

SET 35 • HSF-IF.A.1

BACKWARD 1

Decide whether a relation is a function • Grade 8 • Functions • 8.F.A.1

A table pairs x = 1, 2, 3, 2 with y = 4, 5, 6, 7. Is y a function of x? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q1

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q1/5

Each button on a vending machine gives exactly one kind of snack. Is the snack a function of the button? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q2

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q2/5

Which set of ordered pairs is a function?

A. {(5, 1), (5, 2), (6, 3)}

B. {(0, 0), (0, 1)}

C. {(1, 3), (2, 3), (3, 3)}

D. {(1, 2), (1, 3), (2, 4)}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q3/5

True or false?

Two different inputs of a function can have the same output.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q4

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q4/5

True or false?

A function can pair the input 4 with two different outputs.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q5

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q5/5

SET 35 • HSF-IF.A.1

BACKWARD 2

Check whether points are on the graph of an equation • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10

Is $(-2, 5)$ on the graph of $y = x^2 + 1$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q1/5

The point (a, 10) is on the graph of y = 2x + 4.

Find a.

a = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q2/5

Which point is on the graph of $y = \sqrt{x}$?

A. (9, −3)

B. (−9, 3)

C. (9, 3)

D. (3, 9)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q3/5

True or false?

The graph of $y = 4$ is the set of all points with y-coordinate 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q4/5

True or false?

(1, 3) is on the graph of $y = x^2 + 3$.

A. True

B. False

SET 35 • HSF-IF.A.1

FORWARD 1

Describe transformations of the plane as functions of points • High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2

The translation T maps (x, y) to (x + 3, y − 2).

Find T(4, 1).

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q1/5

A reflection across the y-axis maps (x, y) to which point?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q2/5

Which transformation does NOT preserve distance?

A. $(x, y) \rightarrow (x + 5, y)$

B. $(x, y) \rightarrow (-x, y)$

C. $(x, y) \rightarrow (y, x)$

D. $(x, y) \rightarrow (2x, y)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q3/5

True or false?

A transformation of the plane is a function: each input point has exactly one image.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q4/5

True or false?

A rotation of 90° counterclockwise about the origin maps (1, 0) to (0, −1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q5/5

SET 35 • HSF-IF.A.1

FORWARD 2

Restrict a domain to make a function invertible • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d

Explain why $f(x) = x^2$, with all real numbers as its domain, has no inverse function.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q1/5

$f(x) = (x - 1)^2$. Restrict f to the largest interval that contains $x = 3$ on which f is one-to-one, and write the inverse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q2/5

Which function is one-to-one on all real numbers?

A. $f(x) = x^2 - 4$

B. $f(x) = x^3$

C. $f(x) = |x|$

D. $f(x) = x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q3/5

True or false?

$f(x) = x^2$ with domain $x \geq 0$ has the inverse $f^{-1}(x) = \sqrt{x}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q4/5

True or false?

Every function has an inverse function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q5/5

SET 36 • HSF-IF.A.2 • INTERPRETING FUNCTIONS

Use function notation, evaluate functions, and interpret statements in function notation in context

MAIN	Algebra I main question	**HSF-IF.A.2**
BACKWARD 1	Evaluate expressions for given values	**6.EE.A.2.c**
BACKWARD 2	Solve linear equations	**HSA-REI.B.3**
FORWARD 1	Evaluate compositions of functions	**HSF-BF.A.1.c**
FORWARD 2	Read values of an inverse function	**HSF-BF.B.4.c**

SET 36 • HSF-IF.A.2

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

$f(x) = 2x^2 - 3x + 1$.

Find $f(-2)$.

$f(-2) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q1/5

$h(t) = -16t^2 + 64t$ gives a ball's height in feet t seconds after it is hit.

Find h(1), and explain what it means.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q2/5

$g(x) = 5x - 7$.

Find the value of x for which $g(x) = 18$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q3/5

C(n) is the cost in dollars of n tickets.

What does C(4) = 60 mean?

A. 4 tickets cost $60.

B. C equals 4 times 60.

C. Each ticket costs $60.

D. 60 tickets cost $4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q4/5

True or false?

If $f(x) = x^2$, then $f(a + 1) = a^2 + 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q5/5

SET 36 • HSF-IF.A.2

BACKWARD 1

Evaluate expressions for given values • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Evaluate $2x^2 - 3x + 1$ when $x = 3$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q1/5

Evaluate $-16t^2 + 64t$ when $t = 2$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q2/5

Evaluate $3(a - 4)^2$ when $a = 6$.

A. 144

B. 12

C. 6

D. 36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q3/5

True or false?

When $x = -1$, $x^2 - x = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q4/5

True or false?

When $n = 5$, $2n^2 = 100$.

A. True

B. False

SET 36 • HSF-IF.A.2

BACKWARD 2

Solve linear equations • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve $5x - 7 = 18$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q1/5

Solve $3(x + 2) = 2x - 4$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q2/5

Solve 0.5x + 3 = 11.

A. x = 7

B. x = 28

C. x = 16

D. x = 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q3/5

True or false?

$x = -2$ is the solution of $4x + 9 = 1$.

A. True

B. False

True or false?

The solution of $7 - x = 10$ is $x = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q5/5

SET 36 • HSF-IF.A.2

FORWARD 1

Evaluate compositions of functions • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c

$f(x) = x + 4$ and $g(x) = 3x$.

Find $f(g(2))$.

$f(g(2)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q1/5

$f(x) = x^2$ and $g(x) = x - 3$.

Write $f(g(x))$.

$f(g(x)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q2/5

$f(x) = 2x$ and $g(x) = x + 1$.

Find $g(f(5))$.

A. 11

B. 10

C. 6

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-F1-Q3**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q3/5

True or false?

If $f(x) = x - 1$ and $g(x) = x + 1$, then $f(g(x)) = x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q4/5

True or false?

If $f(x) = 2x$ and $g(x) = x^2$, then $f(g(3)) = 36$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q5/5

SET 36 • HSF-IF.A.2

FORWARD 2

Read values of an inverse function • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c

The function f has domain {1, 2, 3, 4}, with f(1) = 3, f(2) = 5, f(3) = 7, and f(4) = 9. Its outputs are all different, so f has an inverse.

Find $f^{-1}(7)$.

$f^{-1}(7) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q1/5

$f(x) = 2x + 1$ is invertible.

Find $f^{-1}(11)$.

$f^{-1}(11) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q2/5

f is a one-to-one function with $f(4) = 10$.

Which statement is true?

A. $f^{-1}(10) = 4$

B. $f^{-1}(4) = 10$

C. $f(10) = 4$

D. $f^{-1}(10) = \frac{1}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q3/5

True or false?

If f is a one-to-one function, the graph of f^{-1} is the reflection of the graph of f across the line y = x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q4/5

True or false?

$f^{-1}(x)$ means $1 \div f(x)$.

A. True

B. False

SET 37 • HSF-IF.A.3 • INTERPRETING FUNCTIONS

Recognize that sequences are functions whose domain is a set of integers

MAIN	Algebra I main question	**HSF-IF.A.3**
BACKWARD 1	Evaluate functions in function notation	**HSF-IF.A.2**
BACKWARD 2	Generate two numerical patterns and relate their terms	**5.OA.B.3**
FORWARD 1	Repeat a dilation and describe the lengths — nearest related	**HSG-SRT.A.1**
FORWARD 2	Find the sum of a finite geometric series	**HSA-SSE.B.4**

SET 37 • HSF-IF.A.3

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3

A sequence is defined by $f(1) = 3$ and $f(n) = f(n - 1) + 4$ for $n \geq 2$.

Find $f(5)$.

$f(5) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q1/5

A sequence is defined by $f(1) = 2$, $f(2) = 3$, and $f(n) = f(n - 1) + 2f(n - 2)$ for $n \geq 3$.

Find $f(5)$.

$f(5) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q2/5

Write an explicit formula for the sequence 5, 10, 20, 40, …, with f(1) = 5.

f(n) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q3/5

A sequence f(n) starts with f(1).

What is its domain?

A. The positive integers 1, 2, 3, ...

B. The numbers from 0 to 1

C. All real numbers

D. All integers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q4/5

True or false?

The graph of the sequence f(n) = 2n + 1 for n = 1, 2, 3, ... is a solid line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q5/5

SET 37 • HSF-IF.A.3

BACKWARD 1

Evaluate functions in function notation • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

$f(n) = 3n - 1$.

Find $f(4)$.

$f(4) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q1/5

$f(n) = 2^n$.

Find f(5).

f(5) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q2/5

$f(n) = n^2 + 1$.

Find $f(3)$.

A. 7

B. 10

C. 16

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q3/5

True or false?

If $f(n) = 10 - n$, then $f(10) = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q4/5

True or false?

If f(n) = 4n, then f(3) = 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q5/5

SET 37 • HSF-IF.A.3

BACKWARD 2

Generate two numerical patterns and relate their terms • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

Pattern A starts at 0 and adds 3. Pattern B starts at 0 and adds 6.

Write the first 5 terms of each pattern.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q1/5

Pattern A starts at 0 and adds 4. Pattern B starts at 0 and adds 8.

Write ordered pairs (A term, B term) for the first 4 terms. What relationship do you see?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q2/5

Pattern A starts at 0 and adds 2. Pattern B starts at 0 and adds 6.

How do their third terms compare?

A. A's is 3 times B's.

B. B's is 4 more than A's.

C. B's is 3 times A's.

D. They are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q3/5

True or false?

Pattern A starts at 0 and adds 5. Pattern B starts at 0 and adds 10. Each term of B is twice the matching term of A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q4/5

True or false?

Pattern A starts at 1 and adds 2. Pattern B starts at 2 and adds 4. Each term of B is 4 times the matching term of A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q5/5

SET 37 • HSF-IF.A.3

FORWARD 1

Repeat a dilation and describe the lengths • High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Geometry standard. Connection: Repeated dilations produce a geometric sequence of lengths, a function defined on the positive integers.

A segment of length 3 is dilated by scale factor 2, then the image is dilated by 2 again, and so on.

Write the length after n dilations.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q1/5 • NEAREST RELATED

A dilation with center (0, 0) and scale factor 3 maps (2, −1) to which point?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q2/5 • NEAREST RELATED

A dilation with scale factor $\frac{1}{2}$ maps a segment of length 10 to a segment of what length?

A. 20

B. 9.5

C. 10

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

A dilation maps a line that does not pass through the center to a parallel line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

A dilation with scale factor 2 doubles the area of a figure.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 37 • HSF-IF.A.3

FORWARD 2

Find the sum of a finite geometric series • High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

Find the sum of the first 5 terms of 2, 6, 18, ...

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q1/5

Write a formula for the sum of the first n terms of 5, 10, 20, ...

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q2

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q2/5

What is 1 + 4 + 16 + 64?

A. 256

B. 85

C. 84

D. 64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-F2-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q3/5

True or false?

The sum of the first 4 terms of 1, 2, 4, 8, ... is 15.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q4/5

True or false?

A geometric series with first term 4, ratio 1, and 5 terms has sum 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q5/5

SET 38 • HSF-IF.B.4 • INTERPRETING FUNCTIONS

Interpret key features of graphs and tables of functions in context

MAIN	Algebra I main question	**HSF-IF.B.4**
BACKWARD 1	Describe a function qualitatively from a situation	**8.F.B.5**
BACKWARD 2	Use vertex form to find a maximum or minimum	**HSA-SSE.B.3.b**
FORWARD 1	Identify key features of polynomial functions	**HSF-IF.C.7.c**
FORWARD 2	Identify amplitude, midline, and period of a periodic model	**HSF-TF.B.5**

SET 38 • HSF-IF.B.4

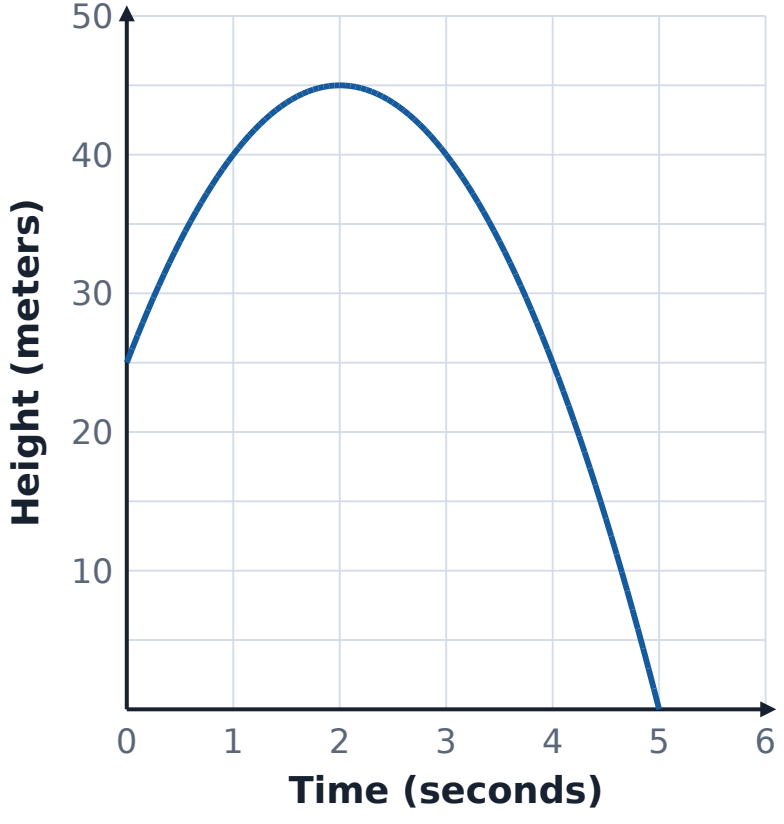
MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4

The graph shows a ball's height h(t) in meters, t seconds after it is thrown from a balcony.

What is the maximum height, and when does it occur?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

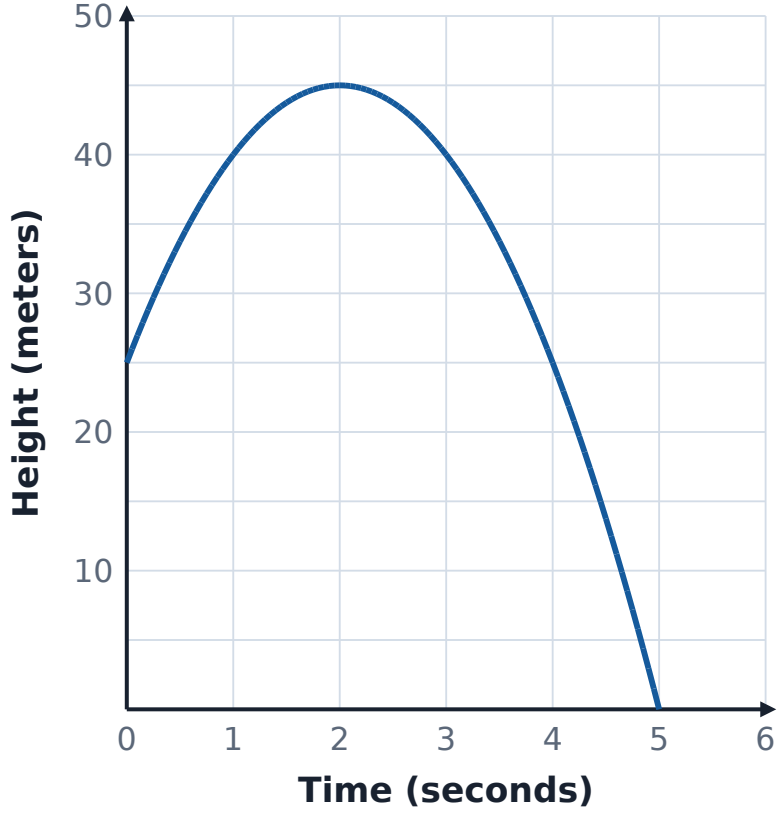
ID S38-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q1/5

The graph shows a ball's height h(t) in meters, t seconds after it is thrown from a balcony (t = 0 is the moment it is thrown).

What is the h-intercept of the graph, and what does it mean?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

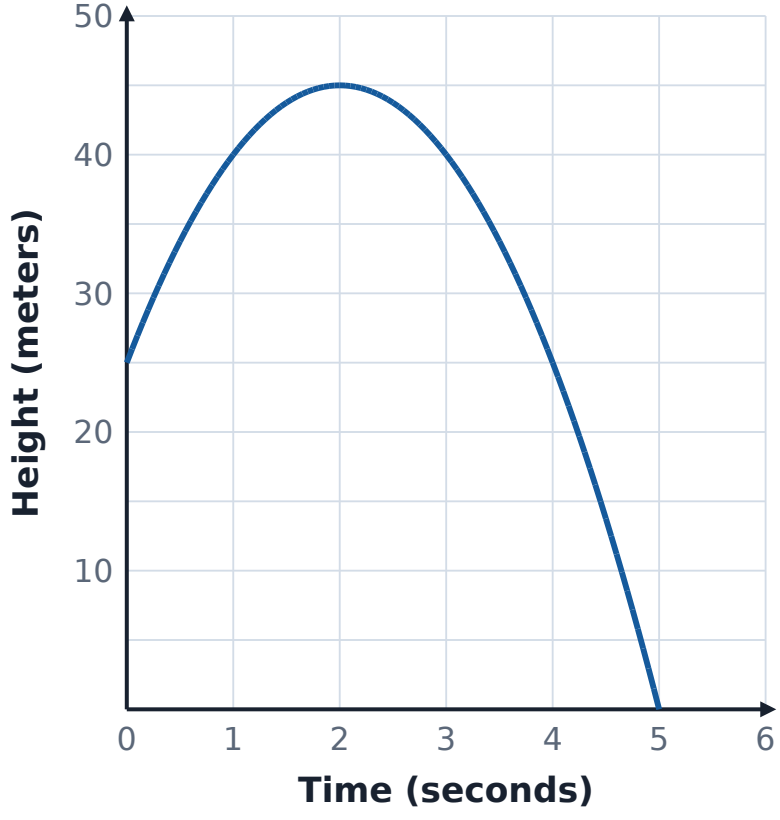
ID S38-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q2/5

The graph shows a ball's height h(t) in meters, t seconds after it is thrown from a balcony.

When does the ball reach the ground? Over what interval is its height increasing?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q3

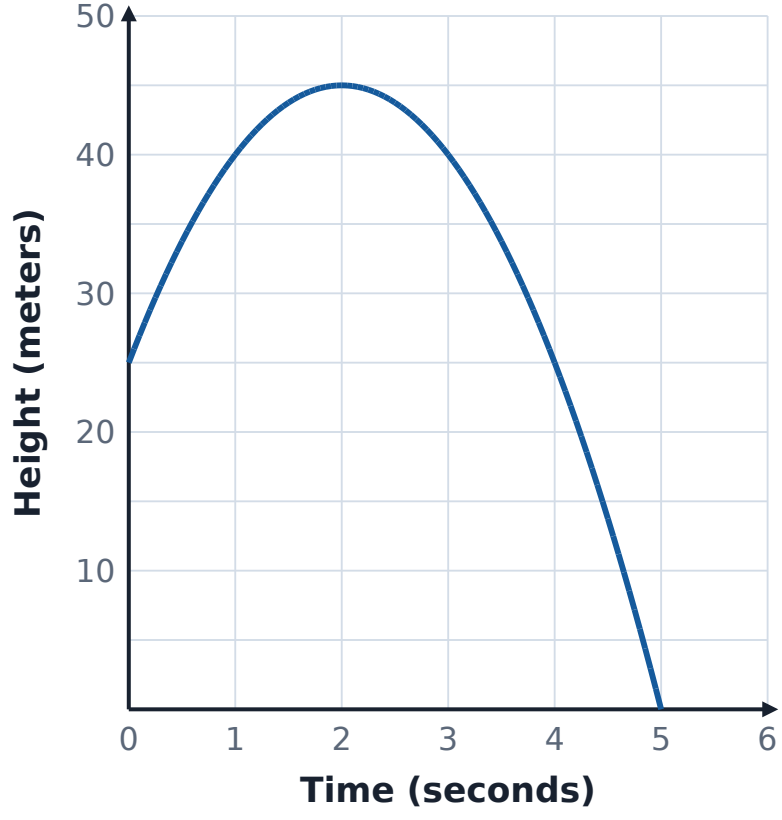
High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q3/5

The graph shows a ball's height h(t) in meters, t seconds after it is thrown from a balcony.

Over what interval is the ball's height decreasing?

A. $0 < t < 2$ **B.** $0 < t < 5$

C. $2 < t < 5$ **D.** $25 < t < 45$



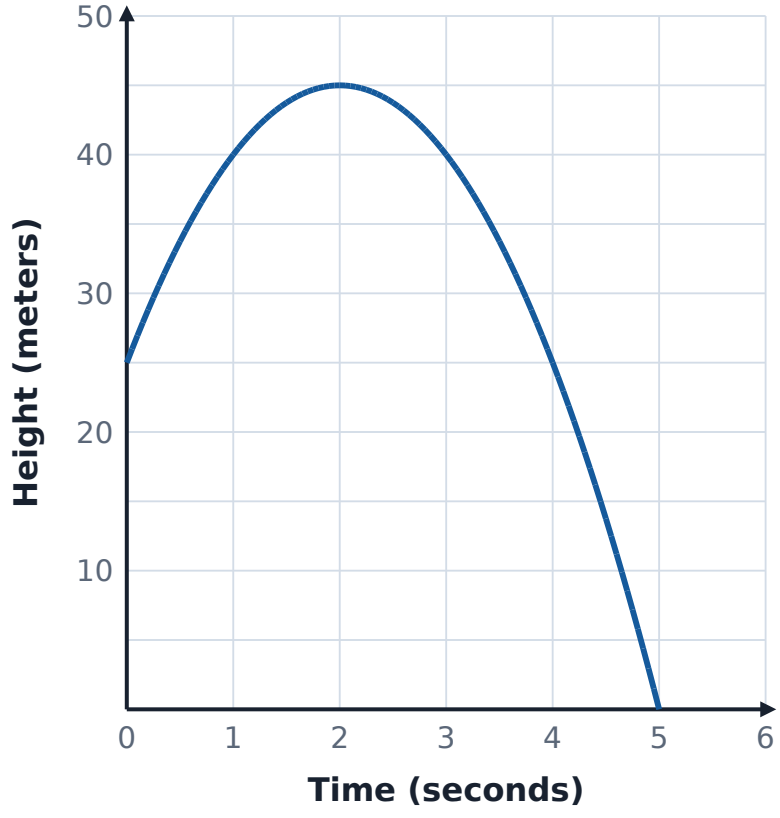
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q4/5

True or false?

The graph shows a ball's height h(t) in meters, t seconds after it is thrown from a balcony. The ball is 25 meters high twice: at t = 0 and at t = 4.



A. True **B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q5/5

SET 38 • HSF-IF.B.4

BACKWARD 1

Describe a function qualitatively from a situation • Grade 8 • Functions • 8.F.B.5

A car speeds up, drives at a constant speed, then slows to a stop.

Describe the graph of its speed over time.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q1

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q1/5

A graph of water level is horizontal from t = 2 to t = 5.

What does that mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q2

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q2/5

Which situation has a graph that first increases and then decreases?

A. Savings with a steady deposit each week

B. The height of a ball thrown up that comes back down

C. The distance a parked car travels

D. The height of a burning candle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q3

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q3/5

True or false?

On a distance-time graph, a steeper segment means a greater speed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q4

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q4/5

True or false?

A decreasing graph always means the quantity is negative.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q5

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q5/5

SET 38 • HSF-IF.B.4

BACKWARD 2

Use vertex form to find a maximum or minimum • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b

Write $-5t^2 + 20t + 25$ in the form $a(t - h)^2 + k$.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q1/5

What is the maximum value of $-(x - 3)^2 + 7$, and for what x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q2/5

What is the minimum value of $2(x + 1)^2 - 8$?

A. -8

B. -1

C. 2

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q3/5

True or false?

$-5(t - 2)^2 + 45$ is greatest when $t = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q4/5

True or false?

The minimum value of $(x - 4)^2 + 1$ is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q5/5

SET 38 • HSF-IF.B.4

FORWARD 1

Identify key features of polynomial functions • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

Describe the end behavior of $f(x) = -2x^3 + x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q1/5

Find the zeros of $f(x) = x(x - 2)(x + 3)$.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q2/5

Which function rises on the right and falls on the left?

A. $f(x) = -x^4$

B. $f(x) = x^3 - 4x$

C. $f(x) = x^2$

D. $f(x) = -x^3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q3/5

True or false?

A polynomial function of degree 4 has at most 4 real zeros.

A. True

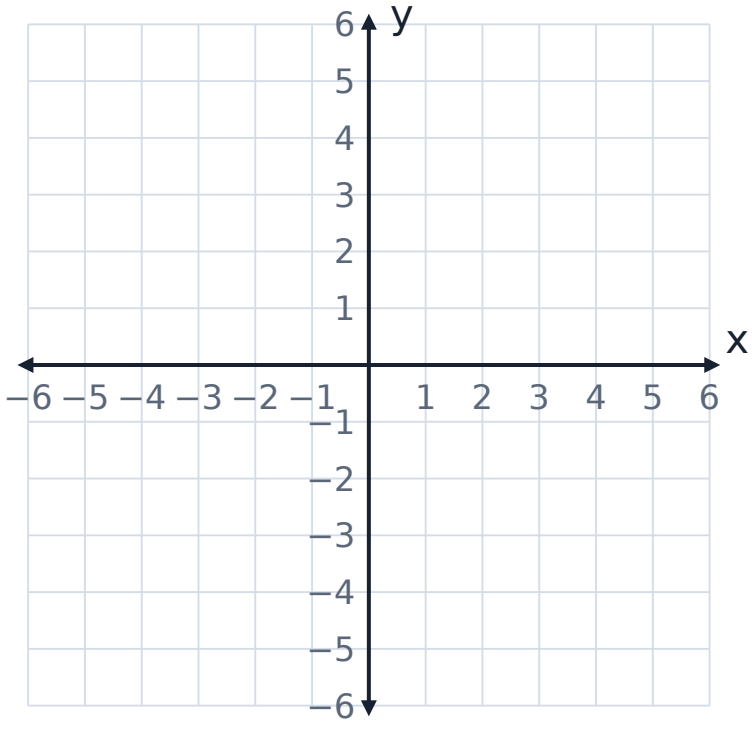
B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q4/5

Graph $f(x) = -x^2(x - 3)$. Mark the zeros and show what the graph does at each zero and at the ends. (Use the grid for x; the y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q5/5

SET 38 • HSF-IF.B.4

FORWARD 2

Identify amplitude, midline, and period of a periodic model • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

A Ferris wheel seat's height is $h(t) = 30 - 25\cos(\pi t \div 5)$ meters after t minutes.

What are the greatest and least heights?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q1/5

What is the period of $h(t) = 30 - 25\cos(\pi t \div 5)$?

Period: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q2/5

What is the midline of $y = 4\sin x + 2$?

A. $y = 4$

B. $y = 2$

C. $y = 6$

D. $y = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q3/5

True or false?

The amplitude of $y = 3\sin x$ is 3.

A. True

B. False

True or false?

The function $y = \sin x$ has period π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q5/5

SET 78 • HSF-IF.B.4 • INTERPRETING FUNCTIONS

Sketch graphs that show key features given in a verbal description

MAIN	Algebra I main question	**HSF-IF.B.4**
BACKWARD 1	Sketch a graph from a description	**8.F.B.5**
BACKWARD 2	Write and graph equations on labeled axes	**HSA-CED.A.2**
FORWARD 1	Sketch polynomial graphs from a description of their features	**HSF-IF.C.7.c**
FORWARD 2	Sketch periodic graphs from a description	**HSF-TF.B.5**

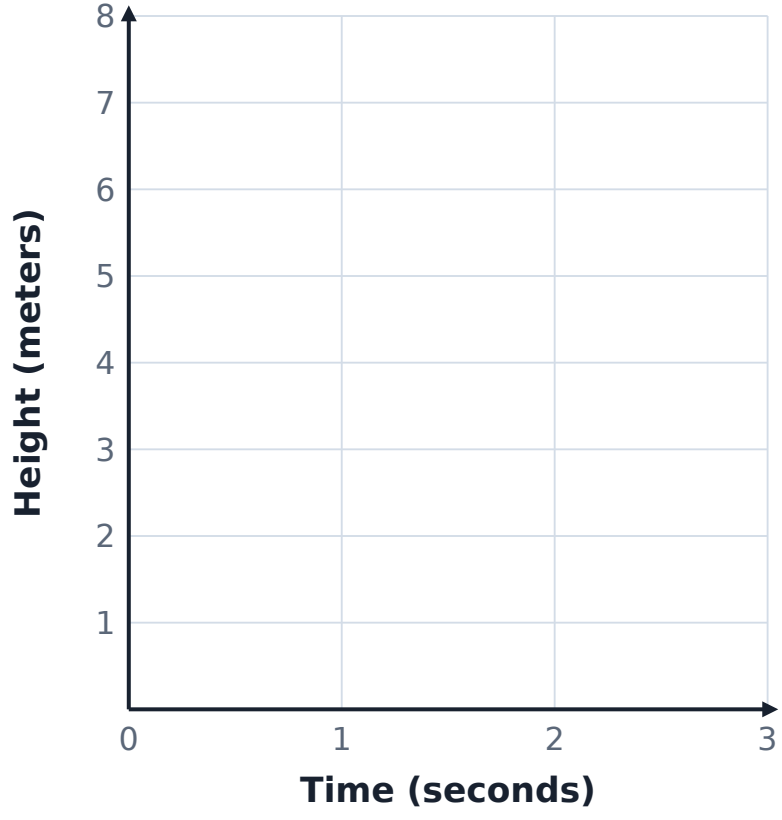
SET 78 • HSF-IF.B.4

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4

A ball is thrown up from a 2-m platform. It reaches its highest point, 7 m, after 1 second, and it lands on the ground after about 2.2 seconds.

Sketch its height against time. Label the starting point, the maximum, and the landing time.



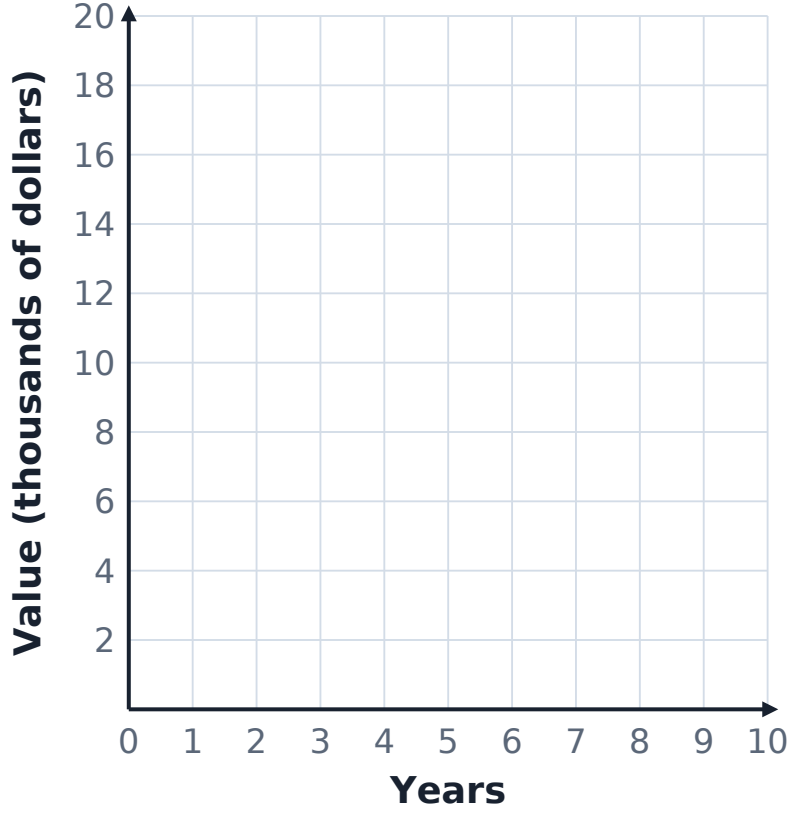
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 78 • MAIN • Q1/5

A car is worth $20,000 when new, and it loses 15% of its value each year, never reaching $0.

Sketch its value for 0 to 10 years. Show the starting value and what happens as time goes on.



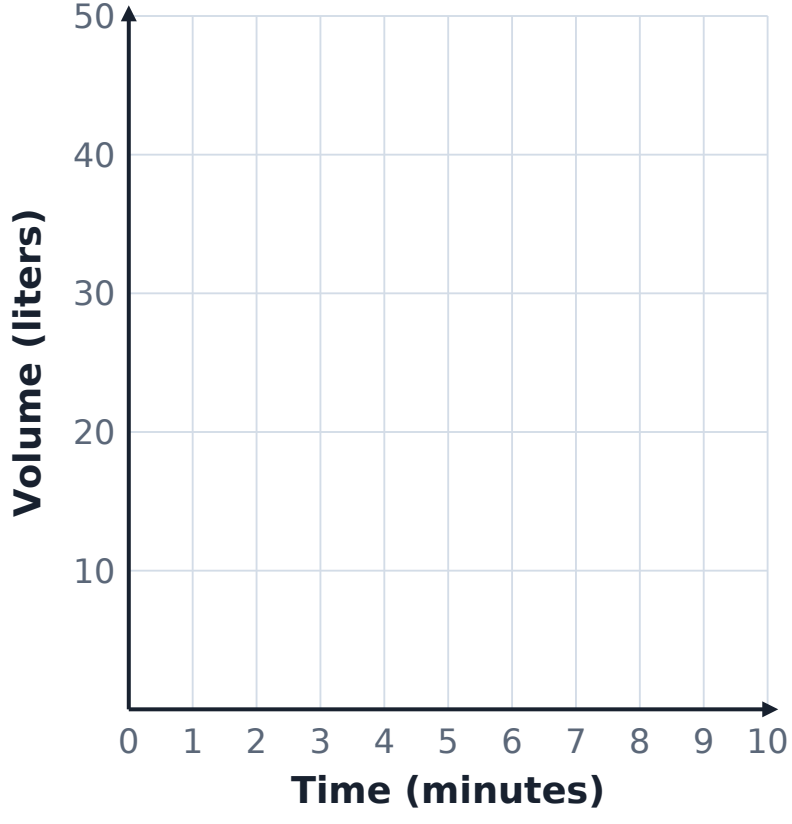
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 78 • MAIN • Q2/5

A 50-liter tank drains at a constant rate until it is empty after 5 minutes, stays empty for 2 minutes, and then refills at a constant rate to 30 liters in 3 minutes.

Sketch the volume over the 10 minutes.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 78 • MAIN • Q3/5

A town's population grew slowly at first and then faster and faster.

Which graph shape fits?

A. Increasing and leveling off

B. Increasing and curving upward

C. A horizontal line

D. Decreasing

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 78 • MAIN • Q4/5

True or false?

A graph of the temperature of a cup of coffee left on a table falls at a constant rate until it reaches 0 °C.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 78 • MAIN • Q5/5

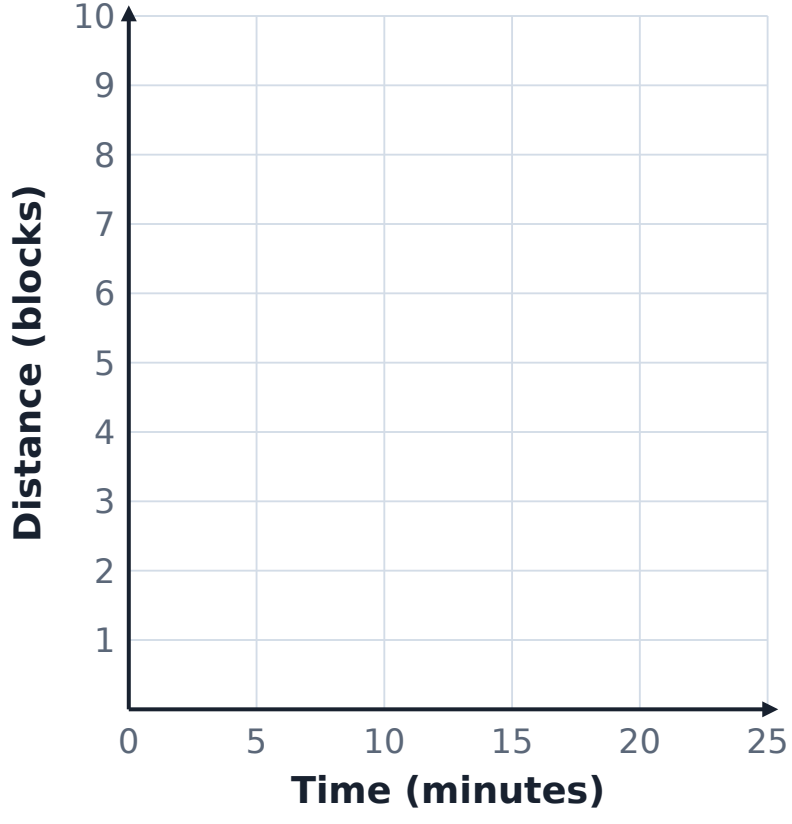
SET 78 • HSF-IF.B.4

BACKWARD 1

Sketch a graph from a description • Grade 8 • Functions • 8.F.B.5

You walk away from home at a steady pace for 10 minutes, rest for 5 minutes, then walk home faster in 10 minutes.

Sketch your distance from home over 25 minutes.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q1

Grade 8 • Functions • 8.F.B.5 • Set 78 • BACKWARD 1 • Q1/5

A graph of a plant's height rises steeply and then levels off. Describe what happened to the plant.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q2

Grade 8 • Functions • 8.F.B.5 • Set 78 • BACKWARD 1 • Q2/5

Which situation has a graph that stays constant (horizontal) for a while?

A. A rising balloon's height

B. A burning candle's height

C. A parked car's distance from home

D. A falling ball's height

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q3

Grade 8 • Functions • 8.F.B.5 • Set 78 • BACKWARD 1 • Q3/5

True or false?

A graph that rises and then falls has a highest point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q4

Grade 8 • Functions • 8.F.B.5 • Set 78 • BACKWARD 1 • Q4/5

True or false?

A horizontal segment on a speed-time graph means the car is stopped.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q5

Grade 8 • Functions • 8.F.B.5 • Set 78 • BACKWARD 1 • Q5/5

SET 78 • HSF-IF.B.4

BACKWARD 2

Write and graph equations on labeled axes • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2

Write an equation for the area A of a square with side s. What kind of graph does it have for s ≥ 0?

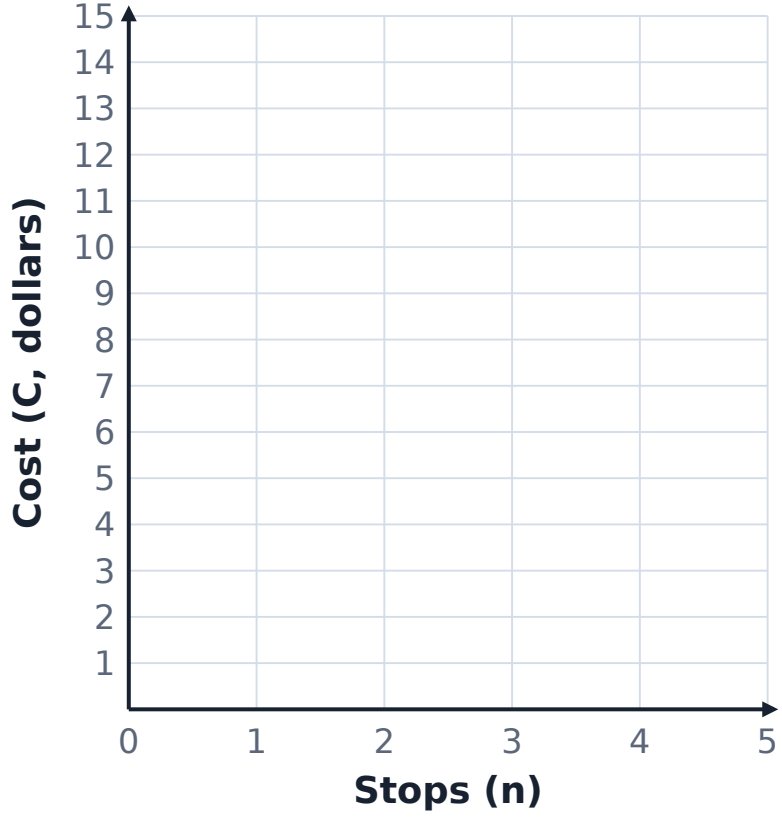
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 78 • BACKWARD 2 • Q1/5

Graph C = 5 + 2n, the cost in dollars of a ride with n stops, for n = 0 to 5.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 78 • BACKWARD 2 • Q2/5

A $1,200 laptop loses $200 of value each year. Which equation gives its value V after t years?

A. $V = 1{,}200 - 200t$

B. $V = 200 - 1{,}200t$

C. $V = 1{,}200 + 200t$

D. $V = 1{,}200(0.8)^{t}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 78 • BACKWARD 2 • Q3/5

True or false?

An equation in two variables can be graphed on a coordinate plane.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 78 • BACKWARD 2 • Q4/5

True or false?

The graph of d = 50t is a curve.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q5

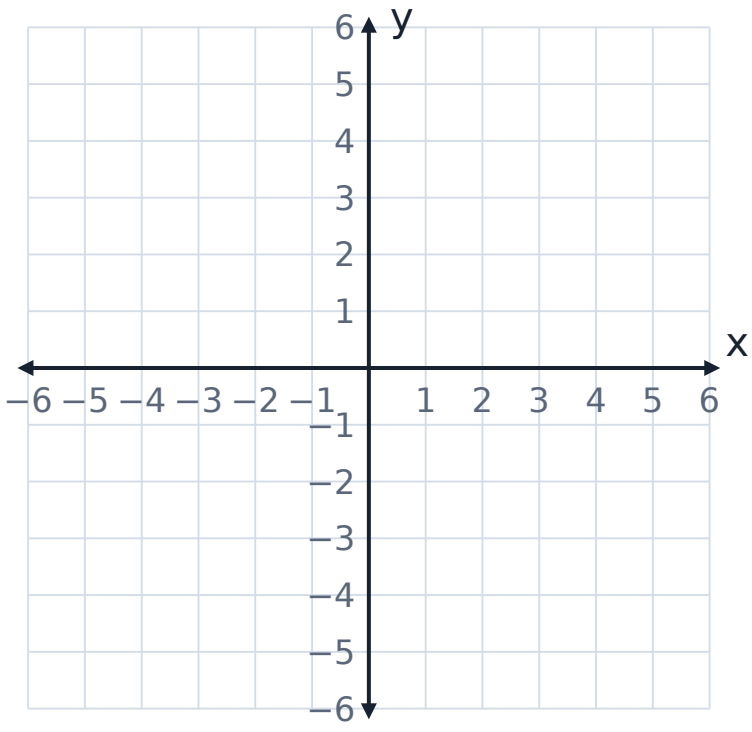
High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 78 • BACKWARD 2 • Q5/5

SET 78 • HSF-IF.B.4

FORWARD 1

Sketch polynomial graphs from a description of their features • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

Sketch a cubic function that falls to the left, rises to the right, and has zeros at −2, 1, and 4. (The y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 78 • FORWARD 1 • Q1/5

A polynomial graph touches the x-axis at $x = 3$ and turns around there. What can you say about the factor $x - 3$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 78 • FORWARD 1 • Q2/5

Which polynomial function has a graph that falls on both ends?

A. $y = -x^4 + 3x$

B. $y = x^3$

C. $y = -x^3$

D. $y = x^4 - 3x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 78 • FORWARD 1 • Q3/5

True or false?

The graph of a degree-5 polynomial with a positive leading coefficient rises to the right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 78 • FORWARD 1 • Q4/5

True or false?

The graph of a cubic function can rise on both ends.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 78 • FORWARD 1 • Q5/5

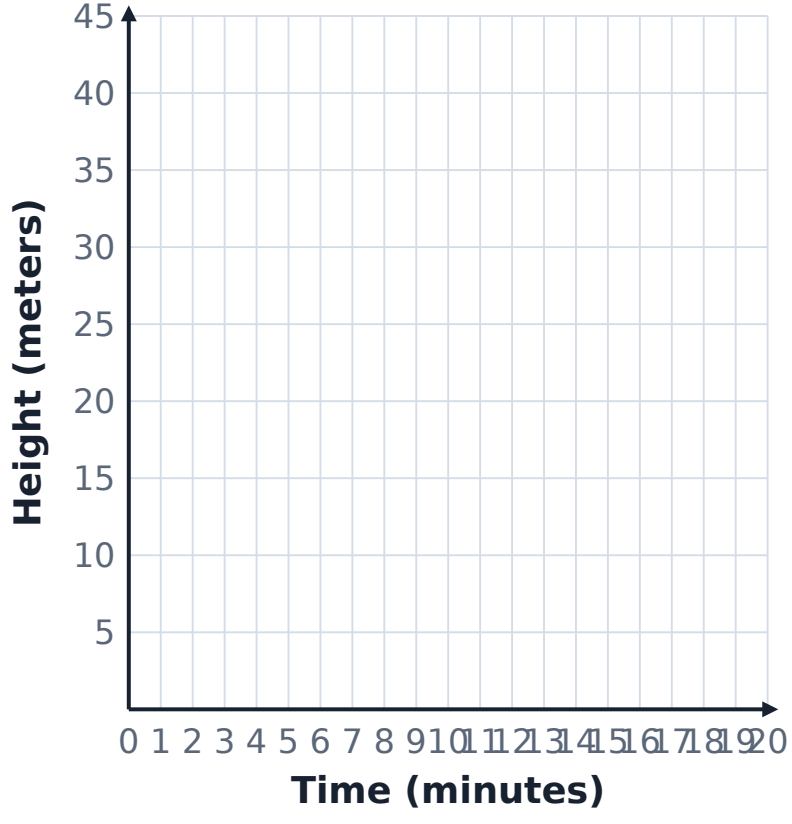
SET 78 • HSF-IF.B.4

FORWARD 2

Sketch periodic graphs from a description • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

A Ferris wheel seat starts at its lowest height, 2 m, reaches its highest height, 42 m, after 5 minutes, and repeats every 10 minutes.

Sketch its height over 20 minutes.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 78 • FORWARD 2 • Q1/5

Daylight lasts 15 hours in late June and 9 hours in late December. Describe a graph of daylight hours over a year.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 78 • FORWARD 2 • Q2/5

Which feature of a periodic graph is the time for one full cycle?

A. The maximum

B. The midline

C. The amplitude

D. The period

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 78 • FORWARD 2 • Q3/5

True or false?

The midline of a periodic graph is halfway between its maximum and minimum values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 78 • FORWARD 2 • Q4/5

True or false?

A tide graph that repeats every 12.4 hours has amplitude 12.4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 78 • FORWARD 2 • Q5/5

SET 39 • HSF-IF.B.5 • INTERPRETING FUNCTIONS

Relate the domain of a function to the quantity it describes

MAIN	Algebra I main question	**HSF-IF.B.5**
BACKWARD 1	Decide whether a solution is viable	**HSA-CED.A.3**
BACKWARD 2	Find where a quadratic model equals 0	**HSA-REI.B.4.b**
FORWARD 1	Use trigonometric ratios as functions of acute angles — nearest r..	**HSG-SRT.C.6**
FORWARD 2	Extend trigonometric functions to all real numbers with the unit ci…	**HSF-TF.A.2**

SET 39 • HSF-IF.B.5

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

$n(k) = 12k$ gives the number of eggs in k full cartons. The store has 30 cartons.

What is a reasonable domain?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q1/5

$h(t) = -16t^2 + 64t$ gives a ball's height in feet from the moment it is hit until it lands.

What is the domain?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q2/5

A taxi fare is $C(m) = 3 + 2.5m$ dollars for m miles, with a limit of 40 miles per ride.

What is the domain?

A. All real numbers

B. $m \geq 3$

C. $3 \leq m \leq 103$

D. $0 \leq m \leq 40$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q3/5

A rectangle has perimeter 20. Its area is $A(w) = w(10 - w)$, where w is the width.

What is the domain?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q4/5

True or false?

P(n) = 50n gives the number of seats on n buses. n = 2.5 is in a reasonable domain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q5/5

SET 39 • HSF-IF.B.5

BACKWARD 1

Decide whether a solution is viable • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3

Solving a problem gives 7.4 buses needed to carry every student.

What is the viable answer? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q1/5

The number of tickets t must satisfy $5t \leq 60$ and $t \geq 0$.

What whole numbers of tickets are viable?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q2/5

Which value is NOT viable for the number of people in a room?

A. 40

B. 0

C. 12

D. −3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q3/5

True or false?

A length of 0.25 meter can be a viable solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q4/5

True or false?

$w(10 - w) = 24$ has solutions $w = 4$ and $w = 6$, but only one of them is a viable width.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q5/5

SET 39 • HSF-IF.B.5

BACKWARD 2

Find where a quadratic model equals 0 • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve $-16t^2 + 64t = 0$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q1/5

Solve $w(10 - w) = 0$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q2/5

Solve $-16t^2 + 64t + 80 = 0$.

A. $t = 4$ or $t = -2$

B. $t = 5$ or $t = -1$

C. $t = 5$ only

D. $t = -5$ or $t = 1$

True or false?

$t = 4$ is a solution of $t^2 - 4t = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q4/5

True or false?

$x^2 + 9 = 0$ has two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q5/5

SET 39 • HSF-IF.B.5

FORWARD 1

Use trigonometric ratios as functions of acute angles • High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Geometry standard. Connection: A trigonometric ratio defined by a right triangle has a domain limited to acute angles, a domain set by the situation.

Right triangle ABC has its right angle at C, with AC = 4, BC = 3, and AB = 5.

Find sin A.

sin A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q1/5 • NEAREST RELATED

For which angle measures θ is sin θ defined as a ratio of sides of a right triangle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q2/5 • NEAREST RELATED

In a right triangle, $\cos A = \frac{12}{13}$.

What is sin A?

A. $\frac{5}{13}$

B. $\frac{12}{5}$

C. $\frac{1}{13}$

D. $\frac{13}{12}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F1-Q3**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

In a right triangle, $\sin A = \cos B$ when A and B are the two acute angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

For an acute angle θ, $\sin\theta$ can be greater than 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 39 • HSF-IF.B.5

FORWARD 2

Extend trigonometric functions to all real numbers with the unit circle • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2

Find cos π.

cos π = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q1/5

Find sin(3π ÷ 2).

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q2/5

With the unit circle, what is the domain of $f(\theta) = \sin \theta$?

A. All real numbers

B. $0 < \theta < \pi \div 2$

C. $\theta \geq 0$

D. $0 \leq \theta \leq 2\pi$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q3/5

True or false?

A negative radian measure gives a clockwise rotation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q4/5

True or false?

sin θ is defined only for acute angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q5/5

SET 40 • HSF-IF.B.6 • INTERPRETING FUNCTIONS

Calculate, interpret, and estimate the average rate of change of a function over an interval

MAIN	Algebra I main question	**HSF-IF.B.6**
BACKWARD 1	Find the rate of change of a linear function	**8.F.B.4**
BACKWARD 2	Compute unit rates with fractions	**7.RP.A.1**
FORWARD 1	Build a periodic model from amplitude, period, and midline	**HSF-TF.B.5**
FORWARD 2	Analyze rational functions: rates of change and asymptotes	**HSF-IF.C.7.d**

SET 40 • HSF-IF.B.6

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6

$f(x) = x^2 + 2x$.

Find the average rate of change from $x = 1$ to $x = 4$.

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q1/5

A town had 12,000 people in 2010 and 15,600 in 2022.

Find the average rate of change, with units.

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q2/5

The table gives the distance d a cart has traveled after t seconds.

Over which 2-second interval was its average speed greatest? Give that speed.

t (seconds)	0	2	4	6
d (meters)	0	30	100	150

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q3/5

What is the average rate of change of $g(x) = 3^x$ from $x = 0$ to $x = 2$?

A. 4

B. 3

C. 8

D. 4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q4/5

True or false?

A linear function has the same average rate of change over every interval.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q5/5

SET 40 • HSF-IF.B.6

BACKWARD 1

Find the rate of change of a linear function • Grade 8 • Functions • 8.F.B.4

Find the rate of change of the line through (2, 5) and (6, 17).

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q1/5

A tank had 50 liters at t = 0 and 20 liters at t = 10 minutes, draining steadily.

What is the rate of change?

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q2/5

What is the rate of change of $y = 7 - 4x$?

A. 7

B. -4

C. $-\frac{4}{7}$

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q3/5

True or false?

A negative rate of change means the quantity is decreasing.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q4/5

True or false?

The rate of change from (0, 3) to (2, 3) is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q5/5

SET 40 • HSF-IF.B.6

BACKWARD 2

Compute unit rates with fractions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A runner goes $\frac{3}{4}$ mile in $\frac{1}{8}$ hour.

What is the speed in miles per hour?

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q1/5

A car uses 12 gallons to go 300 miles.

How many miles per gallon?

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q2/5

A recipe uses $\frac{2}{3}$ cup of sugar for $\frac{1}{3}$ of a batch.

How much sugar is used per batch?

A. $\frac{1}{2}$ cup

B. 1 cup

C. 2 cups

D. $\frac{2}{9}$ cup

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q3/5

True or false?

120 km in 1.5 hours is 80 km per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q4/5

True or false?

$18 for 4 pounds is $4 per pound.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q5/5

SET 40 • HSF-IF.B.6

FORWARD 1

Build a periodic model from amplitude, period, and midline • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

A Ferris wheel has diameter 40 m, its center is 25 m above the ground, and it turns once every 8 minutes. A rider starts at the lowest point.

Write a model $h(t) = A\cos(Bt) + D$ for the rider's height after t minutes.

$h(t) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q1/5

$h(t) = -20\cos(\pi t \div 4) + 25$ gives a rider's height in meters after t minutes.

Give the amplitude, the period, and the midline.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q2/5

Daily high temperatures in a town vary from a low of 4 °C to a high of 28 °C over a 365-day year.

Which values fit a sinusoidal model?

A. Amplitude 24, midline 16, period 365 days

B. Amplitude 16, midline 12, period 365 days

C. Amplitude 12, midline 14, period 182.5 days

D. Amplitude 12, midline 16, period 365 days

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q3/5

True or false?

A model $y = A\cos(Bt) + D$ with period 12 hours has $B = \pi \div 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q4/5

True or false?

A tide that rises from 1 m to 5 m and falls back again has amplitude 4 m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q5/5

SET 40 • HSF-IF.B.6

FORWARD 2

Analyze rational functions: rates of change and asymptotes • High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d

$f(x) = 12 \div x$.

Find the average rate of change from $x = 2$ to $x = 6$.

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q1/5

Give the vertical and horizontal asymptotes of $f(x) = 1 \div (x - 3) + 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q2/5

Where is $f(x) = (x + 1) \div (x - 4)$ undefined?

A. $x = -4$

B. $x = 1$

C. $x = -1$

D. $x = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q3/5

True or false?

The graph of $y = 1 \div x$ gets closer to the x-axis as x gets large.

A. True

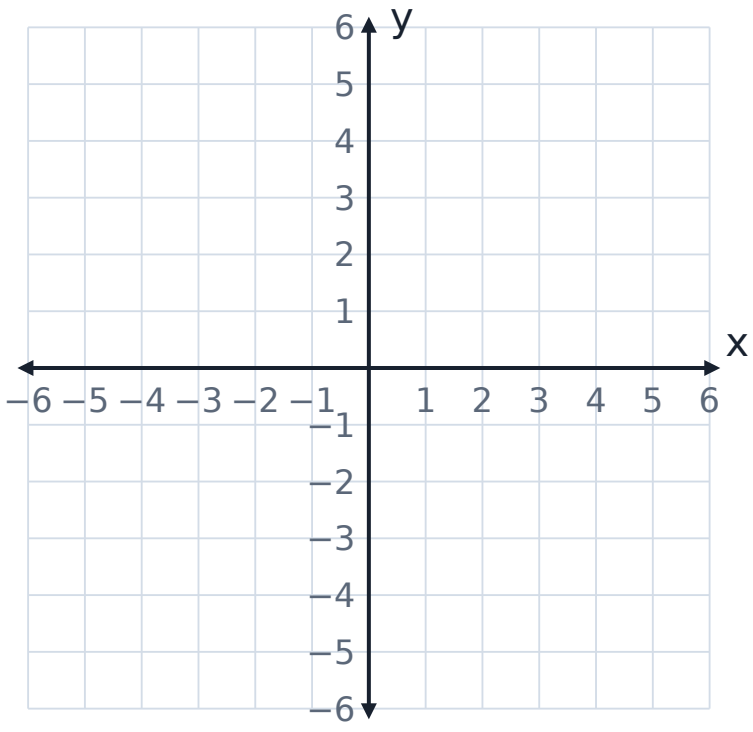
B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q4/5

Graph f(x) = 1 ÷ (x − 2) + 1. Draw both asymptotes as lines and mark the intercepts.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q5/5

SET 79 • HSF-IF.B.6 • INTERPRETING FUNCTIONS

Estimate the average rate of change of a function from its graph

MAIN	Algebra I main question	**HSF-IF.B.6**
BACKWARD 1	Find a rate of change from two points	**8.F.B.4**
BACKWARD 2	Read a unit rate from a graph	**7.RP.A.2.b**
FORWARD 1	Estimate average rates of change from a polynomial graph	**HSF-IF.C.7.c**
FORWARD 2	Estimate average rates of change from a rational-function graph	**HSF-IF.C.7.d**

SET 79 • HSF-IF.B.6

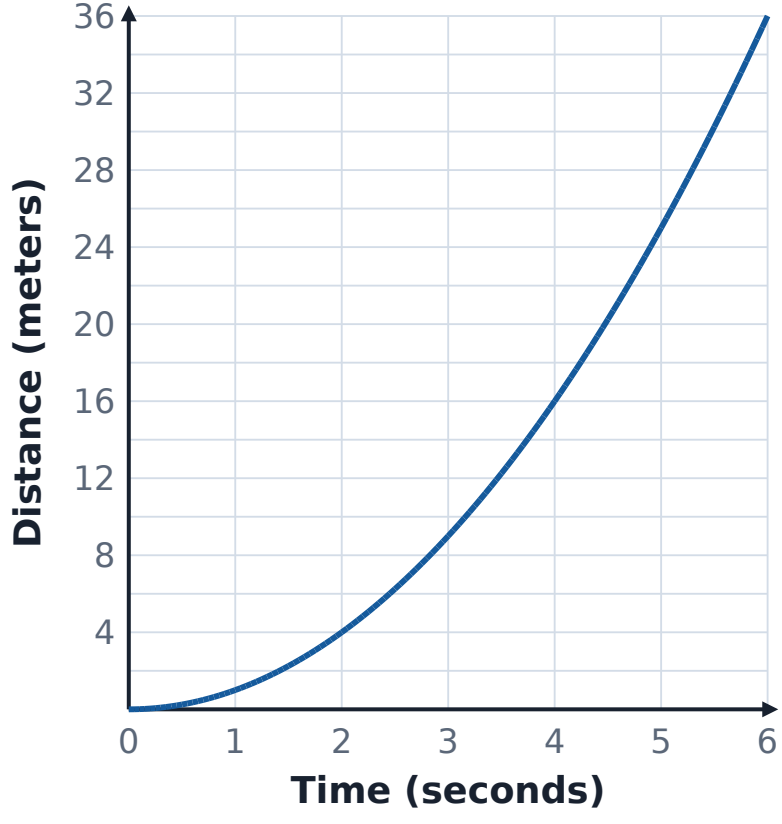
MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6

The graph shows the distance d, in meters, that a cart has rolled after t seconds.

Estimate the average rate of change from t = 1 to t = 3, with units.

Rate: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

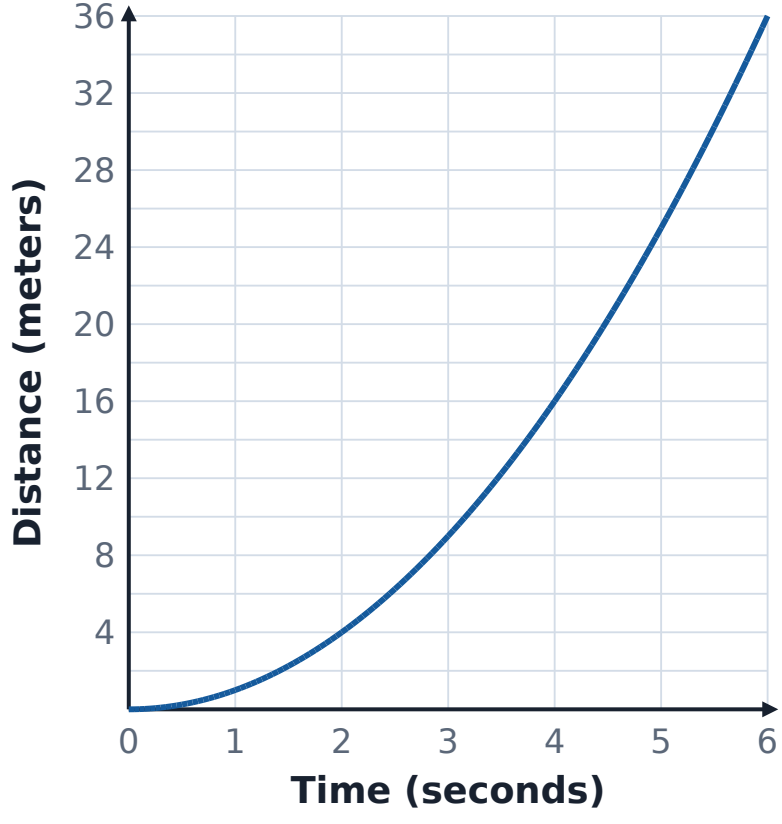
ID S79-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 79 • MAIN • Q1/5

The graph shows the distance d, in meters, that a cart has rolled after t seconds.

Estimate the average rate of change from t = 3 to t = 5. Is the cart speeding up?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

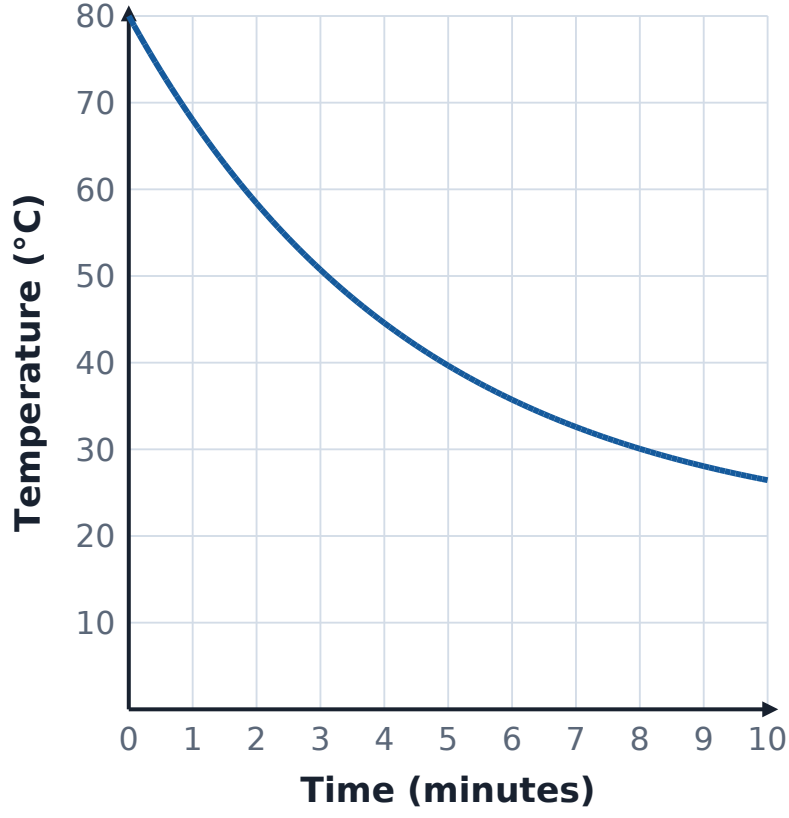
ID S79-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 79 • MAIN • Q2/5

The graph shows the temperature of a cup of coffee.

Estimate the average rate of change of the temperature over the first 5 minutes, with units.

Rate: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 79 • MAIN • Q3/5

The graph shows the temperature of a cup of coffee.

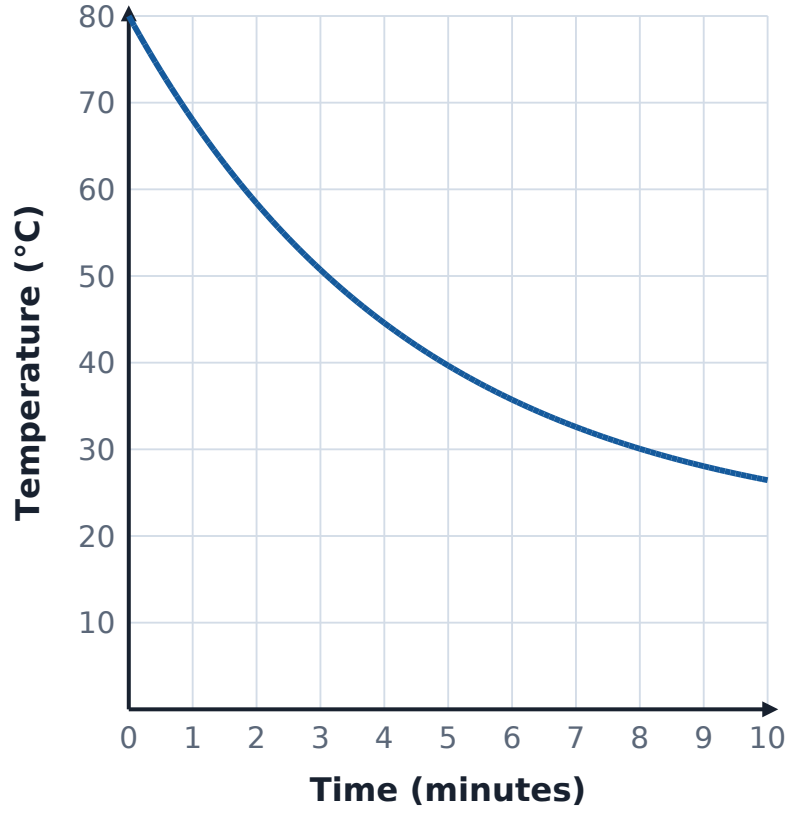
Over which 2-minute interval is the coffee cooling fastest on average?

A. 4 to 6 minutes

B. 8 to 10 minutes

C. 0 to 2 minutes

D. 6 to 8 minutes



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

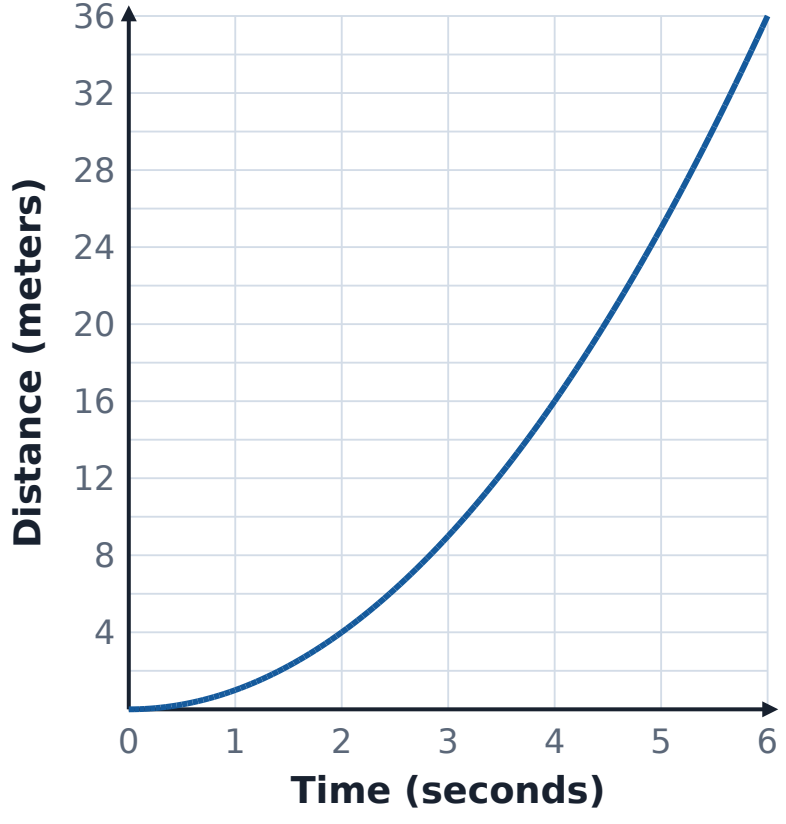
ID S79-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 79 • MAIN • Q4/5

True or false?

The average rate of change of d from t = 0 to t = 6 equals the slope of the segment joining (0, 0) and (6, 36) on the graph.

A. True **B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 79 • MAIN • Q5/5

SET 79 • HSF-IF.B.6

BACKWARD 1

Find a rate of change from two points • Grade 8 • Functions • 8.F.B.4

A line passes through (2, 10) and (6, 30). Find its rate of change.

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 79 • BACKWARD 1 • Q1/5

A straight-line graph of a hike shows 3 km at 1 hour and 9 km at 3 hours. What is the rate of change, with units?

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 79 • BACKWARD 1 • Q2/5

What is the slope of the line through (0, 50) and (5, 25)?

A. 5

B. -5

C. 0.2

D. -25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 79 • BACKWARD 1 • Q3/5

True or false?

The slope between two points on a line is (change in y) ÷ (change in x).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 79 • BACKWARD 1 • Q4/5

True or false?

The line through (1, 4) and (3, 4) has rate of change 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 79 • BACKWARD 1 • Q5/5

SET 79 • HSF-IF.B.6

BACKWARD 2

Read a unit rate from a graph • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

The graph of a proportional relationship passes through (4, 12). What is the unit rate?

Unit rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 79 • BACKWARD 2 • Q1/5

A proportional cost graph passes through (0, 0) and (5, 7.5). What is the cost per item?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 79 • BACKWARD 2 • Q2/5

A proportional graph of miles against gallons passes through (3, 90). What is the unit rate?

A. 3 miles per gallon

B. 270 miles per gallon

C. 90 miles per gallon

D. 30 miles per gallon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 79 • BACKWARD 2 • Q3/5

True or false?

On a proportional graph, the point (1, r) shows the unit rate r.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 79 • BACKWARD 2 • Q4/5

True or false?

A proportional graph passes through (2, 5), so its unit rate is 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 79 • BACKWARD 2 • Q5/5

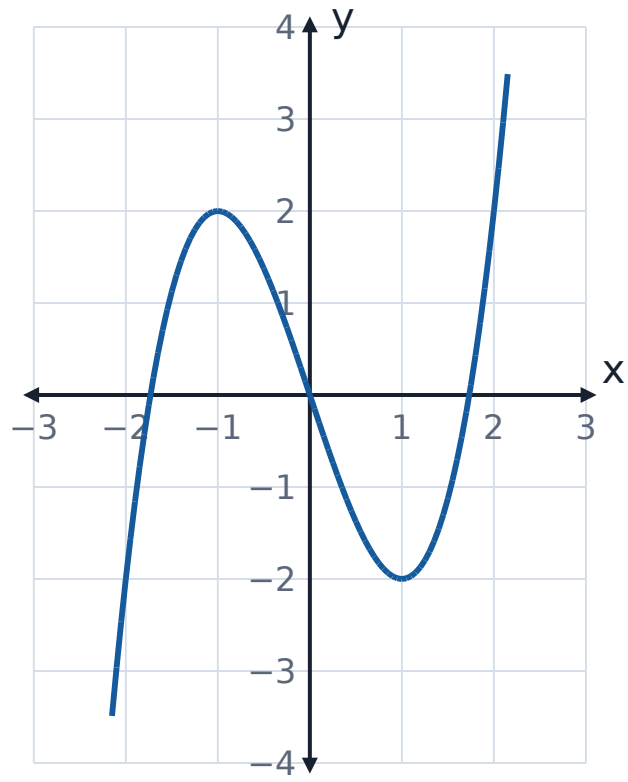
SET 79 • HSF-IF.B.6

FORWARD 1

Estimate average rates of change from a polynomial graph • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

The graph shows $f(x) = x^3 - 3x$.

Estimate the average rate of change from $x = -1$ to $x = 1$.



Rate: _____

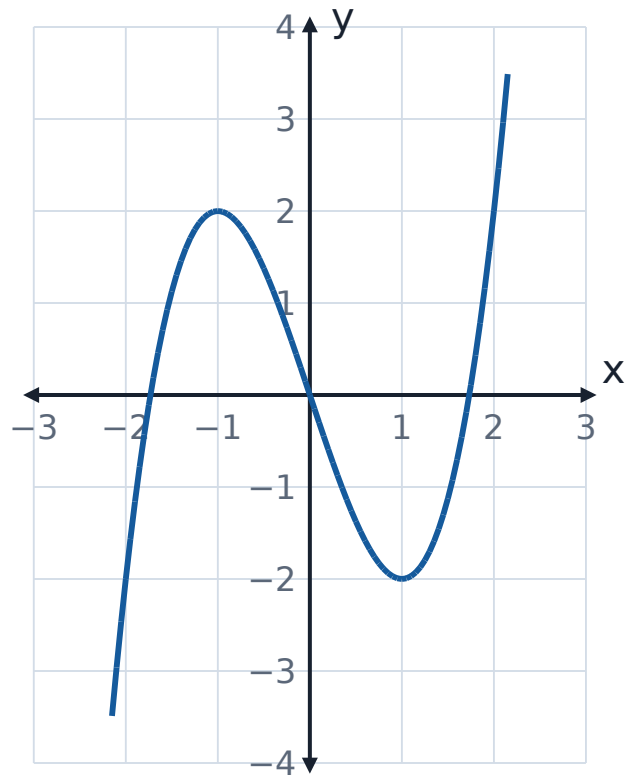
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 79 • FORWARD 1 • Q1/5

The graph shows $f(x) = x^3 - 3x$.

Estimate the average rate of change from $x = 1$ to $x = 2$.



Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 79 • FORWARD 1 • Q2/5

The graph shows $f(x) = x^3 - 3x$.

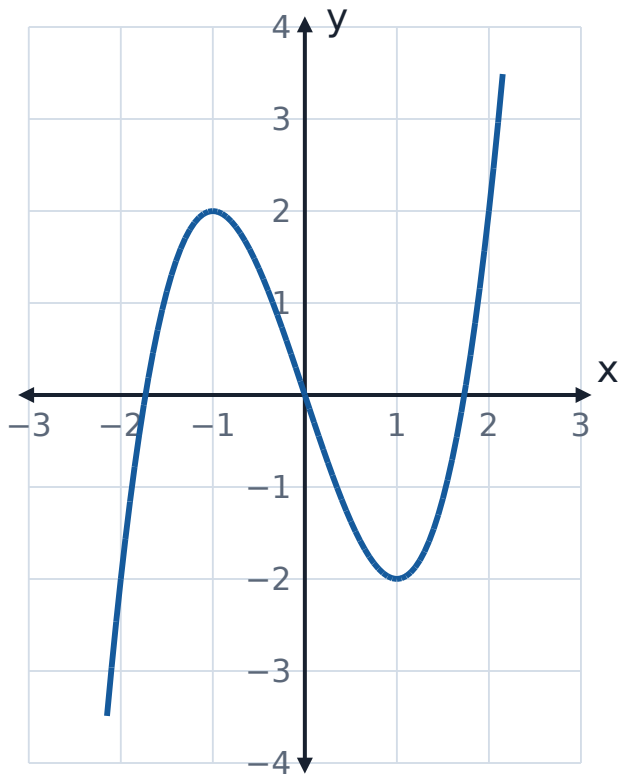
On which interval is f increasing?

A. $-1 < x < 1$

B. $1 < x < 2$

C. $-1 < x < 0$

D. $0 < x < 1$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q3

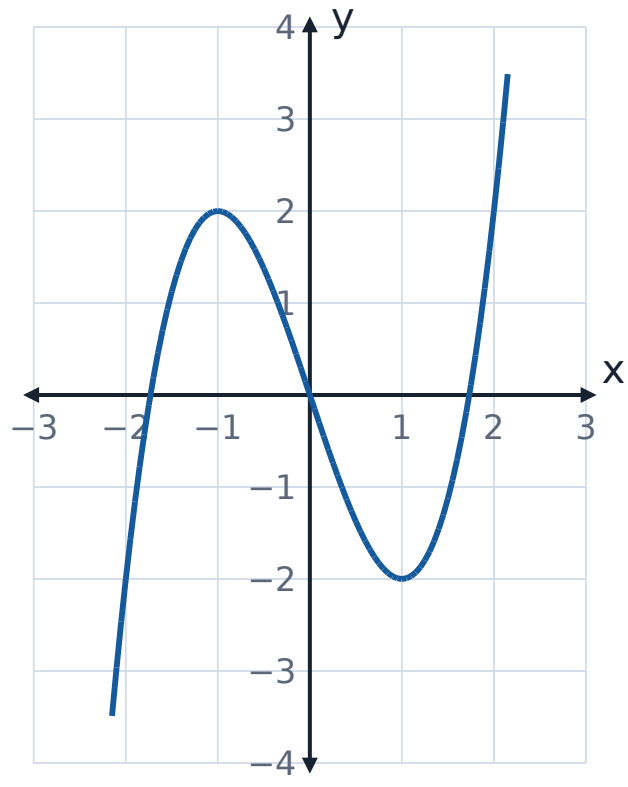
High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 79 • FORWARD 1 • Q3/5

True or false?

The average rate of change of $f(x) = x^3 - 3x$ from $x = -2$ to $x = 2$ is 1.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 79 • FORWARD 1 • Q4/5

True or false?

A positive average rate of change on an interval means the function increases at every point of the interval.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 79 • FORWARD 1 • Q5/5

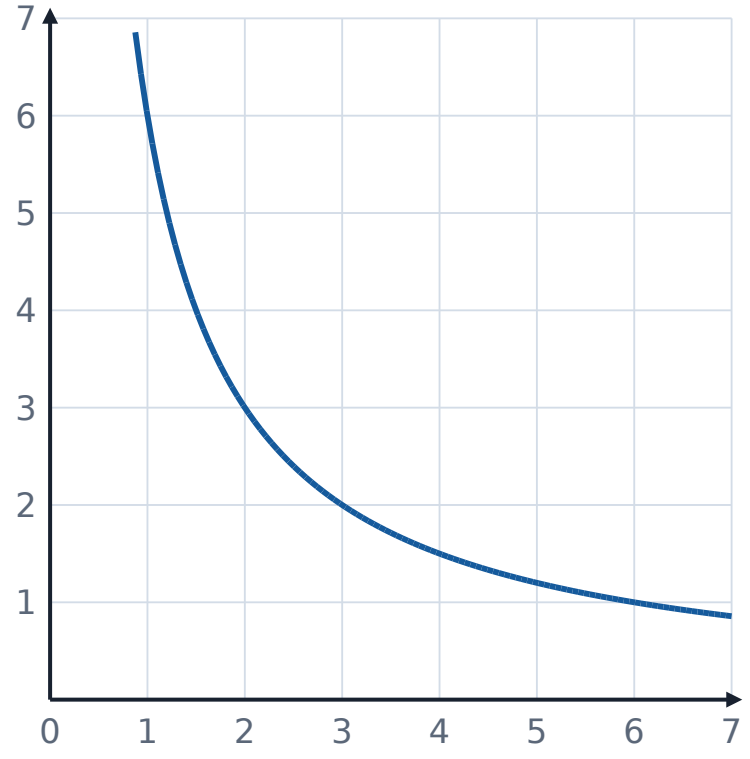
SET 79 • HSF-IF.B.6

FORWARD 2

Estimate average rates of change from a rational-function graph • High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d

The graph shows f(x) = 6 ÷ x.

Estimate the average rate of change from x = 1 to x = 3.



Rate: _____

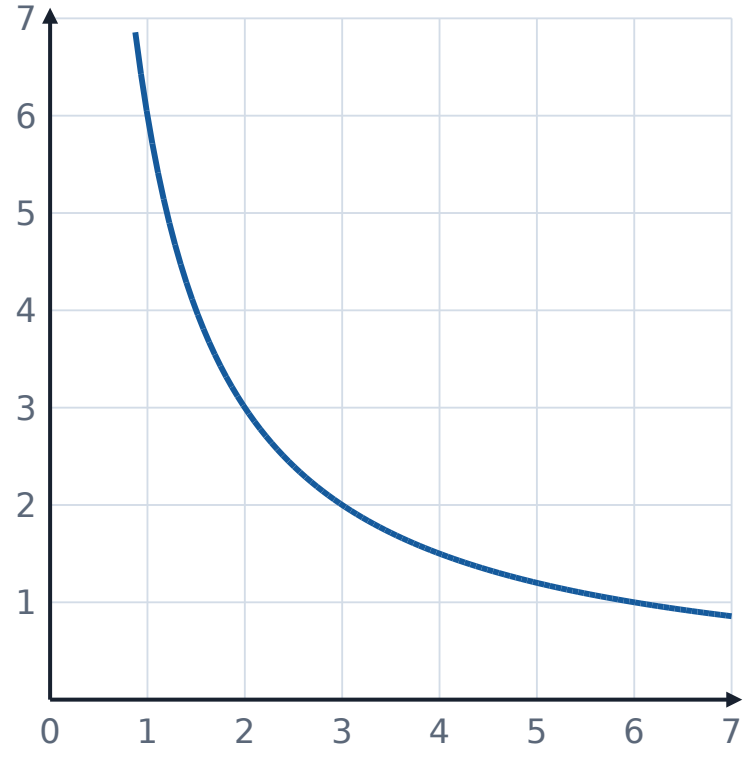
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 79 • FORWARD 2 • Q1/5

The graph shows f(x) = 6 ÷ x.

Estimate the average rate of change from x = 3 to x = 6.



Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 79 • FORWARD 2 • Q2/5

The graph shows f(x) = 6 ÷ x.

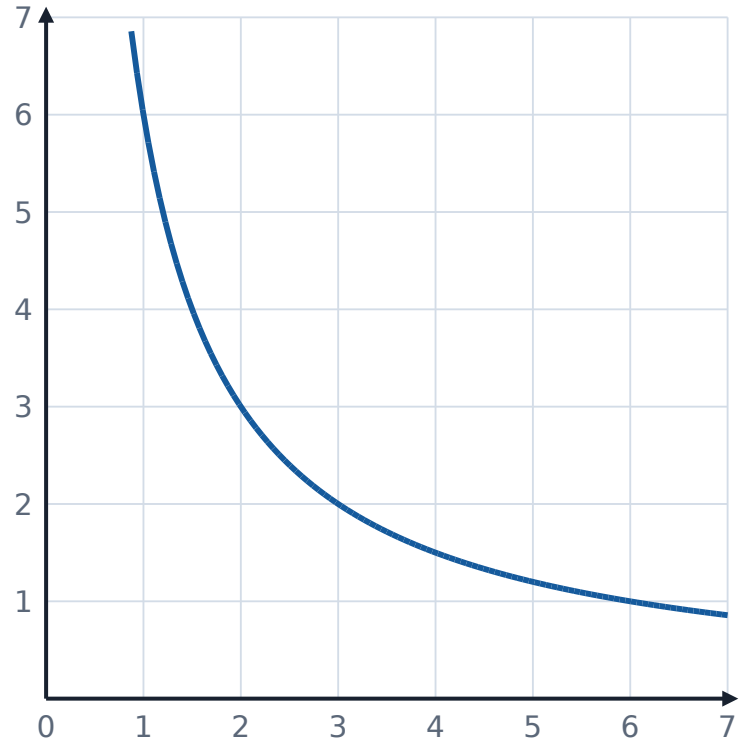
What happens to its average rate of change on intervals farther to the right?

A. It gets closer to 0.

B. It becomes positive.

C. It stays −2.

D. It gets more negative.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 79 • FORWARD 2 • Q3/5

True or false?

The graph of $y = 6 \div x$ has the horizontal asymptote $y = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 79 • FORWARD 2 • Q4/5

True or false?

The graph of $y = 6 \div x$ passes through (0, 6).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 79 • FORWARD 2 • Q5/5

SET 41 • HSF-IF.C.7.a • INTERPRETING FUNCTIONS

Graph linear functions and show their intercepts

MAIN	Algebra I main question	**HSF-IF.C.7.a**
BACKWARD 1	Find slope and write y = mx + b	**8.EE.B.6**
BACKWARD 2	Write linear equations in two variables from a situation	**HSA-CED.A.2**
FORWARD 1	Find a point that partitions a segment in a given ratio	**HSG-GPE.B.6**
FORWARD 2	Find and verify inverses of linear functions	**HSF-BF.B.4.b**

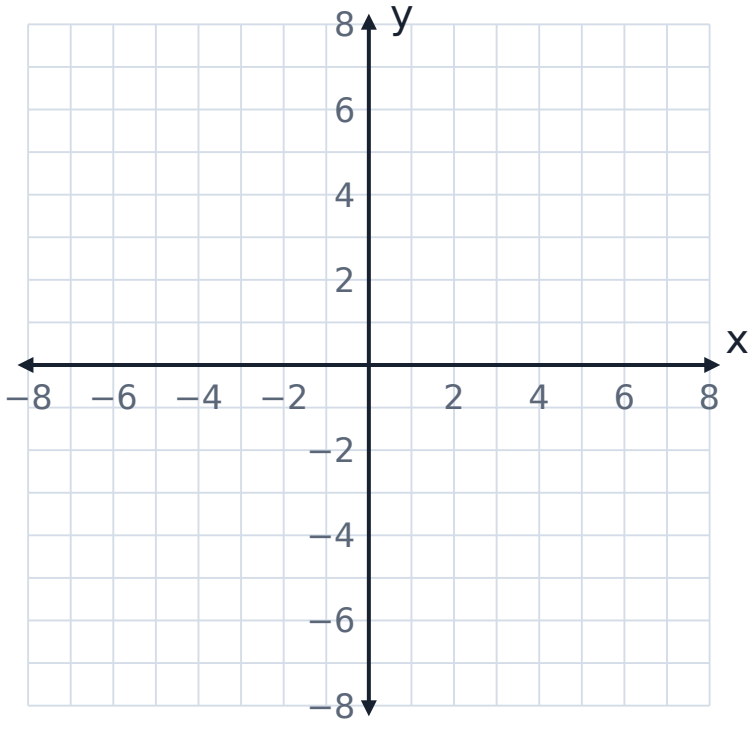
SET 41 • HSF-IF.C.7.a

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a

Graph $f(x) = -\frac{2}{3}x + 4$ on the coordinate plane.

Give the x-intercept and the y-intercept.



Intercepts: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q1/5

Find the x- and y-intercepts of $3x - 5y = 15$.

Intercepts: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q2/5

Write the equation of the line through $(-2, 7)$ and $(4, -5)$ in slope-intercept form.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q3/5

Which describes the graph of $y = -3$?

A. A line with slope 1 through $(-3, -3)$

B. A vertical line through $(-3, 0)$

C. A line with slope -3 through the origin

D. A horizontal line through $(0, -3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q4/5

True or false?

The graph of $2x + y = 6$ has x-intercept 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q5/5

SET 41 • HSF-IF.C.7.a

BACKWARD 1

Find slope and write y = mx + b • Grade 8 • Expressions & Equations • 8.EE.B.6

Find the slope of the line through (1, 2) and (3, 8).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q1/5

A line has slope 2 and y-intercept −1.

Write its equation.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q2/5

What is the y-intercept of $y = 5x - 4$?

A. $\frac{4}{5}$

B. 4

C. 5

D. -4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q3/5

True or false?

Any two distinct points on a nonvertical line give the same slope.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q4/5

True or false?

The line $y = 4x$ passes through $(1, 5)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q5/5

SET 41 • HSF-IF.C.7.a

BACKWARD 2

Write linear equations in two variables from a situation • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2

A gym charges $25 to join plus $15 per month.

Write an equation for the cost C after m months.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q1/5

A candle 20 cm tall burns 2 cm per hour.

Write an equation for its height h after t hours.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q2/5

Ana earns $8 per hour plus $5 in tips.

Which equation gives her earnings E for h hours?

A. E = 8h + 5

B. E = 5h + 8

C. E = 13h

D. E = 8(h + 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q3/5

True or false?

d = 60t gives the distance traveled at 60 miles per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q4/5

True or false?

$y = 100 - 4x$ describes a quantity that grows by 4 each step.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q5/5

SET 41 • HSF-IF.C.7.a

FORWARD 1

Find a point that partitions a segment in a given ratio • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6

Find the midpoint of the segment from $(-2, 3)$ to $(6, 7)$.

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q1/5

Find the point $\frac{1}{4}$ of the way from A(0, 0) to B(8, 12).

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q2/5

Point P divides the segment from A(1, 2) to B(7, 11) in the ratio 1 : 2 (AP : PB).

Find P.

A. (5, 8)

B. (3, 5)

C. (4, 6.5)

D. (2, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q3/5

True or false?

The midpoint of (0, 0) and (10, 4) is (5, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q4/5

True or false?

The point $\frac{1}{3}$ of the way from (0, 0) to (9, 6) is (6, 4).

A. True

B. False

SET 41 • HSF-IF.C.7.a

FORWARD 2

Find and verify inverses of linear functions • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b

Find the inverse of $f(x) = 3x - 6$.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q1/5

Show that f(x) = 2x + 1 and g(x) = (x − 1) ÷ 2 are inverses.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q2/5

Which is the inverse of $f(x) = x \div 4 + 1$?

A. $f^{-1}(x) = 4x + 1$

B. $f^{-1}(x) = 4(x + 1)$

C. $f^{-1}(x) = 4x - 4$

D. $f^{-1}(x) = x \div 4 - 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q3/5

True or false?

If $f(g(x)) = x$ and $g(f(x)) = x$ for every x, then f and g are inverses.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q4/5

True or false?

The inverse of $f(x) = x + 5$ is $f^{-1}(x) = x + 5$.

A. True

B. False

SET 42 • HSF-IF.C.7.a • INTERPRETING FUNCTIONS

Graph quadratic functions and show intercepts, maxima, and minima

MAIN	Algebra I main question	**HSF-IF.C.7.a**
BACKWARD 1	Factor to find the zeros of a quadratic	**HSA-SSE.B.3.a**
BACKWARD 2	Interpret key features of a function	**HSF-IF.B.4**
FORWARD 1	Relate a parabola's equation to its focus and directrix	**HSG-GPE.A.2**
FORWARD 2	Use factors to describe the graph of a polynomial	**HSF-IF.C.7.c**

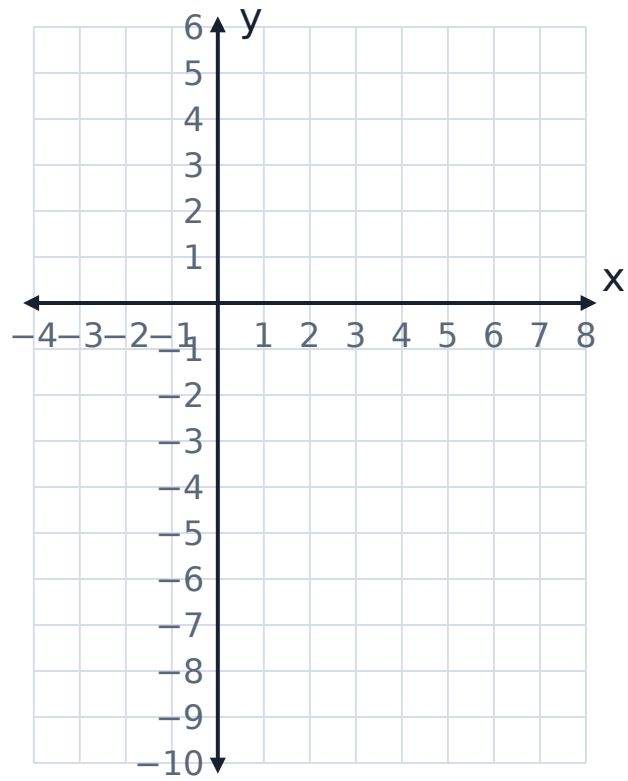
SET 42 • HSF-IF.C.7.a

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a

Graph $f(x) = x^2 - 4x - 5$.

Give the vertex, the x-intercepts, and the y-intercept.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q1/5

Find the vertex and the axis of symmetry of $y = 2x^2 + 8x + 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q2/5

Does $y = -3x^2 + 6x - 1$ have a maximum or a minimum? Find it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q3/5

What are the x-intercepts of $y = (x + 3)(x - 1)$?

A. (3, 0) and (−1, 0)

B. (−3, 0) and (1, 0)

C. (−3, 0) and (−1, 0)

D. (0, −3) only

True or false?

The graph of $y = x^2 + 4$ has no x-intercepts.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q5/5

SET 42 • HSF-IF.C.7.a

BACKWARD 1

Factor to find the zeros of a quadratic • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a

Factor $x^2 - 4x - 5$, and give its zeros.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q1/5

Find the zeros of $2x^2 - 8$.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q2/5

What are the zeros of $x^2 + 6x + 8$?

A. 2 and −4

B. −2 and −4

C. 2 and 4

D. −2 and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q3/5

True or false?

The zeros of $(x - 7)(x + 2)$ are 7 and -2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q4/5

True or false?

The zeros of $x^2 - 9x$ are 0 and -9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q5/5

SET 42 • HSF-IF.C.7.a

BACKWARD 2

Interpret key features of a function • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4

A table has x = −2, −1, 0, 1, 2 and y = 5, 2, 1, 2, 5.

Describe its symmetry and its minimum.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q1/5

A profit function P(x) is positive for $10 < x < 50$, where x is the number of items sold.

What does that mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q2/5

The highest point of a graph is (3, 40).

Which statement describes it?

A. A maximum of 40 at x = 3

B. A maximum of 3 at x = 40

C. A minimum of 3 at x = 40

D. A y-intercept of 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q3/5

True or false?

A function is increasing on an interval when its outputs get larger as its inputs get larger.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q4/5

True or false?

The x-intercepts of a graph are where the output is greatest.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q5/5

SET 42 • HSF-IF.C.7.a

FORWARD 1

Relate a parabola's equation to its focus and directrix • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2

Find the focus and directrix of $y = \frac{1}{8}x^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q1/5

Derive the equation of the parabola with focus $(0, -1)$ and directrix $y = 1$ from the distance definition.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q2/5

A parabola has focus (0, 3) and directrix $y = -3$.

Where is its vertex?

A. (0, 0)

B. (0, 3)

C. (0, −3)

D. (3, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q3/5

True or false?

The parabola $x^2 = 12y$ opens upward.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q4/5

True or false?

The parabola with focus (0, 1) and directrix $y = -1$ is $y = x^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q5/5

SET 42 • HSF-IF.C.7.a

FORWARD 2

Use factors to describe the graph of a polynomial • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

$f(x) = (x + 1)(x - 2)(x - 4)$.

Find the zeros and the y-intercept.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q1/5

At most how many turning points can the graph of a degree-3 polynomial have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q2/5

$f(x) = (x - 3)^2(x + 1)$.

What does the graph do at $x = 3$?

A. It touches the x-axis and turns around.

B. It is undefined.

C. It has a vertical asymptote.

D. It crosses the x-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q3/5

True or false?

$f(x) = x^3 - 8$ has exactly one real zero.

A. True

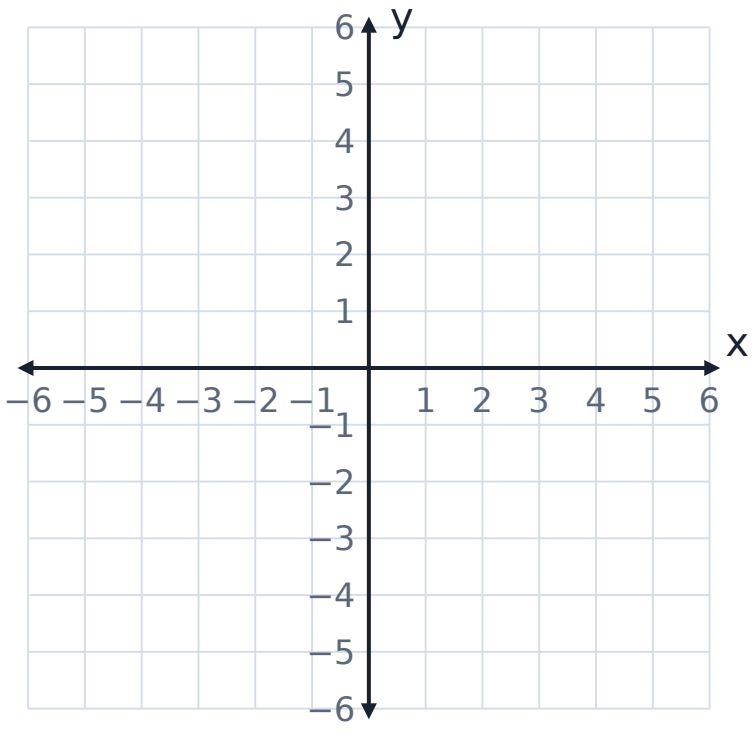
B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q4/5

Graph $f(x) = (x + 2)(x - 1)^2$. Mark the zeros and the y-intercept, and show what the graph does at each zero. (Use the grid for x; the y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q5/5

SET 43 • HSF-IF.C.7.b • INTERPRETING FUNCTIONS

Graph absolute value functions and show their key features

MAIN	Algebra I main question	**HSF-IF.C.7.b**
BACKWARD 1	Find the domain and range of a function	**HSF-IF.A.1**
BACKWARD 2	Understand absolute value as distance from 0	**6.NS.C.7.c**
FORWARD 1	Solve radical equations and check for extraneous solutions	**HSA-REI.A.2**
FORWARD 2	Read values of inverses of power and root functions	**HSF-BF.B.4.c**

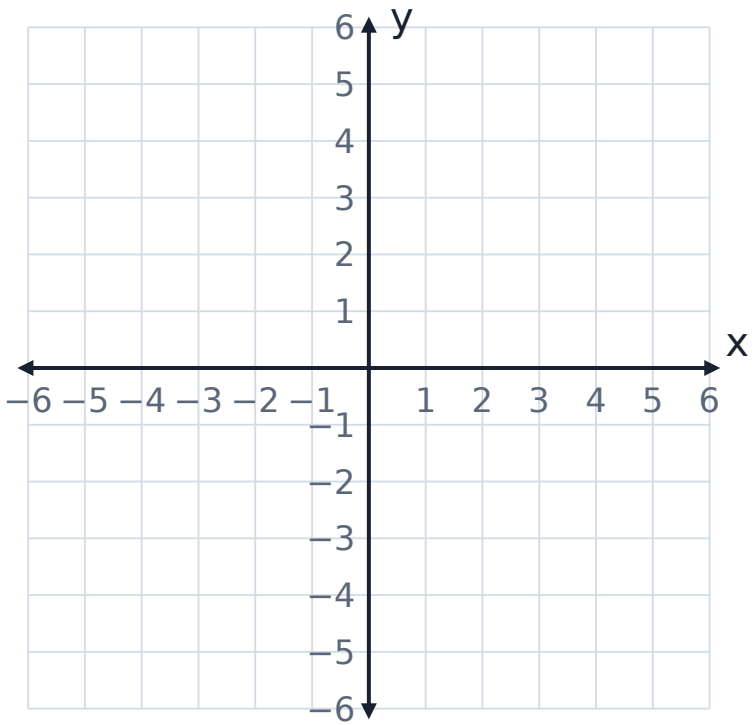
SET 43 • HSF-IF.C.7.b

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b

Graph $f(x) = |x - 2| - 3$.

Give the vertex and the x-intercepts.



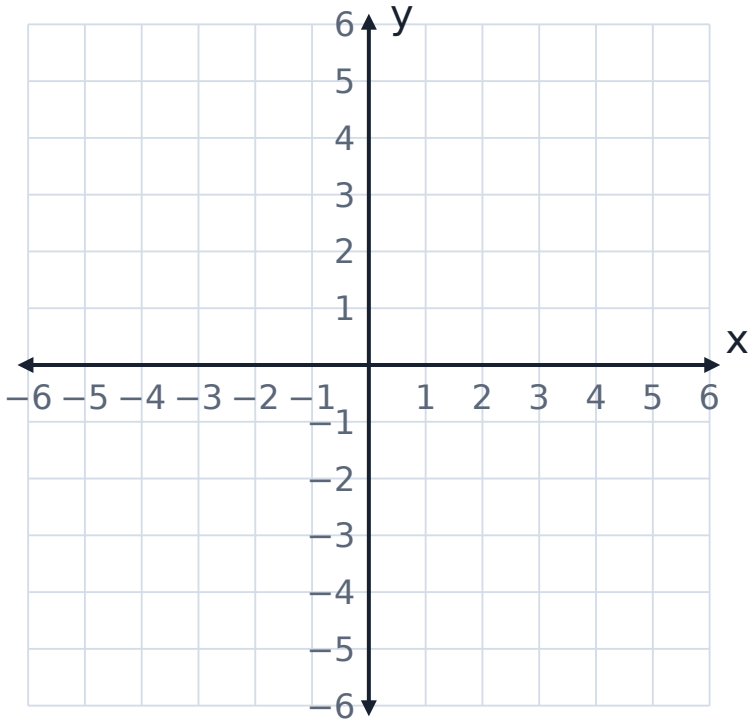
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q1/5

Graph $g(x) = -2|x + 1| + 4$, and label its vertex and its x-intercepts.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q2/5

Describe how the graph of $y = |x + 3| + 1$ is related to the graph of $y = |x|$, and give its range.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q3/5

Which function's graph is a V opening downward with vertex (2, 5)?

A. $y = |x - 2| + 5$

B. $y = -|x + 2| + 5$

C. $y = -|x - 5| + 2$

D. $y = -|x - 2| + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q4/5

True or false?

The graph of $y = 3|x|$ is narrower than the graph of $y = |x|$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q5/5

SET 43 • HSF-IF.C.7.b

BACKWARD 1

Find the domain and range of a function • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1

Over the real numbers, what is the domain of $f(x) = \sqrt{(x - 5)}$?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q1/5

What is the range of f(x) = |x|?

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q2/5

Which value is NOT in the domain of $g(x) = \sqrt{(6 - x)}$?

A. 7

B. -3

C. 0

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q3/5

True or false?

Every real number is in the domain of $f(x) = \sqrt[3]{x}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q4/5

True or false?

The range of $f(x) = x^2$ is all real numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q5/5

SET 43 • HSF-IF.C.7.b

BACKWARD 2

Understand absolute value as distance from 0 • Grade 6 • The Number System • 6.NS.C.7.c

What is $|-9|$?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q1/5

Which is farther from 0, −7 or 5? Explain with absolute value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q2/5

Which value equals $|3 - 8|$?

A. 5

B. -5

C. 11

D. -11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q3/5

True or false?

$|-4| = 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q4/5

True or false?

$|0| = 0$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q5/5

SET 43 • HSF-IF.C.7.b

FORWARD 1

Solve radical equations and check for extraneous solutions • High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

Solve $\sqrt{(x - 2)} = 4$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q1/5

Solve $\sqrt{(x + 6)} = x$. Name any extraneous solution.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q2/5

Solve $\sqrt[3]{(x + 1)} = 2$.

A. $x = 7$

B. $x = 3$

C. $x = 9$

D. $x = 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F1-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q3/5

True or false?

$\sqrt{x} + 3 = 0$ has no real solution.

A. True

B. False

True or false?

$x = 16$ is a solution of $\sqrt{x} = -4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q5/5

SET 43 • HSF-IF.C.7.b

FORWARD 2

Read values of inverses of power and root functions • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c

$g(x) = x^3$.

Find $g^{-1}(27)$.

$g^{-1}(27) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q1/5

The point (4, 2) is on the graph of $f(x) = \sqrt{x}$.

Which point must be on the graph of f^{-1}?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q2/5

$f(x) = x^2$ with domain $x \geq 0$.

What is $f^{-1}(49)$?

A. -7

B. 24.5

C. 2,401

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q3/5

True or false?

The inverse of $f(x) = x^3$ is $f^{-1}(x) = \sqrt[3]{x}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q4/5

True or false?

f is a one-to-one function. If (3, 10) is on the graph of f, then (3, 10) must be on the graph of f^{-1}.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q5/5

SET 74 • HSF-IF.C.7.b • INTERPRETING FUNCTIONS

Graph square root and cube root functions

MAIN	Algebra I main question	**HSF-IF.C.7.b**
BACKWARD 1	Find domains and ranges of root functions	**HSF-IF.A.1**
BACKWARD 2	Evaluate square roots and cube roots	**8.EE.A.2**
FORWARD 1	Solve radical equations and check the solutions	**HSA-REI.A.2**
FORWARD 2	Find inverses of square root and cube root functions	**HSF-BF.B.4.c**

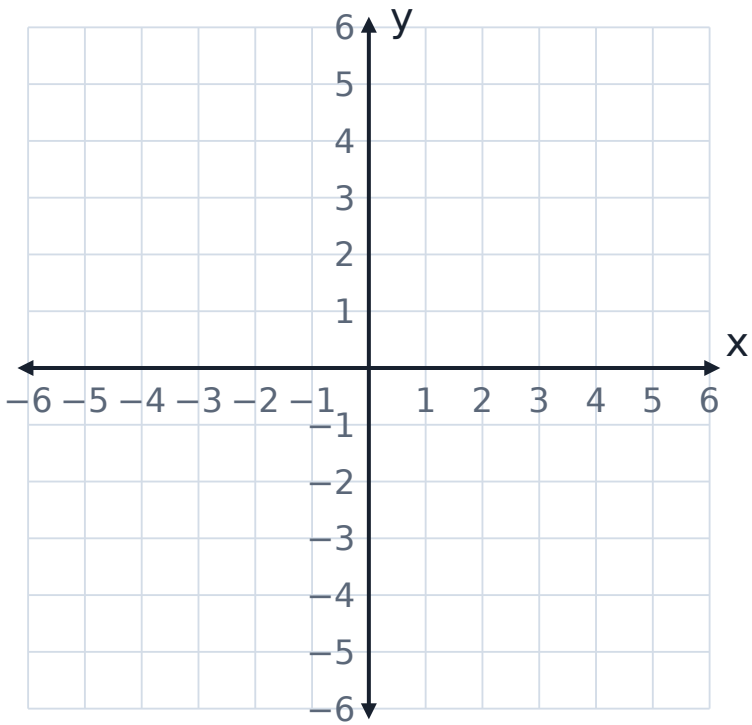
SET 74 • HSF-IF.C.7.b

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b

Graph $f(x) = \sqrt{(x + 4)}$.

Give its domain and range.



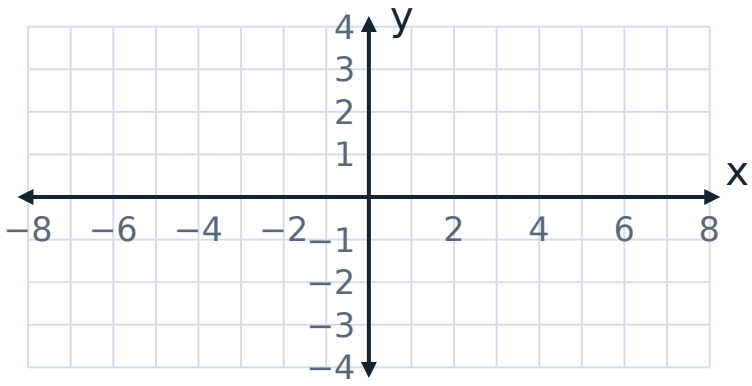
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 74 • MAIN • Q1/5

Graph $g(x) = \sqrt[3]{x}$ for $-8 \le x \le 8$. Plot the points where x is a perfect cube, then draw the curve.



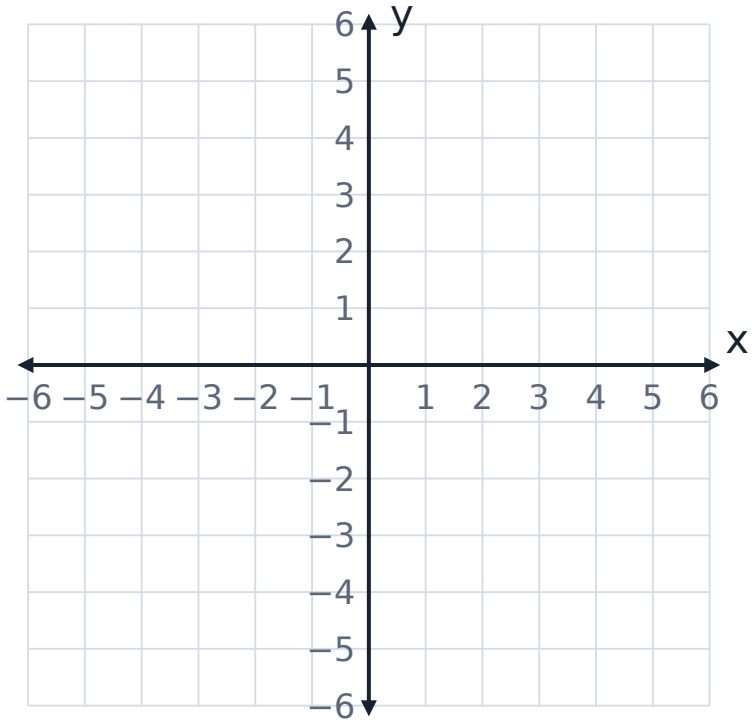
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 74 • MAIN • Q2/5

Graph $h(x) = -\sqrt{x} + 3$.

Give its starting point and its range.



Answer: _____

Which describes the graph of $y = \sqrt[3]{x}$?

A. It starts at (0, 0) and goes only to the right.

B. It has the horizontal asymptote $y = 0$.

C. It passes through $(-8, -2)$, (0, 0), and (8, 2).

D. It is V-shaped with vertex (0, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 74 • MAIN • Q4/5

True or false?

The graph of $y = \sqrt{(x - 1)}$ starts at $(-1, 0)$.

A. True

B. False

SET 74 • HSF-IF.C.7.b

BACKWARD 1

Find domains and ranges of root functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1

Over the real numbers, what is the domain of $g(x) = \sqrt{(2x - 6)}$?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 74 • BACKWARD 1 • Q1/5

What is the range of $h(x) = \sqrt{x} + 2$?

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 74 • BACKWARD 1 • Q2/5

What is the domain of $y = \sqrt[3]{(x - 5)}$?

A. All real numbers

B. $x \leq 5$

C. $x > 5$

D. $x \geq 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 74 • BACKWARD 1 • Q3/5

True or false?

The range of $y = -\sqrt{x}$ is $y \leq 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 74 • BACKWARD 1 • Q4/5

True or false?

$x = -1$ is in the domain of $y = \sqrt{x}$ over the real numbers.

A. True

B. False